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Research

This Month in Geopolitics: July 2026

Key events, themes and trends to watch in the month ahead

July 2026

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Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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At a Glance: Geopolitical events to watch in July 2026

Our monthly Look-Ahead covers a *non-exhaustive* list of key events, issues and themes to watch in the month ahead. The below map intends to serve as an executive summary, providing a list of the issues at a glance. These are covered in more detail in subsequent slides.



In the spotlight for July 2026: US-Iran peace talks and China-EU trade tensions



US-Iran peace talks (all of July)

The US and Iran are keen to press ahead with the framework agreement even if this doesn't lead to a full-scale peace agreement. This process is unlikely to be without friction. There are many moving parts in negotiations and we should watch closely for disagreements over implementation, Gulf states' requirements for security guarantees, and domestic political opposition from key parties. A critical factor to watch will be the potential for Israeli aspirations in Lebanon to upend the deal. We suspect that there is significant US pressure on President Netanyahu to cease operations and accept the accrued gains to prevent a derailing of the deal.

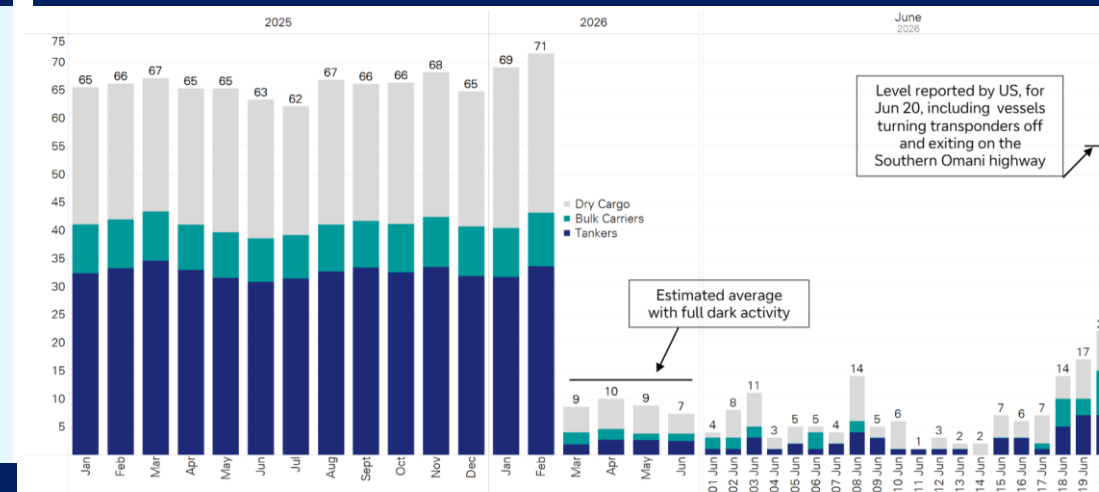
Shipping levels through the Strait of Hormuz have increased (see top chart), but **traffic will likely take time to recover to pre-war levels** given the challenges of de-mining, a lack of trust between the US and Iran, and Iranian [threats](#). Infrastructure repair to ports, terminals and energy infrastructure, as well as the provision of security guarantees are other factors to watch. While crude oil prices have recovered to pre-war levels, full operational normalization, involving infrastructure repairs, [is not expected](#) until 2027 and beyond.

China-EU trade tensions (all of July)

While efforts have been initiated to avoid a trade war, the EU continues to implement new measures against China. On 29 JUN, the EU and China agreed to a 3-month dialogue with a commitment to reconvene in October on four "workstreams" to make progress on trade disagreements, including export controls, trade and investment balancing, and a joint monitoring mechanism to improve transparency and manage trade frictions.

At the same time, the EU Council **has reportedly tasked the European Commission with developing new trade defence tools** implicitly aimed at addressing trade imbalances with China. The trade imbalance, along with perceptions that China's currency is [undervalued](#), is a growing political concern for the bloc (see bottom chart). **Two new EU measures implicitly aimed at China entered into force on 1 JUL.** The first is a temporary €3 (US\$3.4) customs duty per item levied on low-cost shipments entering the bloc. The measure intends to address imports from Chinese e-commerce companies and will remain in force until 1 JUL 2028. **New rules intended to protect the EU steel market from global overcapacity** entered into force, which include a revised quota system. China is the world's largest steel producer, accounting for around 50% of global output.

Blue-water commercial cargo tankers exit the Strait of Hormuz (from 2025) version count

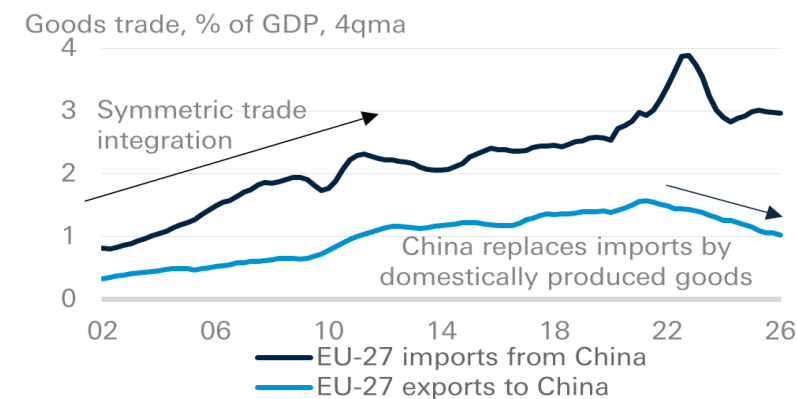


Source: dbDataInsights

Note: Pre-crisis figures do not account for 'dark activity' (estimated 3-6 vessels per day with transponders off, mainly Iranian tankers). Following the crisis, we implemented a methodology to estimate partial dark activity by cross-referencing Persian Gulf port departures with confirmed arrival records outside the SoH, and media reports. Based on this, our recorded average daily vessel crossings from March to mid-June were nine. However, if we were to triangulate all estimated dark activity, average daily crossings during this period would likely have been in the low-teens. Dark activity appears to have picked up significantly since the MoU was signed as the U.S. Central Command said 55 commercial ships passed through the strait on 20 June.

For a detailed analysis of the impact of the Iran conflict and the outlook for the Strait of Hormuz, please see [Twenty Miles That Shook the World](#)

Structural trend of a widening EU trade deficit with China



Key events, issues and themes to watch in July 2026

Listed in chronological order



US trade policy (all of July)

July could see the introduction of new US tariffs against dozens of trading partners. The expiry of US 10% global Section 122 tariffs **on 24 July**, and their expected replacement by Section 301 levies, should keep pressure on countries to strike bilateral deals favourable to the US, spanning trade, investment, industrial and technology concessions. Countries with existing framework agreements with the US [could see](#) the rates of new levies capped at previously agreed levels. **New US Section 301 tariffs on China** could increase bilateral tensions, although we assess that a return to significant trade escalation remains unlikely as long as both leaders continue to signal a [commitment to strategic stability](#).

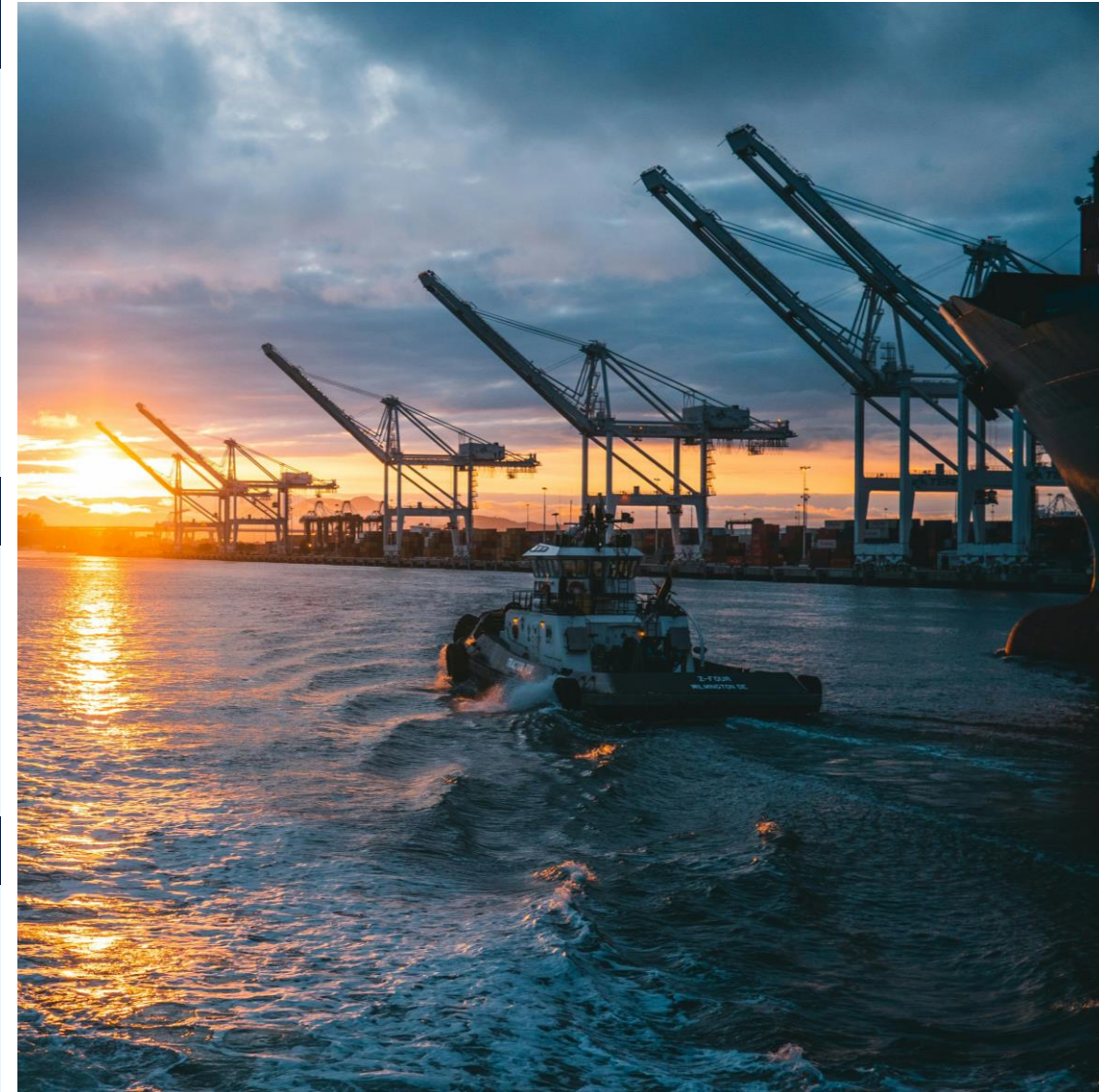
Under a separate [Section 232 investigation into critical minerals](#), after 13 July, the US may choose to impose tariffs, price floors or other trade measures in response to minerals agreements judged to be incomplete or ineffective.

US-Mexico-Canada trade talks (1 July onward)

The renewal talks are expected to be challenging and potentially protracted. 1 July marks the start of the US-Mexico-Canada (USMCA) trade deal's mandatory six-year joint review process. The three countries are required to assess whether to extend the agreement for another 16 years, review it or allow it to move into annual review mode if consensus is not reached. **Officials have signaled that the complex negotiations may continue beyond the initial review date.** President Trump's statements in June expressing skepticism over the deal and its value to the US could add another layer of uncertainty to the talks.

Japanese Prime Minister Takaichi's expected to visit India (1-3 July)

The visit will likely aim to strengthen bilateral trade and investment ties. Prime Minister Takaichi's first official visit to India will likely include a bilateral meeting with Prime Minister Modi. Areas of focus in talks could include increased cooperation in semiconductors, strengthening supply chain resilience, as well as security cooperation. Both countries are members of the Quad initiative, and bilateral ties have strengthened in recent years as New Delhi and Tokyo diversify their international ties amid rising geopolitical uncertainty. **Japan is India's fifth-largest foreign investor**, particularly in sectors such as automotive manufacturing, digital technology, clean energy and infrastructure.



Source: [Ronan Furuta](#) on [Unsplash](#)

Key events, issues and themes to watch in July 2026

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AI/digital summits in Geneva (6 -10 July)

Switzerland will host three UN-backed summits during July. The UN will hold its inaugural **Global Dialogue on AI Governance** (6-7 July), intended to establish shared principles on AI governance and improve interoperability across emerging national regulatory frameworks. The summit takes place alongside two meetings organized by the International Telecommunication Union (ITU). The annual **AI for Good Global Summit** (7-10 July) aims to translate AI policy discussions into technical standards, public-private partnerships and scaled deployment of AI solutions. The **World Summit on the Information Society** (WSIS; 6-10 July) centres on promoting global digital cooperation.

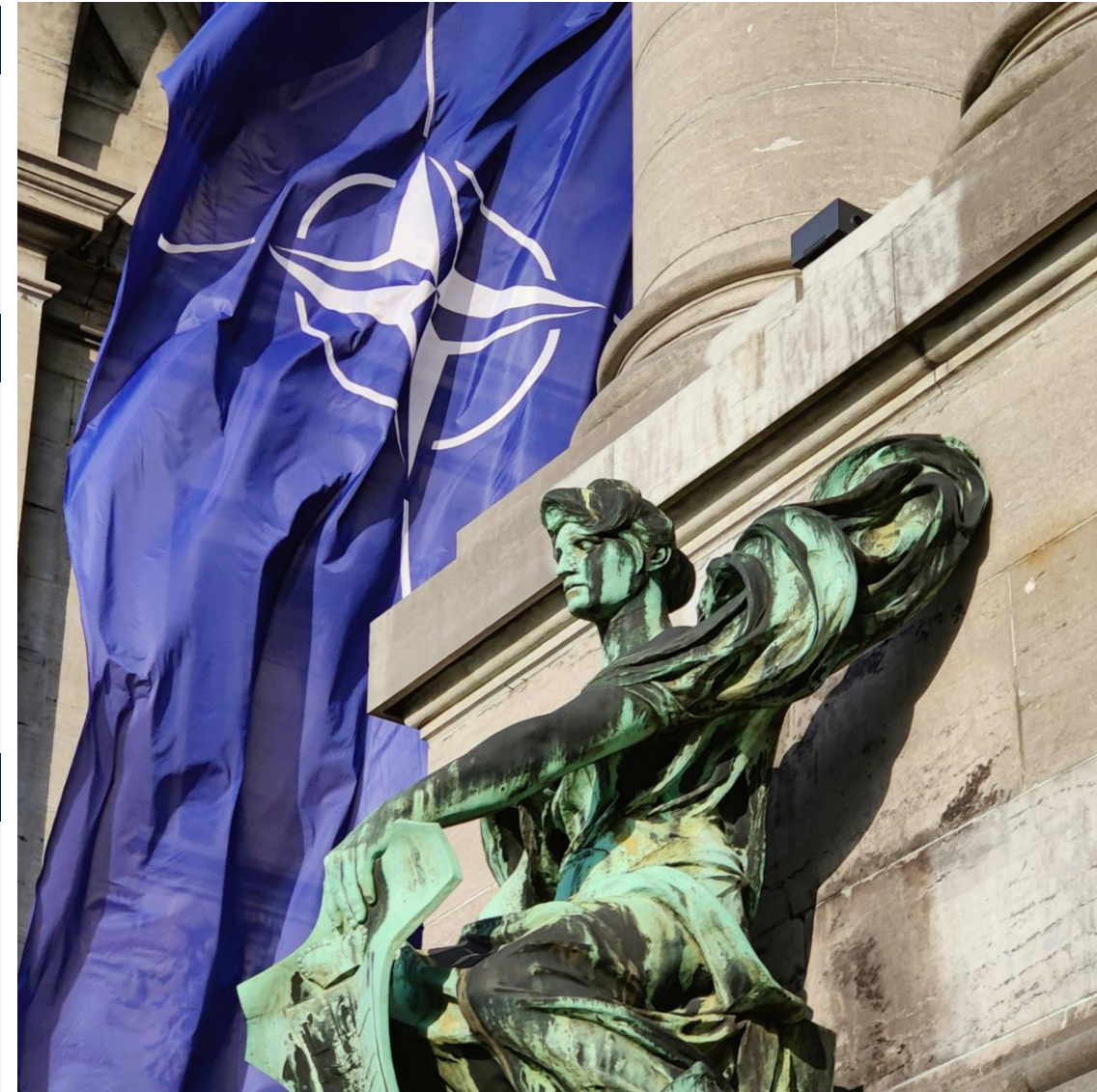
NATO leaders' summit in Ankara (7-8 July)

Pressure on European allies to carry more of the security burden will likely intensify as US stockpile and resource constraints bite after the Iran war. President Trump's demands will likely headline the NATO Leaders Summit. **It is slated to assess the credibility of national defence spending plans** and advance the emerging "NATO 3.0" agenda under which European NATO members take on more responsibility for their own conventional defence.

Ahead of the meeting, **the US has announced some military pullbacks from Europe** and reportedly notified allies of further planned reductions to its NATO Force Model contributions. To slow the withdrawal of hard-to-replace US capabilities (such as air defence, long-range weaponry and strategic airlift), European leaders will stress their commitment to NATO's new 3.5%-of-GDP target for "core" defence spending and their readiness to help clear the Strait of Hormuz.

India's Prime Minister Modi could visit Indonesia, Australia and New Zealand (6 -11 July)

Media reports indicate that Modi could visit the three countries during July, even though no plans have been officially confirmed. The prime minister's itinerary could reportedly include talks with Indonesian President Prabowo Subianto, Australian Prime Minister Albanese and New Zealand's Prime Minister Luxon. If confirmed, the visit would likely reflect New Delhi's **intensifying economic and security engagement with regional middle powers** amid rising global geopolitical risk and global trade volatility.



Source: Jannik on Unsplash

Key events, issues and themes to watch in July 2026

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Appointment of new UK Prime Minister (mid-July)

UK will see the appointment of its 7th Prime Minister since 2016. Prime Minister Keir Starmer announced his resignation on 22 JUNE. The Labour leadership nomination process opens on 9 JUL. At the time of publication, Any Burnham is the only declared candidate. Assuming no other Labour MP mounts a leadership challenge, Burnham could become the Prime Minister by mid-July, pending the completion of the Labour leadership process and formal appointment by the King.

Burnham's key priorities appear to be significant devolution of power, regional economic growth, fiscal discipline, and political reform.

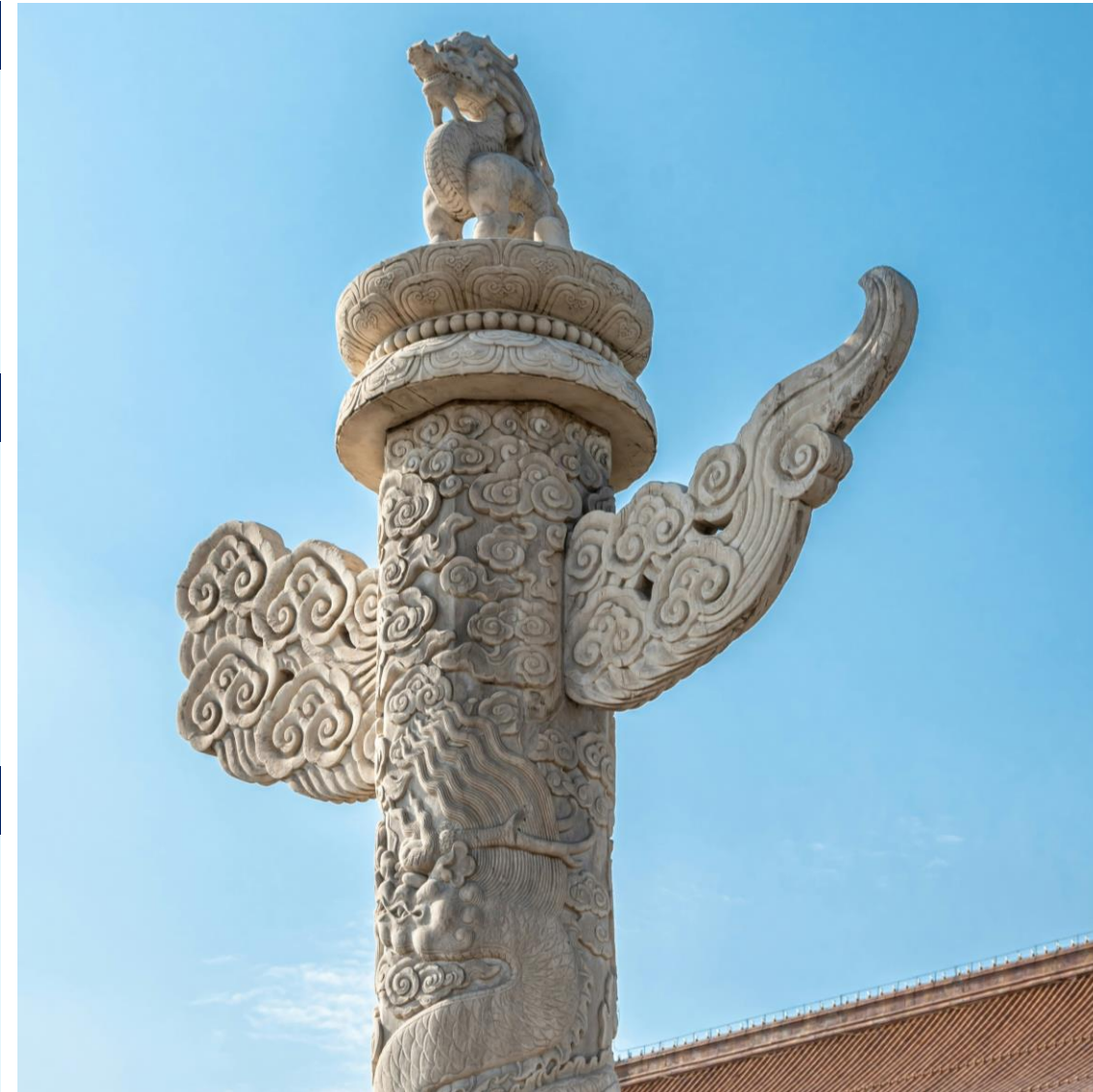
US-India trade negotiations (24 July as a soft deadline)

Trade talks are reportedly nearing completion, reaffirming strategic stability despite recurring bouts of friction. The Modi-Trump meeting on the sidelines of the June G7 summit has reportedly injected renewed momentum into bilateral trade talks, which are now believed to be nearing completion. India's Trade Minister Piyush Goyal has expressed **optimism that an interim deal could be concluded by 24 July**, when the US' global 10% tariffs expire.

But even if negotiations extend beyond this deadline, we maintain our view that a convergence of strategic interests – including long-term competition with China – should keep bilateral ties broadly on track. **Nevertheless, this does not preclude the possibility of recurring bouts of friction over trade, tariffs and [security issues](#).**

Politburo meeting in Beijing (late July)

The July meeting of the Politburo – China's top policy-making body – is one of the more important sessions in this year's political calendar. The meeting will most likely assess the state of the Chinese economy during the first half of 2026 and **set the economic policy direction for the rest of the year**. Post-meeting readouts could provide insights into priorities around industrial policy, support for the property sector, and the implementation of the 15th Five-Year Plan adopted in March 2026.



Source: 典方 on [Unsplash](#)

Appendix 1



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I, robot... you, unemployed: How workers are responding to AI fears

July 1, 2026

While media headlines this year have been filled with stories of AI-driven layoffs, less attention has been paid to how workers are responding to the new AI-centric world they find themselves in. Are people so fearful of displacement by AI that they are pulling back on spending and increasing savings? Or are they embracing its potential to enhance their careers and lifestyles?

This research report uses our proprietary dbDataInsights household survey¹ to examine how workers are responding to fears that they will lose their jobs to AI. The analysis shows that, contrary to the idea that worker's fear could slow down the economy as they spend less to boost precautionary savings, individuals most concerned about AI-driven job loss are not retreating. Instead, they are taking some proactive steps.

- **Proactive upskilling:** Workers who are more concerned about losing their job to AI are more likely to undertake AI-related training – potentially in an attempt to complement, rather than be substituted by, AI.
- **Increasing wage demand:** They are more likely than other workers to seek out a pay rise.
- **Confident spending:** They are more likely to be currently spending more, in direct contrast to the idea that greater concern about job security will act as a drag on spending.

The survey also reveals who these concerned individuals are. They tend to be younger, more educated, have a better understanding of AI, and are already using AI in their professional and personal lives.

The findings suggest that workers are not resigned to the fear of losing their jobs to AI. They are attempting to adapt to the new AI world. Supporting this trend – both at the level of individual businesses and broader public policy – could help reduce the social cost of AI adoption that some fear.

Authors

Michael Kirker
Economist

¹ For more background on the dbDataInsights household survey see the [Annex](#) at the end of this report.



Survey background

Our dbDataInsights household survey asks a wide range of questions each month to nationally representative samples in Germany, France, Italy, Spain, the United Kingdom, and the United States. The analysis in this report draws exclusively from the survey. Respondents to the survey are asked directly about their fears of losing their job to AI:

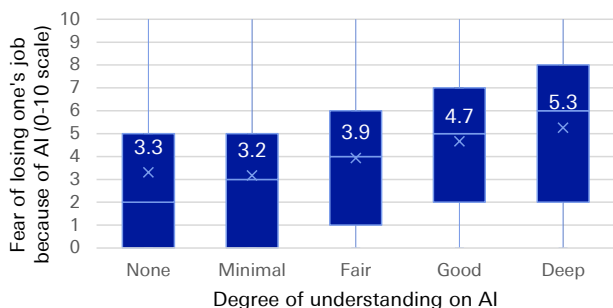
Using a scale of 0 to 10, where 0 is 'Not at all concerned' and 10 is 'Extremely concerned', how concerned are you that you will need to look for a new job within the following time frames because of AI?

This report will focus specifically on the one-year ahead horizon, although results are similar if we consider the two- or five-year ahead horizon instead. We began asking this question in our June 2025 survey round, and the level of concern has been broadly stable since then, both over time and similar across countries. Therefore, this analysis aggregates data from all monthly surveys and countries to maximise statistical power

The fear of losing one's job to AI is not born out of ignorance or irrational fear about AI – it is driven by informed concern. In our survey, those that have a greater understanding of AI show higher levels of concern about losing their jobs.

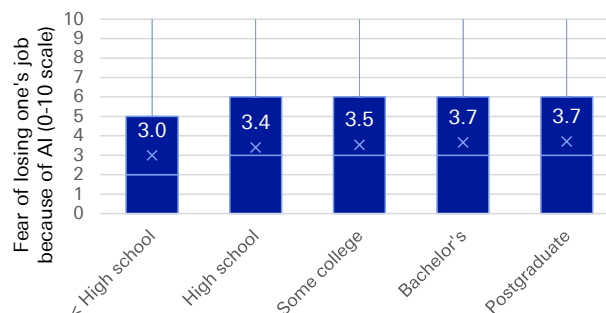
When asked to describe their own knowledge level of AI, respondents who reported higher levels of understanding generally had a higher concern about losing their job to AI than those with less self-reported knowledge (Figure 1). This relationship is not likely to be driven by bias in people's self confidence. **On average, those with higher levels of education also showed higher levels of fear (Figure 2),** and **those who have actively used AI in the past three months either at home or at work – and thus should have greater knowledge about the latest AI developments – also showed greater concern on average about losing their job to AI (Figure 3).**

Figure 1: Those with a higher self-report understanding of AI are also those with the highest level of fear about losing their jobs over the next year



Source : Deutsche Bank Research, dbDataInsights
Notes: Numerical values denote the average fear (on a 0-10 scale) of each group

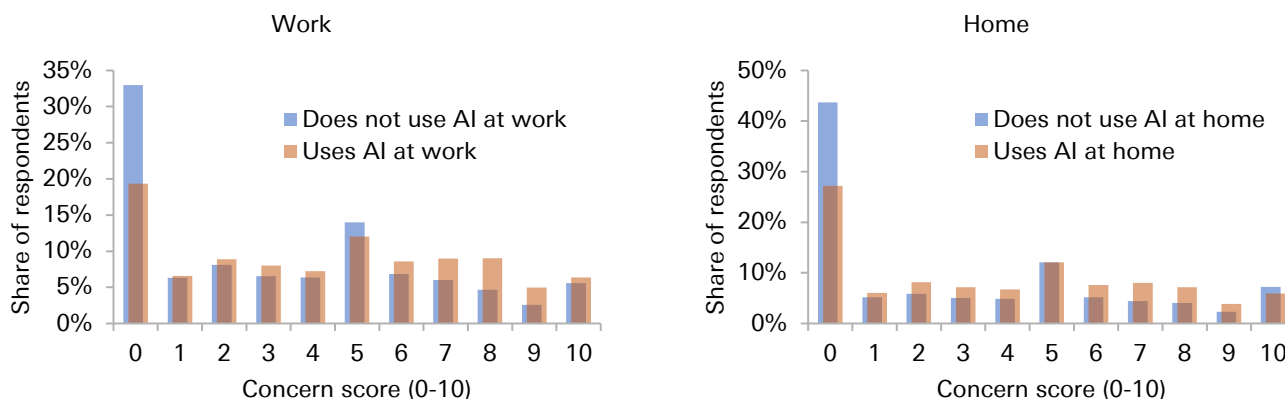
Figure 2: The more educated respondents show more concern about losing their job to AI over the next year



Source : Deutsche Bank Research, dbDataInsights
Notes: Numerical values denote the average fear (on a 0-10 scale) of each group



Figure 3: The distribution of concern about losing one's job to AI over the next year is higher for those respondents who have used AI at work or at home within the last three months



Source : Deutsche Bank Research, dbDataInsights

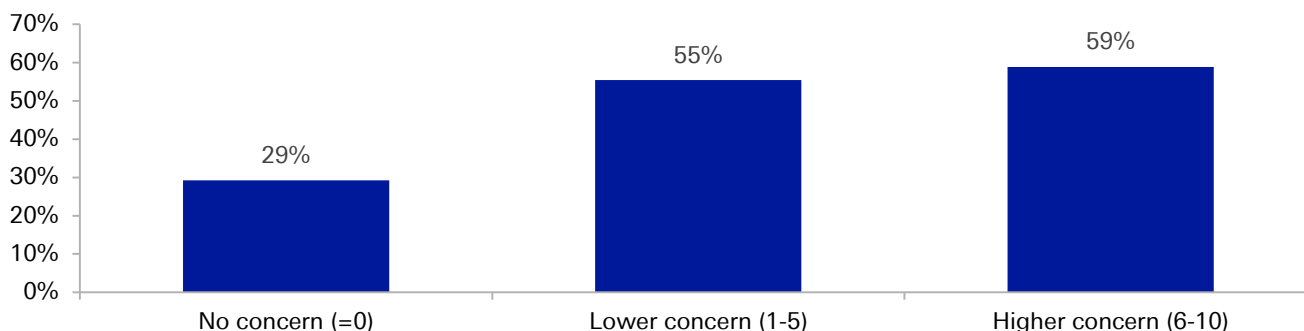
The proactive worker: Upskilling to become a complement to AI rather than a substitute

Worker's fear of losing their job to AI over the next year translates into direct action by the workers. They are not taking the transition to a new AI world lying down. One way in which they are doing this is by upskilling – adapting to become a complement to AI investment rather than a substitute for it.

Up until December 2025, our survey asked respondents what AI learning activities they may have been engaged in: (1) Watching videos to learn about AI; (2) Completing a course to learn more about AI; (3) Reading articles to learn more about AI; or (4) none of these. If the respondent has done any of these types of training (1-3) then they are classified as having undertaken training.

As [Figure 4](#) shows, the proportion of respondents undertaking some form of AI upskilling/training is increasing with the level of concern about losing one's job to AI over the next year.

Figure 4: The proportion of respondents undertaking AI training/upskilling is higher for those with higher levels of concern about losing their jobs to AI over the next year



Source : Deutsche Bank Research, dbDataInsights



To work out if the higher likelihood of undertaking training is associated with the level of concern, and not driven by differences in the types of people who are more likely to be concerned, a logit regression analysis is carried out. The regression includes controls for a range of other factors of respondents in an attempt to isolate the effect of concern about job loss on the likelihood of undertaking AI training/upskilling.² The results are shown in the table of [Figure 5](#).

Figure 5: Logit model: Impact of concern about losing one's job to AI on the likelihood of doing AI training

	Share upskilling	Average marginal effect
Full sample	55.3%	1.67pp (SE 0.12)
Only those concerned (concern>0)	61.6%	0.51pp (SE 0.17)

Source : Deutsche Bank Research, dbDataInsights.
Notes: pp = percentage points

The regression is run on the full sample, and then again on the subset who say they have some level of concern (concern > 0 on the 0-10 scale). The reason for this is that The change in behaviour from going from "not at all concerned" (a score of 0) to a score of 1 could be quite different from the change in behaviour of someone who is already concerned becoming slightly more concerned (say moving from a 4 to a 5).

The regression analysis shows that a one-unit increase in AI concern (moving one point higher on the 0-10 scale) correlates with a statistically significant average marginal effect of around 1.7 percentage points (pp) for the full sample, and 0.51pp for the concerned subsample. **This means a one-point rise in an individual's concern about losing their job to AI over the next year is, on average, associated with a 0.5-1.7pp increase in the probability of that person undertaking some form of AI training.**

To put that into context, the average level of concern across our survey has risen slightly over half a point over the last year. Therefore, the average respondent has become 0.25-0.9pp more likely to be undertaking some form of AI because of their rising concern about job loss.

This finding is a positive development. Workers are not waiting to be replaced by AI. They are actively working to become AI complements rather than substitutes. If AI does lead to the productivity gains business leaders hope for, tomorrow's labour force are more likely to have the skills and knowledge needed to take advantage of the AI, thereby further boosting the productivity gains from AI.

The pay-rise paradox: Demanding more in the face of disruption

Conventional economic theory would suggest that the because AI is a technology that threatens to replace workers, the adoption of AI by businesses should weaken the bargaining power of workers, making them less likely to push for pay rises.

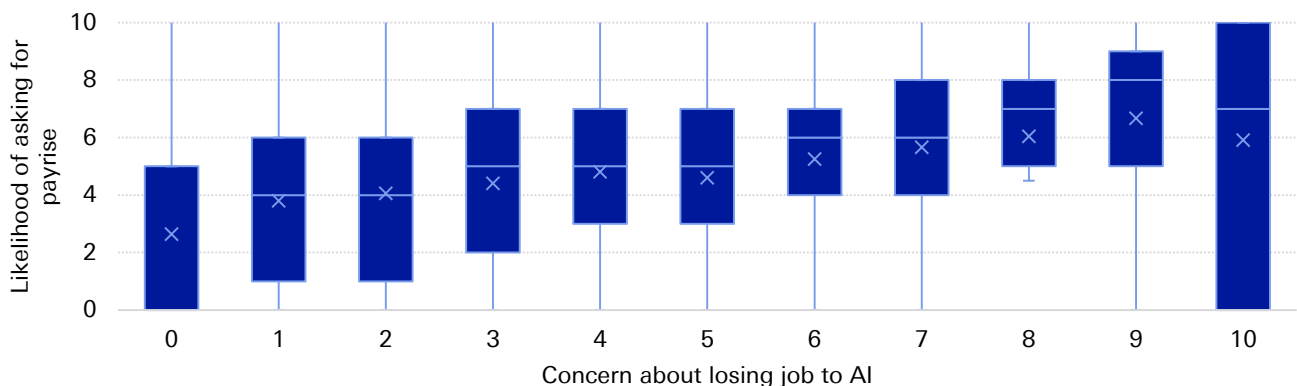
² For more on the regression framework used in this report, see [Annex 2](#).



However, the data from our dbDataInsights survey shows the opposite: workers who are more concerned about losing their job to AI are, in fact, more likely to ask for higher wages.

Figure 6 illustrates this paradoxical finding. In our survey, we ask respondents how likely (on a scale of "0 – not at all likely" to "10 – extremely likely") they are to ask their employer for a pay rise in the next three months. Looking at the distribution of responses to this question for each different level of concern about losing one's job to AI over the next year there is a clear pattern of **respondents with higher levels of concern being the most likely to ask for a pay rise in the next three months.**

Figure 6: Paradoxically, workers who report higher levels of concern about losing their jobs to AI are also saying they are more likely to ask for a pay rise in the next three months



Source : Deutsche Bank Research, dbDataInsights

To explore this relationship further, two different forms of regression analysis are used. The first is a panel OLS regression. The results are presented in the table of Figure 7. This approach makes the simplifying assumption that the likelihood of asking for a pay rise is a continuous variable (rather than the discrete 0-10 buckets). The advantage of making this assumption is the coefficient from the regression has a rather simple and intuitive interpretation.

Figure 7: Panel OLS: The impact of concern about AI on the likelihood of asking for a pay rise

	Likelihood of asking for a wage increase (0-10)		Coefficient
	Mean	Median	
Full sample	4.22	5	0.29 (SE 0.01)
Only those concerned (concern>0)	5.01	5	0.30 (SE 0.02)

Source : Deutsche Bank Research, dbDataInsights

The second approach is an ordered logit regression. The results of this are presented in the table of Figure 8. This approach accounts for the discrete buckets for the likelihood of asking for a wage rise, and only requires that the buckets be in ascending order (and not necessarily equally spaced apart).



Figure 8: Ordered Logit: The impact of concern about AI on the likelihood of asking for a pay rise

	Likelihood of asking for a wage increase (0-10)		Odds ratio for a 1pt increase
	Mean	Median	
Full sample	4.22	5	1.21 (95% CI: 1.20, 1.21)
Only those concerned (concern>0)	5.01	5	1.22 (95% CI: 1.20, 1.22)

Source : Deutsche Bank Research, dbDataInsights

Both regression approaches point to the same conclusion. **An increase in the concern about losing one's job in the next year is associated with a higher likelihood of asking for a wage rise in the next three months, even after we control for the characteristics of respondents that are likely to influence their behaviour.**

The panel OLS regression coefficient indicates that a one-point increase in concern about losing one's job to AI (on our 0-10 scale) is associated, on average, with a 0.3 point increase in the likelihood of asking for a pay rise.

So, that would suggest that if an individual's concern about losing their job rose by 3 points on the 0-10 scale, then the likelihood of them asking for a pay rise in the next three months would increase by nearly one point on its 0-10 scale.

The ordered logit gives us an odds ratio of around 1.21-1.22. **That tells us that each one-point increase in the concern of losing one's job to AI is associated with a 21-22% higher likelihood that the individual reporting a higher likelihood of asking for a pay rise in the next three months (than someone with a lower level of concern).**

Why do we find such a counter-intuitive result? It is difficult to say with any certainty. But there are a few theories that could explain the pattern:

- **Omitted Variable Bias:** The regression analysis includes a range of other variables (like the respondents age, whether they are using AI, country-time fixed effects), that attempt to control for factors that could also influence the respondents willingness to ask for a wage rise, and could thus confuse our estimate of the effect from AI job loss concern. But it is possible that the regression does not control for every possibility.

It's possible that unobserved respondent characteristics influence both concern about losing one's job to AI and the propensity to ask for a wage increase. For instance, ambitious or career-driven individuals might be more inclined to seek pay rises while at the same time are more attuned to both the threats (and opportunities) presented by AI, leading to higher concern as well.

- **Strategic Preparation for Future Redundancy:** Although not reported here, the level of concern about losing one's job to AI over the next year is not too dissimilar to the concern over the next five years. Perhaps workers expect that the time horizon for being replaced by AI is quite long. This could lead them to strategically maximize their current earnings, including seeking higher wages today, in an attempt to bolster future severance payouts or raise their incomes today to build up their savings in anticipation of a future job loss.



The confident consumer: Spending and saving in the AI era

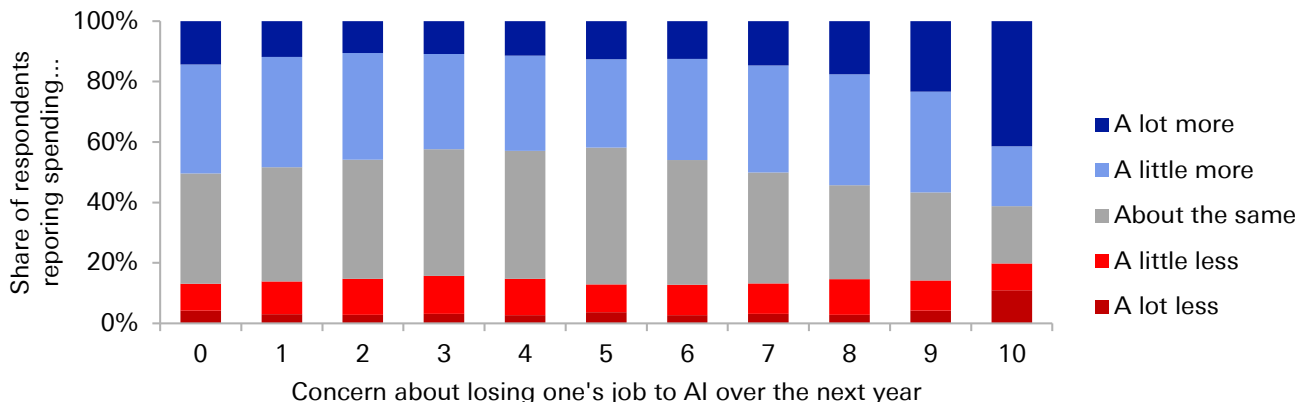
Consumption smoothing behaviour tells us that if workers become more concerned about losing their job in the future, they should reduce their consumption today and save more, thereby having more money available for when they are unemployed.

Thus while spending on AI data centres can boost GDP and economic activity in the near term, at the same time worries about job security can act as a drag on spending, thereby lowering GDP and economic activity.

The results from our household survey present mixed results regarding this behavioural prediction of households. **While workers concerned about losing their job to AI state an intention to save more (a prudent response to uncertainty), they are also more likely to report spending more than this time last year (something that goes against consumption smoothing).**

In the survey, we ask respondents if there are current spending: a lot more; a little more; about the same; a little less; or a lot less, relative to a year ago. [Figure 9](#) shows how the distribution of responses varies by the level of concern about losing one's job to AI. While all groups are more likely to report spending a little or a lot more than last year (a consequence of higher inflation rates), those with a greater level of concern about losing their job are also the ones more likely to report spending more than this time last year.

Figure 9: Respondents who are more concerned about losing their job to AI are also more likely to report spending more than this time last year.



Source : Deutsche Bank Research, dbDataInsights



[Figure 10](#) reports the results from the ordered logit regression of spending behaviour onto the fear of job loss (and other controls).³ **The results suggest that each one-point increase in the concern of losing one's job to AI is associated with a 3-6% higher likelihood that the individual reports a higher degree of spending relative to last year.** The result is statistically significant.

Figure 10: Ordered logit: The impact of concern about AI on spending growth

	Odds ratio for a 1pt increase
Full sample	1.032 (95% CI: 1.02, 1.04)
Only those concerned (concern>0)	1.060 (95% CI: 1.05, 1.07)

Source : Deutsche Bank Research, dbDataInsights

This positive correlation between AI concern and increased spending contradicts the initial hypothesis that AI-driven job loss concerns would dampen consumer spending. Spending less would imply a negative correlation (i.e., an odds ratio below 1).

Savings and consuming are two sides of the same coin. It is difficult to accurately measure savings behaviour in household surveys. However, we do ask respondents about their intentions to save. Specifically, respondents are asked: *'Thinking 12 months ahead, do you anticipate your current savings amount(s) will increase, decrease, or remain about the same?'* Response options in the survey included: *'Increase significantly,' 'increase slightly,' 'stay about the same,' 'decrease slightly,' or 'decrease significantly.'*

The ordered logit regression analysis was repeated on the savings intention of the survey respondents. The results are presented in the table of [Figure 11](#).

The results suggest that each one-point increase in concern about losing one's job to AI over the next year is associated with a 6-9% higher likelihood that the individual reports a higher intention to save more over the next year.

Figure 11: Ordered logit: The impact of concern about losing one's job to AI on savings intentions

	Odds ratio for a 1pt increase
Full sample	1.06 (95% CI: 1.05, 1.07)
Only those concerned (concern>0)	1.09 (95% CI: 1.08, 1.10)

Source : Deutsche Bank Research, dbDataInsights

³ Inflation expectations is included as an additional control here as it could impact on the decision to bring forward the timing of consumption spending. We also include a control related to the question "are you better or worse off financially than a year ago" as being better off financially than a year ago could explain why the respondent is spending more today than last year.



Thus, a higher concern for losing one's job to AI is, on average, directly associated with an intention to increase savings in the near future. This positive correlation aligns with the concept of precautionary savings, where the perceived heightened risk of AI-induced job in the future loss motivates consumers to save more today.

Thus while the reported spending behaviour of respondents contradicts to predictions that concerned households should be spending less and saving more, those respondents also report a greater intention to save more going forward. But whether they actually follow through on these intentions is not something we can observe. Therefore, the weight of the evidence from our survey suggests that the uncertainty shock from AI fear is not (yet) dampening consumption.

Conclusions and implications

The standard economic theory tells us that workers/households should fear AI. As a substitute for their labour, it weakens their bargaining power and makes it more likely they will be laid off.

But concerns about AI-driven job loss are not creating a passive, fearful workforce poised to retreat from the economy. Instead, our dbDataInsights household survey reveals a powerful paradox: the individuals who are most anxious are also the most proactive, turning their concern into decisive economic action.

- Fear of job-loss from AI is not born out of ignorance. Our survey shows that those who are most informed about AI show the greatest fear of losing their job.
- Workers who are fearful of AI are responding by investing in their own skills and trying to learn more about AI, turning themselves from a type of labour AI can substitute for, to a type of labour that is a complement to AI.
- Rather than weakening worker's bargaining power, fear of AI seems to be encouraging workers to ask for higher wages.
- The confidence also extends to their role as consumers. Higher fear of losing one's job to AI is not associated with workers spending less (savings for the future when they are unemployed). Instead they seem to be spending more.

The key insight from the findings here is that fear of AI is not (yet) acting as a drag on households as standard theory would predict. While businesses are spending big on AI models and data centres, workers are actively trying to adapt to the new world we are moving into. Proactive policies can support workers adapting to AI and lead to productivity gains that go well beyond simply replacing labour with AI tools.

But it must be kept in mind that these findings present only a snapshot of a highly dynamic situation. For now, the 'AI Anxiety Paradox' shows a workforce that is resilient and adaptive. However, these behaviours are rooted in the anticipation of change. Should widespread, large-scale AI-driven layoffs materialise, this dynamic could fundamentally alter, potentially shifting today's proactive ambition back towards a precautionary retreat.



The author would like to thank Anthony Chaimowitz from the dbDataInsights team for his invaluable assistance in administering the survey and collating the data.

Annex 1: dbDataInsights household survey

The dbDataInsights household survey provides valuable insights into household/consumer attitudes and expectations regarding personal finances, labour market conditions, and buying conditions. The survey is conducted by Deutsche Bank and is used by economists and sector analysis in their research on the macroeconomy and specific sectors.

The survey covers households/consumers in Germany, France, Italy, Spain, the United Kingdom, and the United States. For each country, a national-representative sample of 550 households is surveyed each month (650 households in the US). Results are collected throughout the month to provide a high-frequency look at how expectations and attitudes are evolving in response to the latest developments in the economy.

The survey has been running since 2018 and is continuously evolving to address relevant issues. The survey is conducted online. Although the individual respondents change each month, the panel remains nationally representative each month.

Annex 2: Regression analysis framework

This Annex describes in more detail the regression analysis framework used throughout the report.

To assess the correlation between AI concern and behavior, a series of panel regressions is run using micro-level survey data. In these regressions, the variable of interest (e.g., savings intentions, spending) is regressed against the self-reported concern level, incorporating fixed effects to control for individual respondent characteristics.

Standard fixed effects in most of the analysis account for:

- **Age:** Categorized by ranges (18-24, 25-34, 35-44, 45-55, 55-64, 65+)
- **Using AI:** Binary indicator of whether the respondent has used AI in the last 3 months, either at work or at home.
- **Income Brackets:** Low, medium, and high.
- **Country-Month Interactions:** To capture broader macroeconomic trends (e.g., economic shifts, political events).

Other fixed effects are added where appropriate. For example when examining worker's savings intentions, inflation expectations of the respondents are also included.

All results in the paper are based on concern of losing one's job to AI over the next year. However, using their concern of losing their job over the next two and five years produce similar results.

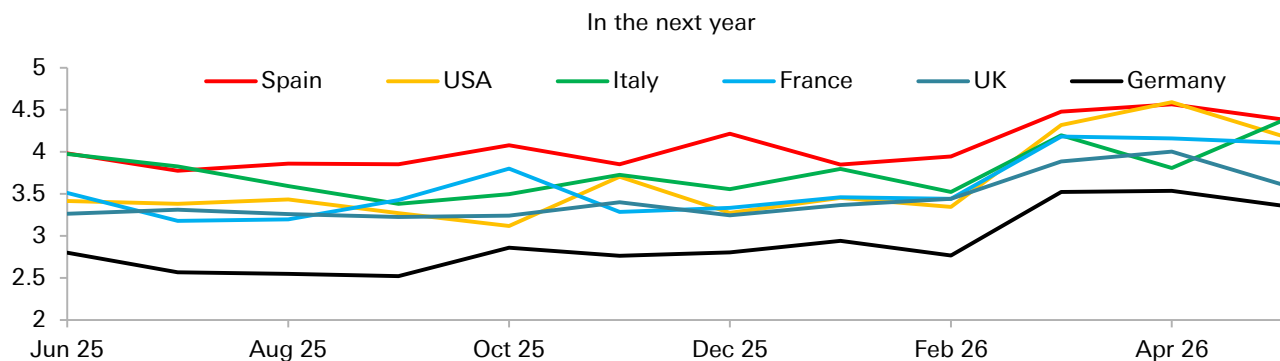


Annex 3: Further results about what type of respondents are worried about AI job loss

[Figure 12](#) plots average level of concern in each country we survey (Germany, France, Italy, Spain, the UK, and the US) for each month. A few patterns stand out here:

- **Up until March, the average level of concern was quite stable.** Between June 2025 and February 2026 the average level of concern in all countries remained broadly stable. This period we saw release of some major new LLM models.⁴ But the stability of concern shown in [Figure 12](#) suggests the improvement of these models did little to affect the average concern of workers, either because the improvements did not change the opinion of respondents, or respondents may have been ignorant of these latest developments.⁵
- **A shift occurred in March, with average concern rising significantly in all countries.** This increase may be linked to broader geopolitical anxieties, such as the Iran war, which could heighten general economic and financial well-being concerns, making respondents more sensitive to AI-related job loss risks.
- **Across all surveyed nations, Germany consistently records the lowest average level of concern, while Spain registers the highest or near highest.** France, Italy, and the UK show broadly similar levels of concern, though a recent surge in the US has brought its average closer to Spain's level.

Figure 12: Average concern (scale 0-10) of losing your job to AI



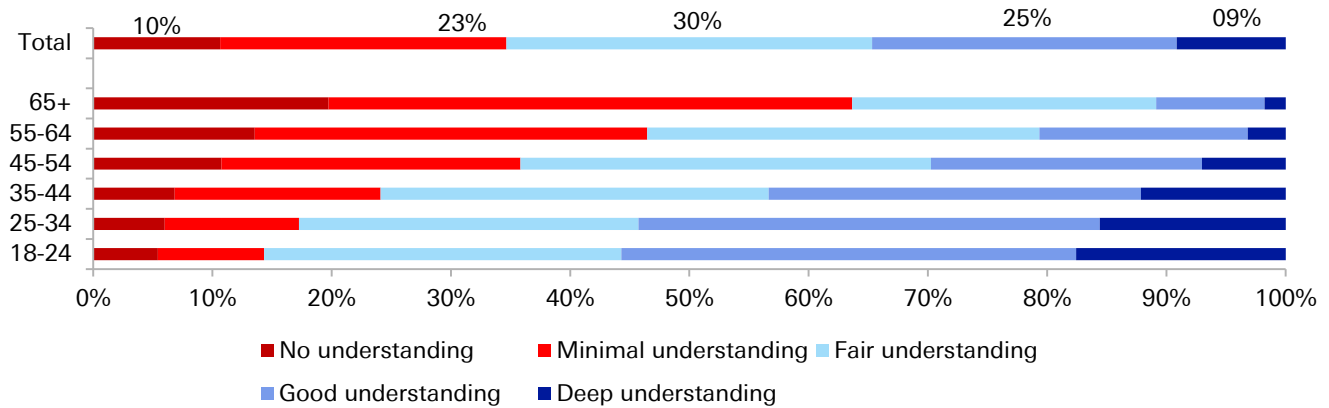
Source : Deutsche Bank Research, dbDataInsights

⁴ Including OpenAI's GPT-5 (August 2025), Google's Gemini 3.1 Pro (February 2026), etc.

⁵ However, as shown in [Figure 13](#), a majority of respondents report at least "a fair understanding of AI", suggesting that the new model developments would have likely to have been known by a significant proportion of respondents.



Figure 13: Self reported understanding of AI by age group – The young claim to understand AI better



Source : Deutsche Bank, dbDataInsights

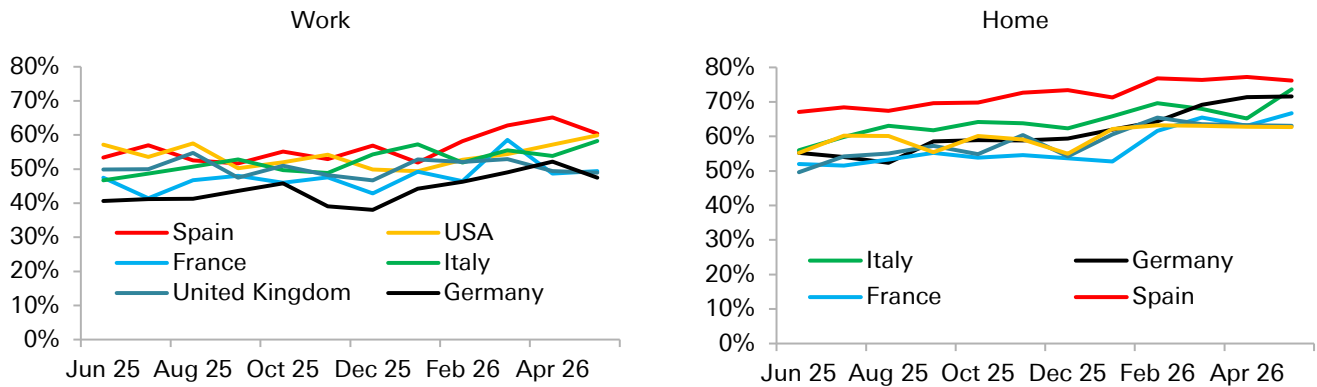
Younger individuals also report a higher self-assessed understanding of AI (Figure 13). Younger individuals tend to be more tech-savvy. Their deeper understanding of AI's capabilities likely influences their heightened concern about job loss.

Our survey indicates a strong association between AI usage and elevated concerns about losing one's job to AI. Figure 14 shows the percentage of respondents who have used AI in the last three months, both at work and at home. A few interesting patterns stand out in Figure 14.

- **The proportion of respondents using AI at work is broadly comparable between the US and the European countries we surveyed.** This finding offers some reassurance amidst European political concerns about falling behind in the 'AI race,' suggesting that even without developing as many cutting-edge models of their own, European businesses are effectively leveraging existing AI tools.
- **While US respondents show similar rates of AI usage at work and home, in Europe, home usage generally surpasses work usage.** This could signal a greater appetite among European workers for integrating AI into their professional lives, provided they are granted more autonomy with the tools.
- **AI adoption, both at work and at home, is steadily increasing.** Across all surveyed countries, the percentage of respondents using AI in the past three months shows a consistent upwards trend.



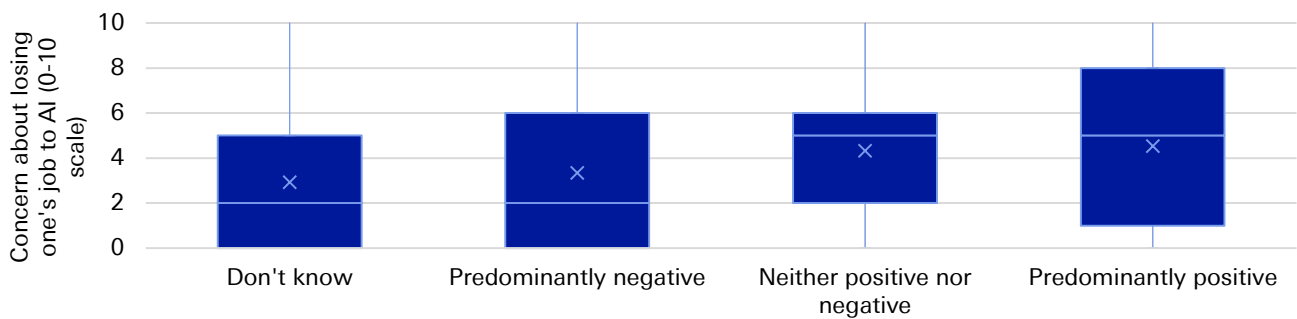
Figure 14: Share of respondents who have used AI in the last 3 months at...



Source : Deutsche Bank Research, dbDataInsights

Those with a more positive outlook on AI's societal impact also demonstrate the highest concern about losing their own jobs (Figure 15). This pattern suggests that techno-optimism/pessimism may not be the primary determinant of individual job security concerns.

Figure 15: Those thinking AI will have a more positive social impact are the most concerned about losing their jobs to AI



Source : Deutsche Bank Research, dbDataInsights



Appendix 1

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AI IPOs 101: why they'll change the boom forever

July 01, 2026

Four reasons why the expected IPOs of OpenAI and Anthropic matter so much

The expected wave of initial public offerings of the world's leading AI companies will change the AI boom forever.

It will bring unprecedented **transparency**, **funding** and **accountability** to the industry – as well as finally giving investors a **pure-play benchmark** to value the likely winners and losers of a revolutionary technology.

The [news](#) at the end of last week that [OpenAI](#) is considering delaying its own share sale, originally expected for later this year, indicates just how much is at stake. The company, which lit the fuse for the AI frenzy with its launch of ChatGPT in November 2022, as well as its now slightly larger rival [Anthropic](#), which makes Claude, confidentially filed for IPOs in early June. Those share sales would reportedly aim to raise about \$60bn apiece and value each company at more than \$1 trillion.

They and their peers need to **strike while the iron is hot** to secure computing power, distribution and capital. But there is no guarantee that there will be enough investor appetite to absorb hundreds of billions of dollars in rapid-fire demand.

The AI boom is fast approaching a **moment of truth**, as rapid growth and soaring valuations collide with ballooning capital expenditure, a public backlash and the challenges of real-life adoption. Challenges abound, including Chinese open-source models that promise similar performance for a fraction of the price. Whoever comes out on top could define the era.

In this report, we look at:

- Why have an IPO?
 - Which AI-related companies are likely to have IPOs?
 - OpenAI and Anthropic in a nutshell
 - Four ways that IPOs will change the AI boom forever
1. Greater transparency for financials – and business models
 2. More diverse funding sources – and more spending
 3. Greater accountability for their actions – and the industry
 4. New era for comparing valuations to peers – and the past

Authors

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Research Analyst



Why have an IPO?

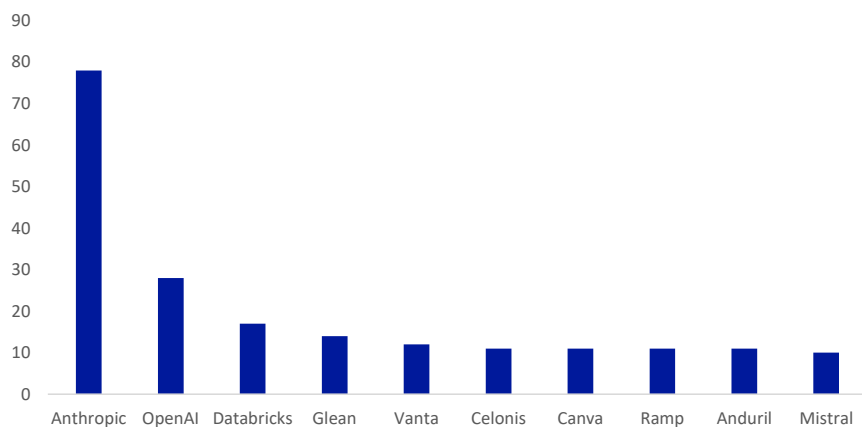
1. **Access new sources of funding** to invest, reduce debt and improve financial flexibility.
2. **Create liquidity** to give existing investors and employees a way to cash out and enable acquisitions with publicly traded shares.
3. **Improve public credibility and profile**, lending additional weight with consumers, businesses and lawmakers.

Which AI-related companies are likely to have IPOs?

The two leading contenders to sell shares to the public this year are Anthropic, given a 78 percent chance, according to bets on Polymarket, and OpenAI, now down to 28 percent.

Other AI-related companies that may go public include data analytics platform Databricks, enterprise data platform Glean, and compliance tool Vanta, according to Polymarket bets. Others include business process platform Celonis; graphic design tool Canva; finance operations platform Ramp; defence company Anduril; and Europe's only notable frontier-model maker Mistral.

Figure 1: Implied probability of select AI-related IPOs by 2027



Source : Polymarket, Deutsche Bank Research

OpenAI in a nutshell

- San Francisco-based OpenAI was **founded in 2015** as a non-profit research laboratory led by **Co-Chairs Elon Musk and now-CEO Sam Altman**, the former head of Y-Combinator. It was backed by some of the biggest investors in Silicon Valley, including Peter Thiel and Reid Hoffman. (Musk severed ties with OpenAI in 2018, triggering a legal dispute that concluded in May in OpenAI's favour.)
- Its **mission** was "to ensure that artificial general intelligence – AI systems that are generally smarter than humans – benefits all of humanity".
- On November 30, 2022, OpenAI launched **ChatGPT**, a generative AI chatbot based on its GPT-3.5 large language model, which became a



runaway success with 100 million users in three months. GPT-4, released in March 2023, was so far ahead of the competition that it took a year or more for serious challengers to gain a foothold. However, GPT-5, launched last August, was quickly matched amid stiffer competition.

- The non-profit board voted to remove **Altman** as CEO in November 2023. He was reinstated four days later, after a revolt by investors and employees.
- Its **most important strategic relationship has been with Microsoft**. When OpenAI transitioned into a “capped-profit” entity in 2019, it was able to obtain capital investment including \$1bn from Microsoft, which later added another \$10bn in early 2023. Their exclusive partnership and revenue-sharing agreements enabled Microsoft to be an early leader in AI and secured funding and computing power for OpenAI.
- The companies **scaled back their partnership** in late April 2026, though Microsoft retains a 27 percent stake in OpenAI. (OpenAI’s for-profit LLC became a public benefit corporation controlled by the non-profit in May 2025.)
- Today, **more than one in ten people on the planet use ChatGPT every week**. The name is to chatbots what PostIts are to sticky labels. It reportedly surpassed \$25bn in annualised revenue in March and has indicated to investors that it will achieve its first annual profit in 2030.
- Although the market is increasingly crowded, OpenAI is still producing **cutting-edge models**, with GPT-5.5 excelling at agentic coding. Brand new GPT-5.6 won the same badge of honour as the latest Claude model last Friday by being restricted to a small group of users over US government concerns that it was so powerful that it could pose a security risk. It also has a strong developer ecosystem building on top of its Application Programming Interface (API).

Anthropic in a nutshell

- San Francisco-based Anthropic was **founded in 2021** by seven former employees of OpenAI, including **Dario Amodei, its CEO**, and his sister Daniela Amodei, its president. Anthropic is a public benefit corporation whose [mission](#) is “to ensure that the world safely makes the transition through transformative AI”.
- Led by its **Claude models**, the company has focused heavily on the enterprise market, positioning itself as a bastion of high standards of safety and reliability.
- **Google parent Alphabet and Amazon** have invested billions of dollars in Anthropic. Like OpenAI, it has struck complex agreements that commit it to use chips and data centres from its investors in return, including a commitment [reported](#) by The Information to spend about \$200bn with Google Cloud over five years.
- This has been a **breakout year** for Anthropic. The release of its Cework and Claude Code agent tools in January and February prompted a “SaaSocalypse” slump in stocks of software-as-a-service companies, amid concerns that their utility-like business models could be largely disintermediated.
- The **Claude website had almost one billion visits in May**, still behind ChatGPT at 5.6 billion and Google’s Gemini at nearly three billion, but way up from 200 million in January, according to data from Similarweb. Meanwhile, Claude Code now accounts for slightly more than half of the business-focused coding market.
- **Anthropic’s revenue** reached an annual recurring rate of \$47bn in May,



overtaking OpenAI. CEO Amodei [said](#) that an 80-fold increase in usage in the first quarter on an annualised basis was “just crazy” and “too hard to handle,” forcing it to limit access for some users and sign up new sources of compute to keep up with demand.

- A **disagreement with the US Administration** came out into the open in February when the Department of Defense announced that it was a supply-chain risk. The company had demanded restrictions on the use of its technology for domestic surveillance and autonomous weapons.
- Then two weeks ago, Anthropic [abruptly switched off its most powerful](#) **Table 5 and Mythos 5 models** worldwide after the US Department of Commerce ordered it to restrict access by foreigners, due to concerns that it could be weaponised by bad actors. It won approval to restore some access at the end of last week.

Four ways that upcoming IPOs will change the AI boom forever

1. Greater transparency for financials and business models

Nothing sharpens an entrepreneur’s mind – and business model – like the prospect of an IPO.

- It is a decision that they may take with some trepidation, given the obligation to open the doors and explain their finances, business model and outlook in detail.

To prepare for an IPO, AI companies will have to produce public-company-grade reporting, credible performance indicators, risk analyses and forecasts.

- They will also have to make any relevant disclosures about a growing number of lawsuits over copyright, alleged negligence regarding safety, and corporate governance.
- Remuneration may also have to be revamped, with public metrics around pay for performance for management and provisions for employees to vest stock.

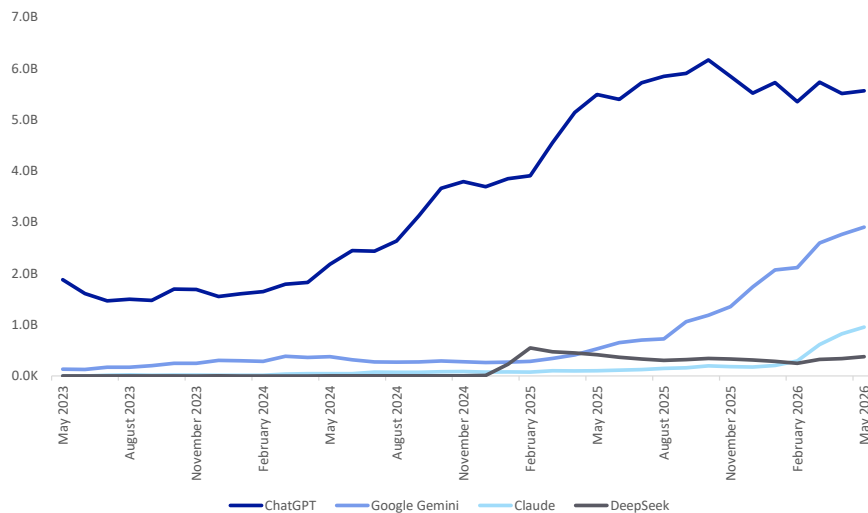
Greater transparency may help investors take a view on how much they consider currently lossmaking AI models to be winner-takes-all, like Amazon or Meta, where network effects create a virtuous circle, or commodities, where one model is viewed as interchangeable with another.

- While OpenAI has the leading chatbot by far with around 900 million monthly users, it has been rapidly pivoting its business model this year to try to attract enterprise users who will pay for it.
- That includes creating the [OpenAI Deployment Company](#) in May with 19 private-equity groups led by TPG to distribute its enterprise products, spearheaded by so-called “forward deployed engineers”. It said previously that enterprise revenue was on track to match consumer revenue by the end of the year.
- Since CEO Sam Altman declared a “code red” in December, OpenAI has scrapped what it called “side quests” such as its video generating tool Sora and is [working](#) on a “unified superapp” that brings its chatbot, coding, browsing and agentic capabilities together, aiming to convert satisfied home users to a similar business offering.



- Anthropic overtook it in business subscriptions in May, according to data from Ramp. And Similarweb data shows monthly visits to ChatGPT fell below a majority of the generative AI market for the first time in May, suggesting consumers are increasingly willing to switch between models.

Figure 2: Top four generative AI platforms: monthly visits worldwide (desktop and mobile web)



Source : Similarweb, Deutsche Bank Research

- OpenAI has indicated that it is generating about \$25bn to \$33bn in annualised revenue, up from a \$4bn annual rate at the end of 2024. By comparison, Anthropic's annualised run rate shot up from \$1bn a year at the end of 2024 to \$47bn in late May. The company previously said that its first profitable full year will be in 2029, one year earlier than OpenAI.

Greater disclosure around IPOs will finally illuminate the business models of the frontier model makers.

- While chips and data centres are easy to measure, actual monetisation of AI is opaque, given that private model makers make only limited disclosures and public hyperscalers rely on AI models for just a small part of their AI revenues.
- The key questions are how intense and rapid adoption will be; is the return on investment justified once businesses have to make hard choices to scale beyond pilots; and will premium all-singing, all-dancing frontier models win out over cheap, stripped-down, task-specific alternatives?

The stakes for pure-play model makers are higher now than ever before, with pressure from above and below.

- On the one hand, the hyperscalers have unsurpassed access to data, distribution and compute, as well as the ability to offer competing models on their platforms and still take a cut.
- On the other hand, ever-smarter open-source models may be good enough for most tasks. GLM 5.2, an open-source model from China's Z.ai, has become a viral hit in the US after rocketing up the leaderboards over the past couple of weeks. It scores 51 in Artificial Analysis's Intelligence Index, not

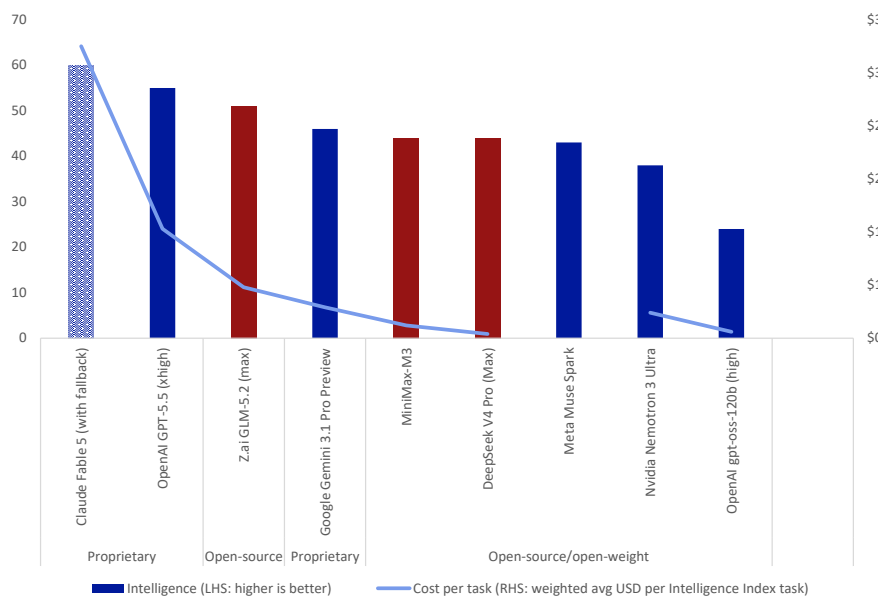


far behind Claude's (currently restricted) Fable 5 and OpenAI's GPT-5.5, yet costs just 48 cents per task in the index, versus \$2.75 and \$1.03 respectively.

Price is becoming a battleground as enterprise customers begin to balk at the true costs of rolling AI out at scale.

- Yes, cost per token, the building block of AI, is falling by between nine and 900 times a year, depending on the task, according to Epoch AI. But that is being offset by ever more compute-hungry agentic applications – as well as by well-meaning initiatives to encourage employees to be creative that have incentivised budget-busting “token-maxxing”.
- The economics of AI suddenly look less attractive once token usage is no longer being subsidised by investors. Anthropic has changed from a flat fee to a usage-based model, and OpenAI's Altman said last month that cost had come out of the blue to become a “huge issue” for customers this year. Both are [reported](#) to be considering price cuts that could erode their profit margins.

Figure 3: Artificial Analysis Intelligence Index of selected AI models vs cost per tasks

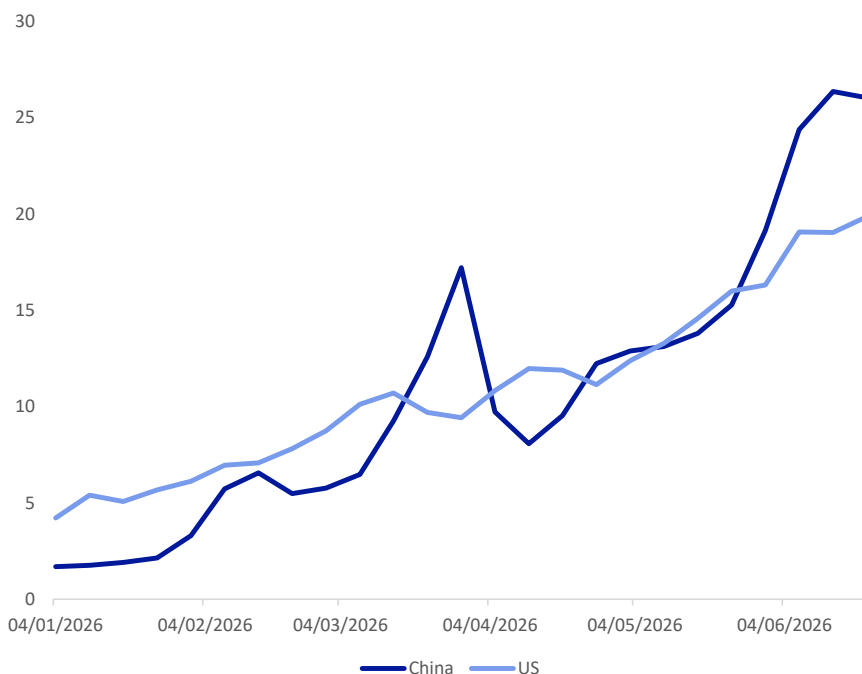


Source : Artificial Analysis Intelligence Index. “Fallback” mechanism routes safety-flagged messages to Opus 4.8 model. Meta Muse Spark price not available. US models in blue; Chinese models in red

- OpenRouter data showing weekly token usage of the top models on its platform indicates that the number of tokens processed by US models has risen fivefold this year to 20 trillion, convincingly outpaced by a thirteenfold increase by Chinese models to 26 trillion. The platform allows customers to access hundreds of LLMs from different providers through a single account.



Figure 4: The number of tokens processed by OpenRouter using Chinese models has overtaken tokens processed using US models this year (trillion)



Source : OpenRouter, Deutsche Bank Research

- Amazon, Walmart and Cisco are among companies that have reined in token use. Uber burned through its AI budget for the year in four months and has now capped spending, with its Chief Operating Officer [saying](#) that “the link is not there yet” between the use of Claude Code and customer innovation.
- Demand is growing for new model architectures that can divert queries to the most efficient model for any given prompt, such as OpenRouter’s new Fusion plugin.

2. More diverse funding sources – and more spending

AI is an arms race for compute, pitting startups against established hyperscalers.

- Specialist model-makers are on the back foot compared with a company like Alphabet parent Google that has ample cloud compute facilities and is generating roughly \$50bn in cash from operations every quarter that it can use to buy even more.
- Private companies cannot tap the same pool of investors forever. Abundant public market funding would break the cycle of relying on the piecemeal multi-billion-dollar investments and complicated financing arrangements among the leading firms of the past few months.

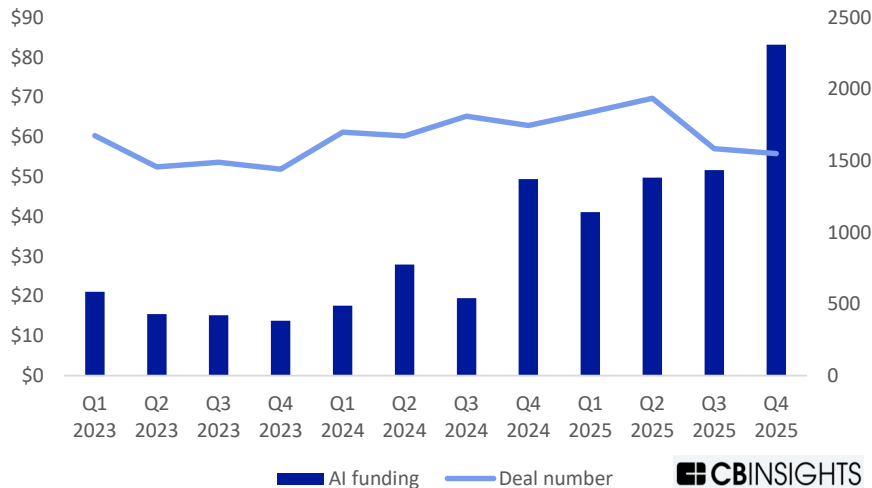
To be sure, this has not been a bad time to be raising private money.

- Three lean years for venture investment reversed last year as investors injected \$425bn, an increase of almost a third from the previous year, into more than 24,000 companies, [Crunchbase](#) data show. AI funding was particularly healthy, exceeding \$225bn globally, albeit concentrated in few



companies, according to [CB Insights](#).

Figure 5: Global AI funding surged in recent quarters, albeit concentrated in fewer companies: quarterly equity funding (\$bn) and deal count



Source : CB Insights, Deutsche Bank Research

- Anthropic said at the end of May that it [raised](#) \$65bn at a post-money valuation of \$965bn, more than double its valuation in February. In one leap, it overtook OpenAI, which [raised](#) \$122bn at an \$852bn post-money valuation in March. If either achieved the mooted market value of \$1trn, it would be comfortably within the world's top 20 most valuable companies.

But growth does not come cheap.

- OpenAI told investors in February that it was targeting some \$600bn in infrastructure investments by 2030, less than half of its earlier estimate of \$1.4tn amid concerns about whether future revenue could justify the outlay. Its emphasis has changed from building its own data centres to just renting them.
- OpenAI's funding round in March was anchored by long-term partners Amazon, Nvidia and SoftBank. Amazon will invest \$50bn in OpenAI in return for OpenAI using 2GW of its Trainium chip capacity and cooperating on new joint products.
- Long-term shareholder and partner Microsoft still took part even though the two have grown more distant and [unwound](#) some exclusive arrangements in April. The startup also took investments from at least 19 institutional investors and funds, as well as some retail investors via banks and exchange-traded funds, and expanded a revolving credit facility to about \$4.7bn.
- Meanwhile, Anthropic's latest round looped in at least two dozen investors and included \$15bn of previously committed investments from hyperscalers including \$5bn from Amazon – after the startup agreed to buy up to 5GW of compute capacity from it and a similar amount from Google and Broadcom.
- In a sign of just how complicated the AI funding org chart is becoming, Google, which already [reportedly](#) owned a 14 percent stake in Anthropic



worth \$3bn, pledged to invest another \$40bn in it in April and more recently was said by Bloomberg to have agreed to backstop lease payments worth \$35bn on five data centres.

IPOs would not just unlock new equity finance.

- Public listings would give the companies and their backers leverage capacity: the ability to raise debt against liquid, transparently valued collateral. Those funds could then be recycled into further expenditure on data centres, acquisitions and other ventures.
- A publicly funded OpenAI or Anthropic would likely not just build bigger models but also seek to become the default foundational platform layer similar to what Windows was for PCs and to Amazon's AWS for the cloud. They could use larger war chests to expand their API services, developer tools and enterprise partnerships, commoditising the layers where rivals (and perhaps some customers) operate.

Financial flexibility matters for all of the players in the land grab for capacity in the AI boom.

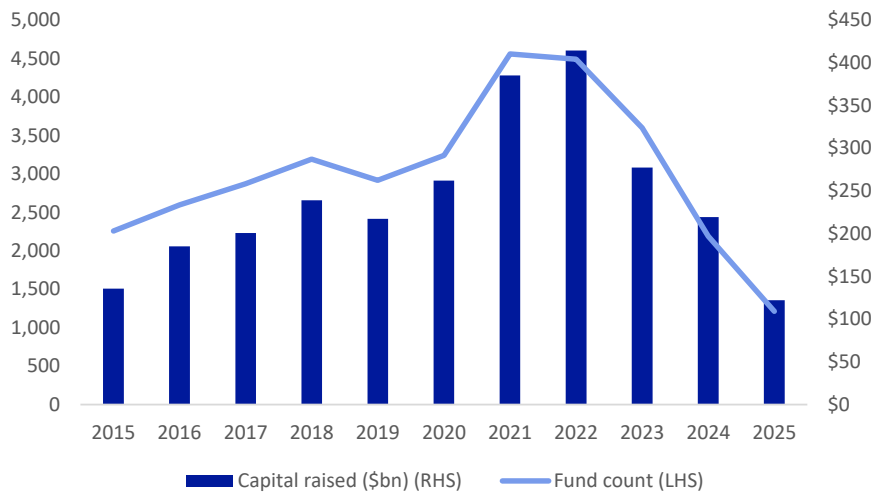
- Capital expenditure by hyperscalers looks set to rise by two-thirds to more than \$750bn this year. Even cash-rich Google turned to the debt markets this year, raising almost \$32bn in less than 24 hours in February, then raising \$84.75bn in the biggest equity capital markets transaction ever on June 1. Nvidia sold \$25bn of high-grade bonds in mid-June.
- And smaller companies are turning to the public markets, most recently with South Korean memory chip maker SK Hynix announcing that it is planning to raise more than 45 trillion won, or almost \$30bn, in a landmark US listing.

An opening of the IPO floodgates after a five-year drought would give companies the currency they need to embark on a spending spree.

- It could even break the inertia of an upstream drought in fundraising for early investments by unblocking trapped liquidity. Global venture capital fundraising fell last year to \$122 billion, little more than a quarter of the peak in 2021 and 2022, as investors shied away from illiquid investments in favour of less risky strategies, PitchBook [data](#) show.



Figure 6: Global private-equity fundraising dropped significantly after 2022, trapping upstream liquidity



Source : [PitchBook](#), Deutsche Bank Research. Data as of Dec 31, 2025

- That could mean a surge in mergers and acquisitions, turning other private companies into targets and giving an exit route to impatient investors. This year has already seen more than 5,000 technology acquisitions or investments pending or completed with a value of about \$850bn, more than double the level for the same period last year, Bloomberg data show.
- Elevated valuations and regulatory hurdles for the biggest companies would put them out of reach of all but a tiny number of strategic buyers (the hyperscalers) or sovereigns. An alternative approach has been “acquihires”, pioneered by Microsoft two years ago when it hired most of the staff of Inflection, including co-founder Mustafa Suleyman, and licensed the use of its AI models for \$650 million.
- But smaller companies have a much greater chance of being acquired.
- Most recently, US chipmaker Onsemi agreed to buy Synaptics, which makes semiconductors for smart devices, for about \$6.2bn in stock late last week. Indeed, Bending Spoons, which buys up struggling software companies, including AOL and Vimeo, itself filed for an IPO last week. The Italian company aims to raise as much as \$1.6bn at a \$19bn valuation early this month.

Last but not least, IPOs will give companies a potent new weapon in the battle for talent: liquid, publicly traded stock options.

- For top-tier researchers and engineers, this would be a far more tangible and immediate form of wealth than illiquid shares in a private rival that might seek to lure them away.
- Such a liquidity shock could force private competitors into responding with ever larger funding rounds at higher valuations on the promise of “jam tomorrow” for star employees – or into going public themselves. They would also have to be on high alert to prevent their existing core teams from being poached by a newly public OpenAI or Anthropic armed with a trillion-dollar market capitalisation.
- Meta’s Mark Zuckerberg sparked a bidding war for top AI researchers last



year, offering compensation packages worth hundreds of millions of dollars in some cases. OpenAI and Anthropic were the most recent players to fish in the small pond of genuine AI experts a couple of weeks ago, hooking two legends of the industry. Shares of Google, where they worked, fell more than 5 percent on the day.

3. Greater accountability for their actions – and for the industry

The AI industry has lost its lustre in recent months.

- Warm initial perceptions of it as a slightly magical composer of wedding speeches have faded and may vanish once Mom and Pop own shares in pure-play AI companies for the first time. Those companies risk becoming a lightning rod for broader fears about the impact of AI on trust, jobs and vulnerable industries.

At the most basic level, public companies are held to higher standards around consistency and accountability.

- For example, OpenAI has attracted criticism for reversing its earlier opposition to advertising in an attempt to create a more sustainable business model. Critics have said conversations could be biased by sales imperatives.

Beyond that, OpenAI and Anthropic have already had a taste of the political challenges in store as public companies.

- Anthropic was blacklisted by the US Department of Defense in February for refusing to let Claude be used for autonomous weapons and mass surveillance. The dispute cost Anthropic a \$200m [contract](#) and is now in the courts. OpenAI was then criticised from the other side after it struck a deal to replace Anthropic. Altman later described the move as being “complex but the right decision with extremely difficult brand consequences and very negative PR for us in the short term”.
- More recently, the US Department of Commerce issued an export control directive late on June 12 to suspend all access by foreign nationals to Anthropic’s Claude Mythos 5 and less powerful Fable 5 models because of security concerns.
- Although it had to temporarily disable access to all customers to ensure compliance, its reputation for having the most powerful models will have benefitted (see our report [AI twist in the tale for Anthropic’s Fable and Mythos](#)). The silver lining may eventually eclipse the cloud amid a branding boost that money can’t buy. However, alternatives to US models are also likely to benefit from concerns that access could be switched off without warning.

Most broadly, the makers of the best-known chatbots may pay a price for being seen as proxies for all of AI when controversy hits the industry.

- Hot topics include cybersecurity. The leaders of the Five Eyes cybersecurity agencies of the US, UK, Australia, Canada and New Zealand warned in a joint [statement](#) last week that frontier models are likely to supercharge cyber threats: “The timeline is not years, it is months”.

The biggest underlying popular concern is what AI will mean for jobs.

- The leaders of both Anthropic and OpenAI have moderated their



predictions of severe disruption in recent weeks, with Amodeli pivoting to say automation would dramatically increase productivity and Altman saying the impact on jobs has been less than he expected.

- One challenge is that AI has become a scapegoat for job cuts at companies ranging from Block (4,000 employees, or nearly half the workforce) to Atlassian (10 percent of roles) and even British American Tobacco (with AI cited as one of a number of factors in a planned one-fifth reduction in its workforce).
- However, the tide has begun to turn on predictions of catastrophic job losses. A slump in entry-level graduate jobs may have been more due to rising interest rates and remote work than AI – and may be reversing. And business leaders like Amazon Web Services' CEO Matt Garman have [said](#) eliminating the pipeline of talent makes no sense.

Nowhere has the AI backlash become more focused than over data centres.

- As fast as these “AI factories” are mushrooming around the US and the world, so are concerns – both justified and unjustified – about them overwhelming the electricity grid, forcing up power prices, and sucking rivers dry for water to cool their chips.
- How the leading model makers, hyperscalers and infrastructure companies handle the issue could determine whether it remains contained or unleashes moratoriums, regulation or swingeing taxes that put the entire boom at risk.
- In the US, home to half of the world’s data centres, opposition to “AI factories” is becoming a rare area of bipartisan consensus, uniting prominent Democratic Representative Alexandria Ocasio-Cortez and independent Senator Bernie Sanders on the left, and populist strategist Steve Bannon on the right.
- Maine recently passed a moratorium on larger data centres while, across the Atlantic, data-centre hub Ireland has introduced a “Bring Your Own Power” policy that requires new facilities to generate their own electricity or secure renewable energy contracts.

Data centers are – literally – close to home for voters.

- That makes them ripe fodder for targeted lobbying and grassroots campaigns in a way that vague fears about long-term job losses are not. The US midterm elections in November will be an important litmus test of the political consequences of local pressure.
- A Gallup poll in May found that 71 percent of Americans would oppose a data centre being built in their area. At least 75 projects worth a record \$130bn were blocked or delayed in the first quarter, [according](#) to Data Center Watch.
- Tech companies have done their bit to respond. Microsoft, for example, said in its most recent data centre announcement last week that it would create thousands of jobs at the facility in Pecos, Texas; pay to ensure it does not cause a rise in electricity prices; and replenish more water than it uses. “We are approaching this project as a new neighbour,” it said.

The IPOs of the biggest AI companies could also have repercussions in the geopolitical contest for technological supremacy, raising their status to that of strategic national assets.



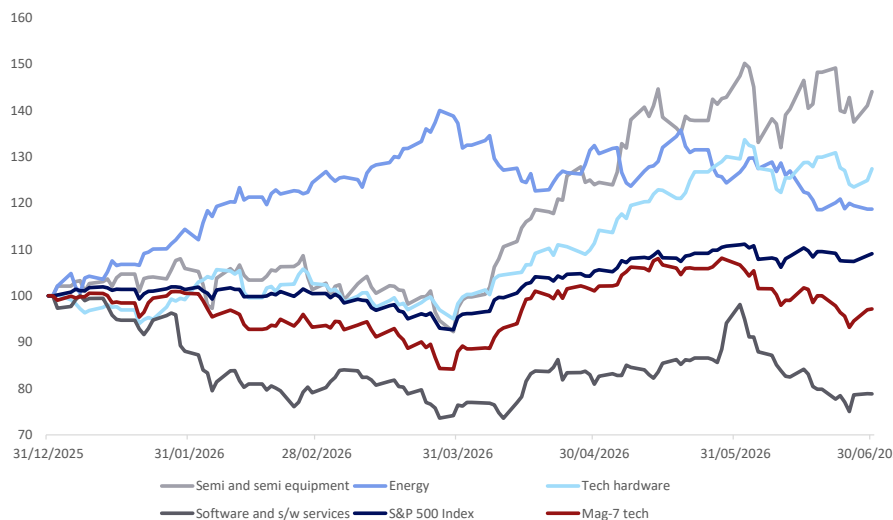
- That could encourage other powers to accelerate support for their own national champions, like China's "AI tigers" and France's Mistral, to avoid becoming dependent on US-controlled foundation models. Other countries may find themselves having to choose to align with one centre of power or the other.

4. New era for comparing valuations to peers – and the past

Part of the anxiety about an AI bubble is the lack of pure-play, public, investable companies to act as a benchmark. A listed OpenAI or Anthropic would fill a huge gap.

- Sentiment about AI has driven much of the market since the launch of ChatGPT, with anticipated gains from investment and productivity helping to explain the two-thirds rise in the S&P 500 Index during that period.
- That rise has been bumpy, soaring and swooning from week to week, as one area after another, such as software, semiconductors or infrastructure companies, has come into favour and then fallen out again. Investors have crowded into trades on a concentrated and shifting list of potential AI enablers and adopters, and away from prospective casualties.
- To give one example, the Magnificent Seven biggest technology stocks have whipsawed over the past month amid concerns about how much of their AI capital expenditure will pay off, shedding as much as 12 percent by late last week. They have tripled over the past three-and-a-half years to account for almost a third of the S&P 500.

Figure 7: S&P 500 Index, Magnificent Seven tech stocks and other select AI-related sectors performance YTD to June 30 (indexed to 100)



Source : Bloomberg Finance LP, Deutsche Bank Research

- That makes investing in AI right now feel a bit like going to a horse race and having to bet on the jockeys and the condition of the track because the horses themselves are unavailable.

Of course, there is no guarantee that any filing ends up in a successful IPO.

- The San Francisco-based co-working company WeWork was valued at



\$47bn in early 2019 but failed to gain traction when it filed its prospectus in August that year. It eventually went public in 2021 at a heavily reduced valuation and ended up filing for bankruptcy in late 2023.

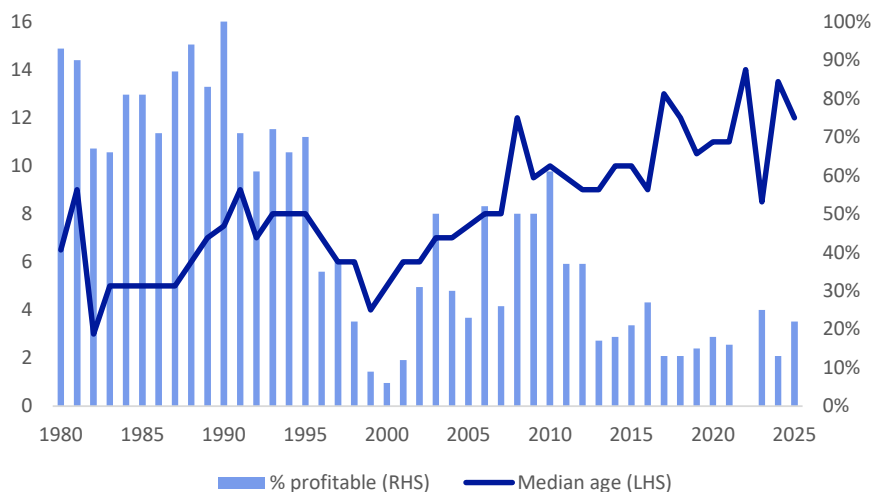
Another way that major IPOs could remove some distortions is by acting as a release valve for private markets, enabling long-awaited exits from investors.

- Private-equity companies are sitting on a record backlog of unsold companies, thanks in part to geopolitical disruption and fears of a “SaaSocalypse”, according to [data](#) from Pitchbook and others. The median age of still-held assets continued to trend upwards from three years in 2022 to four years in 2025 as exit activity stalled.

Despite concerns that the boom turns to bust, there are differences in the IPO markets from the height of the dot-com bubble.

- Overall valuations are clearly elevated by historical standards, with the Shiller Cyclically Adjusted Price-Earnings (CAPE) ratio for the S&P 500 recently surpassing 40, nearing the record 44 it reached at the peak of the dot-com bubble.
- However, the IPOs coming to market are more mature and more profitable.
- Back in 1999 to 2000, the median age of venture-capital-backed tech startups going public shrank to about 4.5 years, according to University of Florida data. Since then, the median age has been relatively stable at ten-and-a-half years and was 12 years last year.
- Only 6 percent of those tech startups going public at the height of the dot-com bubble were profitable, compared with 22 percent since 2010, and they had a median opening price-to-sales ratio of over 40, four times higher than the average since 2010.

Figure 8: Median age and profitability of technology IPOs (1980-2025)



Source : University of Florida, Jay Ritter, Deutsche Bank Research

Perhaps the greatest insurance against a bubble is the fixation of today’s senior investment managers on the dot-com bust that scarred them in their formative years in the market.



- For them, mega IPOs bring back memories of the IPO of Netscape in August 1995, widely regarded as the start of the dot-com bubble. However, there is just as important a lesson in how long it took for that bubble to go bad despite prophecies of doom and gloom right from day one.
- Bloomberg published an article two days after the IPO on August 9, 1995, entitled “Why investors say Netscape shows a market poised for a fall” and warns “Be worried, very worried”. “Netscape was a little insane,” says a trader quoted in the article, after the stock doubled on its first day of trading, giving the lossmaking company a \$2.9bn market value. “It certainly hints at more frothiness in the market and that’s usually the sign of the beginning of the end of a bull market”.
- Yet in the end, the Nasdaq shot up from 1,005 on the day of Netscape’s IPO, not peaking until almost five years later, at over 5,000 on March 10, 2000. It was only then that it dramatically fell back, reaching a trough of about 1,100 in October 2002.
- As for Netscape, it was bought by America Online (AOL) for \$10bn in 1999, and then vanished in the fallout from AOL’s record-breaking \$124bn purchase of Time Warner in 2001.

Yet Amazon, another star of the times, shows the power of patience – and picking a winner.

- Adjusting for stock splits, the shares debuted at 7.5 cents in the company’s \$54m IPO on May 15, 1997. After the roller coaster of the dot-com bubble and crash, the shares took a decade to recapture their early peak – and then continued to rise, reaching almost \$275 a month ago.
- Yet on the day of its IPO, when the stock rose 31 percent, a Pennsylvania-based fund manager told Bloomberg: “I personally am going to stay away from it. I don’t think it’s been proven that one can make a lot of money over the web.”

Recent relevant research

[Assessing AI in finance with Evident Co-CEO Alexandra Mousavizadeh](#) (June 9, 2026)

[ChatIPO: Quick take on OpenAI’s record-breaking IPO plans](#) (May 21, 2026)

[Will Claude’s AI finance agents change your life?](#) (May 6, 2026)

[AI, Iran and the Gulf 101](#) (April 13, 2026)

[What AI says about AI eating itself and the world...](#) (Feb 16, 2026)

[AI Q&A: Why have market vibes suddenly shifted on AI?](#) (Feb 16, 2026)

[Three AI themes for 2026: the honeymoon is over](#) (Jan 20, 2026)



Appendix 1

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Deutsche Bank
Research

Q2 Tech Performance Review: Memory Wins, Chips Dominate

July 1, 2026

Marion Laboure | Camilla Siazon

Deutsche Bank Research Institute

Deutsche Bank
Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.



Apple increases MacBook and iPad prices by 20%

Micron Soars After AI-Fueled Forecast Shatters Estimates

Wall Street ends lower on semiconductor selloff as AI spending concerns mount

Alibaba sues Pentagon over inclusion on Chinese military blacklist

Chinese AI, chip firms are driving an onshore IPO rebound

Anthropic Halts Access to Top AI Models After U.S. Ban on Foreign Use

Apple seeks to buy memory chips from blacklisted Chinese company

SUMMARY: Memory wins, chips dominate

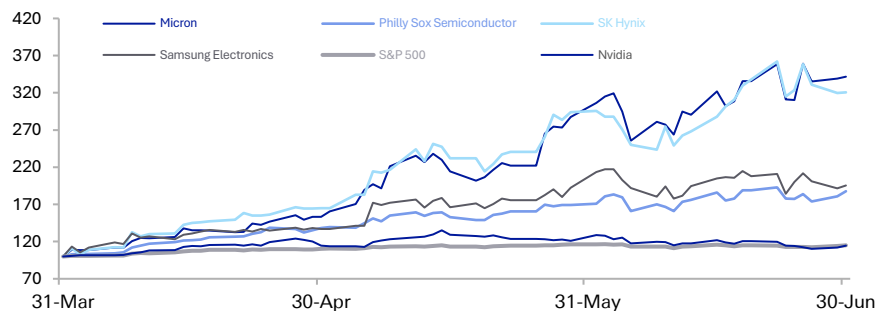


In Q2, equities ran largely uncorrelated from events in Iran, driven by a tech rally despite ongoing concerns over the war. Strong earnings growth fuelled the Mag 7's rise in April (+15%) and May (+7%). Then last month, risk assets sold off despite the June 15 US-Iran peace deal, as worries over rising Fed hawkishness, the memory shortage's inflationary impact, and doubts over AI demand sustainability triggered a sell-off in AI stocks. Still, despite the meltdown over the last few weeks, tech delivered a remarkable run this quarter, with AI excitement significantly outweighing macro and geopolitical concerns around the war. The Mag 7 ended up +12% on the quarter.

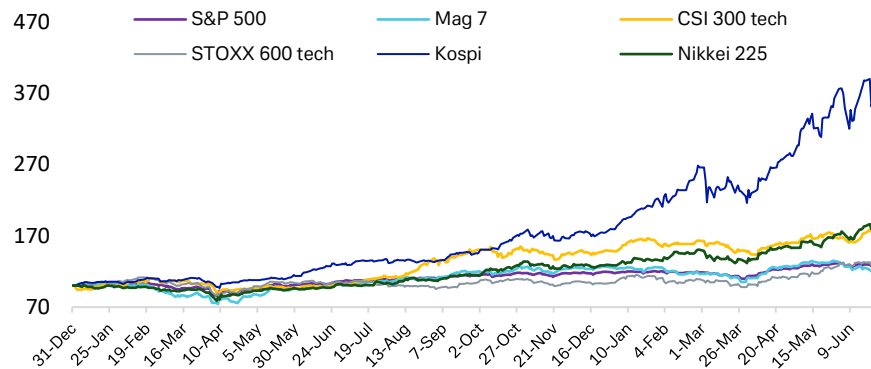
The spotlight has been on memory chipmakers in particular, which drove most of tech's outperformance – and its June selloff. The ongoing memory shortage has led to rising returns for Micron (+242% q/q), SK Hynix (+228% q/q) and Samsung Electronics (+100% q/q), all of which have now joined the \$trn dollar market-cap club. Asian equities followed, with the Nikkei 225 (+37% q/q) and Kospi (+67% q/q) outperforming their Western counterparts.

What will Q3 bring? Micron's results were probably enough to restore confidence in AI-driven demand, particularly in data-centre memory. But volatility under a more hawkish Fed will likely persist, and BIS has warned of AI exuberance risk ending in a lengthy investment bust. In crypto, payments and fintech, expect continued pressure as investors rotate into AI. Tech should remain the underlying volatility driver, with the US most exposed.

Memory chipmakers have had geometric rises...



... pushing up Asian equities in particular

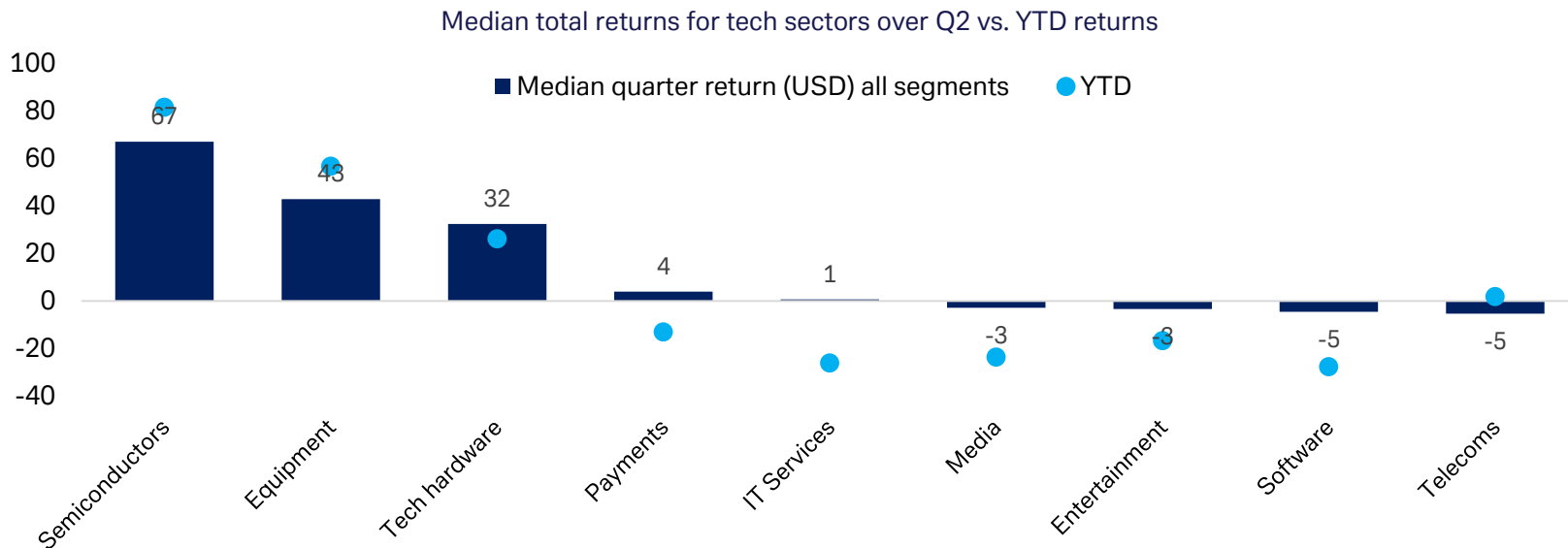


Sources: Deutsche Bank Research, Bloomberg Finance LP. Updated June 30, 2026. Returns in local currency.

1. GLOBAL INDUSTRY BREAKDOWN



Semiconductors and chip-related stocks are tech's clear group-stage winners in Q2 and YTD, led by Chinese and other Asian companies like GigaDevices (+248% q/q) and ACM Research (+210% q/q). Outside of semis, equipment and tech hardware, the rest of the field struggled. Payments and IT services continue to underperform in both AI risk-on and risk-off environments, with this quarter also weighed down by Accenture's (-37% q/q) weak FQ326 earnings in June. Software remains the worst performer, as AI disruption continues to hit companies including Intuit (-39% q/q), Horizon Robotics (-38% q/q) and Sensetime (-27% q/q).

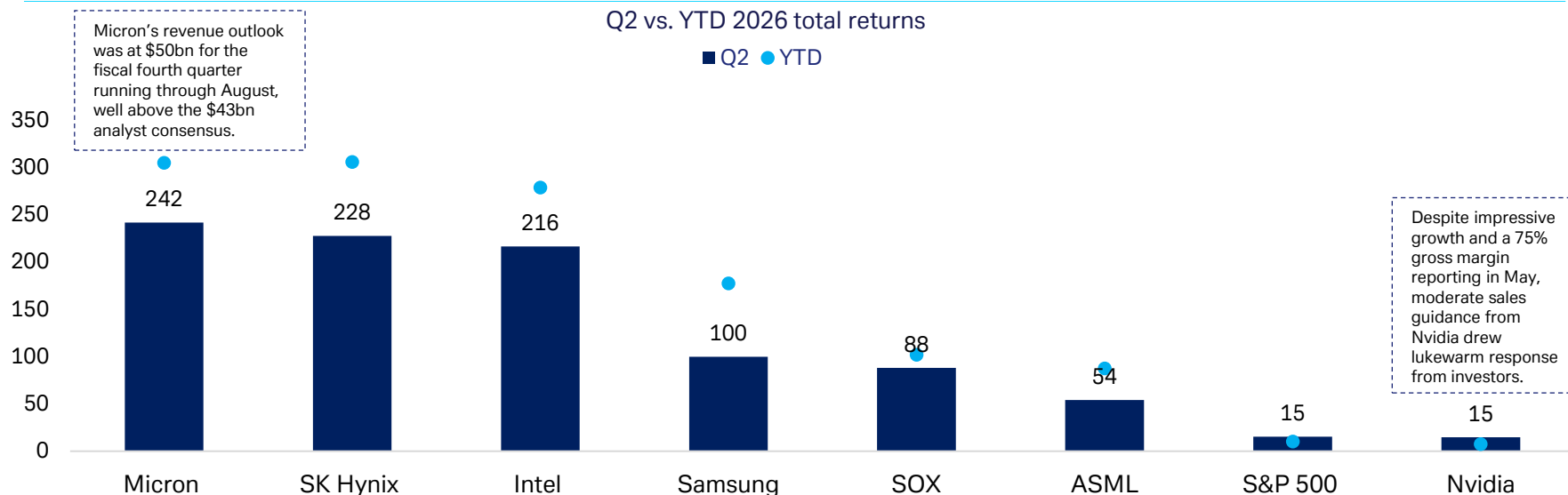


Source: Deutsche Bank Research, Bloomberg Finance LP. Updated June 30, 2026.

2. SEMICONDUCTORS



Memory chipmakers will continue to benefit from persistent industry imbalances, with Micron and SK Hynix the standout performers. Micron's earnings beat ignited hopes about AI-fuelled growth and helped push back against bubble fears. SK Hynix has also confirmed it will [seek over \\$29bn](#) via the issue of ADRs on the Nasdaq. Intel (+216% q/q) surged on a strong earnings beat and accelerating foundry momentum, with reports that Apple is evaluating Intel's 18A process and Google has reportedly placed orders for more than 3mn TPUs for 2028. Elsewhere, the Philadelphia Semiconductor Index (+88% q/q) posted its best quarterly performance since the index began in the early 1990s.

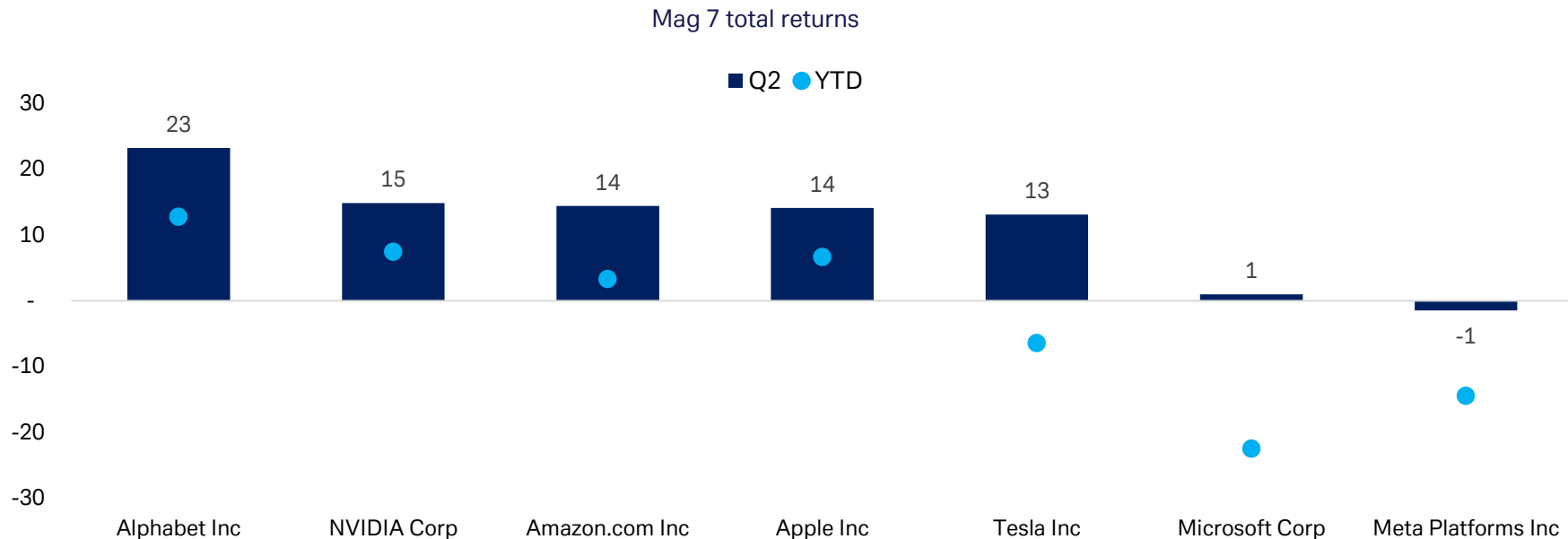


Source: Deutsche Bank Research, Bloomberg Finance LP. Updated June 30, 2026. Returns in local currency.

3. MAGNIFICENT 7



The Mag 7 started Q2 on a high, driven by strong earnings growth against a backdrop of hyperscaler capex expected to exceed \$700bn this year. In June, elevated large-cap tech positioning combined with AI concerns temporarily dragged the index into correction territory, although the group recovered by quarter-end (+12% q/q). Microsoft (+1% q/q) continues to lag YTD, as dual concerns over its ability to compete on AI spending and AI disruption risk persist.



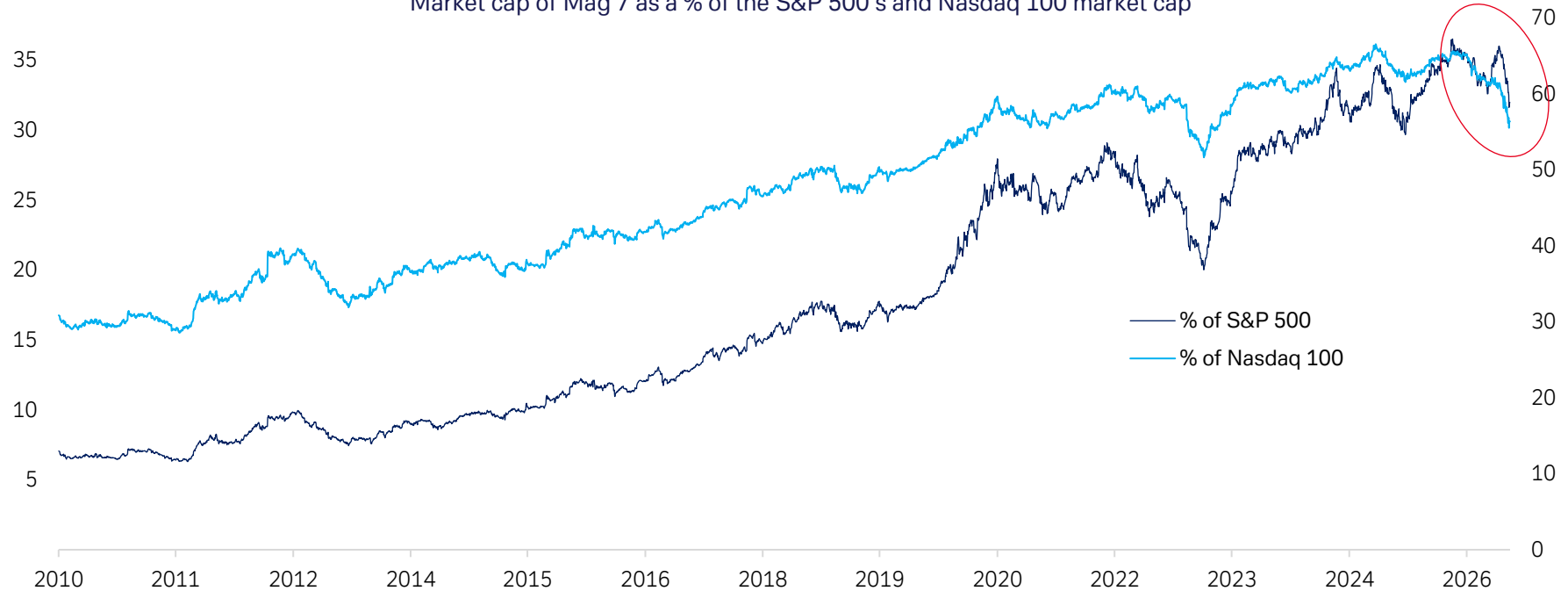
Source: Deutsche Bank Research, Bloomberg Finance LP. Updated June 30, 2026.

3.1 MAGNIFICENT 7



Large-cap tech stocks are down significantly, dragging the S&P 500 down modestly, even as two-thirds of the overall stocks and industries are up, alongside small- and mid-caps. Still, [according to our strategists](#), there is an ongoing rotation away from large-cap tech and into other sectors and regions that have lagged.

Market cap of Mag 7 as a % of the S&P 500's and Nasdaq 100 market cap



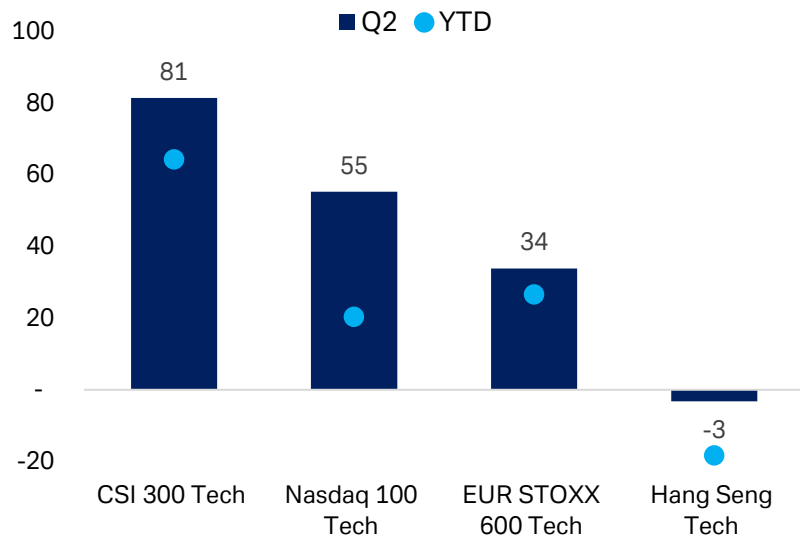
Source: Deutsche Bank Research, Bloomberg Finance LP. Updated June 30, 2026.

4. REGIONAL BREAKDOWN



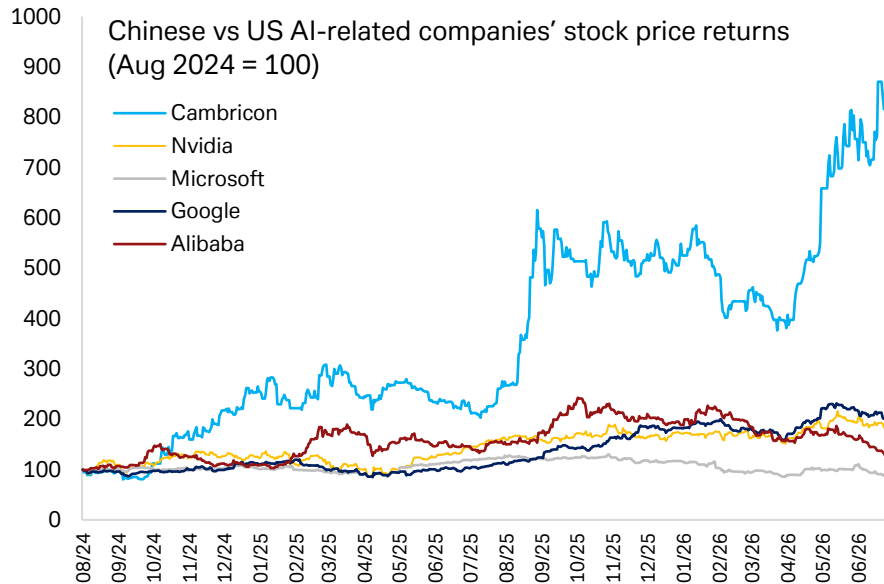
Chinese stocks are leading YTD and quarterly gains, buoyed by local memory chipmakers, while off-shore HK-listed tech continues to suffer from old-guard heavyweights like Xiaomi (-32% q/q), Alibaba (-21% q/q), and Tencent (-10% q/q). Elsewhere, European tech equities saw some growth but suffered substantial outflows during the war, with software companies particularly affected over the last six months.

Top global tech indices – Q2 vs. YTD total returns



Source: Deutsche Bank Research, Bloomberg Finance LP. Updated June 30, 2026.

Chinese vs US AI-related companies' stock price returns (Aug 2024 = 100)



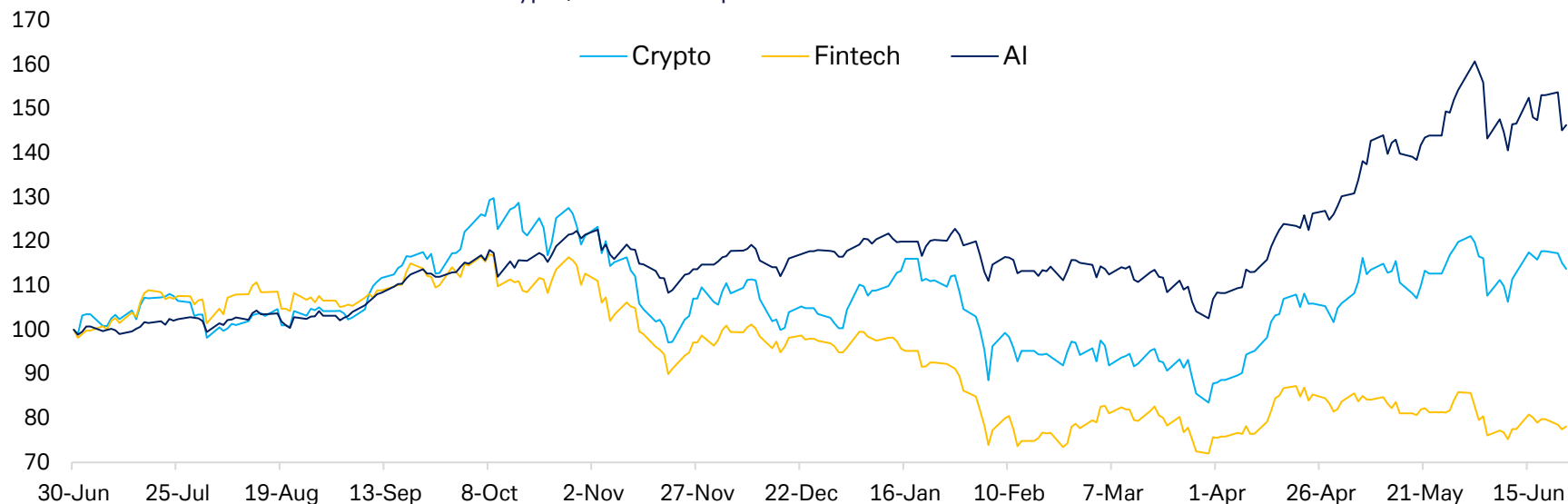
Source: Deutsche Bank Research, Bloomberg Finance LP. Indexed to 100. Updated June 30, 2026.

5. CRYPTO, AI & FINTECH



Crypto extended its struggles in Q2, falling below \$60k in June as a renewed sell-off was driven by a more hawkish Fed outlook, record institutional ETF outflows, uncertainty over the CLARITY ACT, and rotation of risk capital into AI (see [here](#)). Our Fintech and blockchain ETF remains the worst-performing segment YTD (-17%), weighed down by ongoing pain at companies like Canton Strategic Holdings (-14%) and Circle (-34%). Fintech sentiment remains weak more broadly, with the steady demand deterioration coupled with long-term fears around AI disintermediation.

Crypto, AI & Fintech performance rebased to June 2025

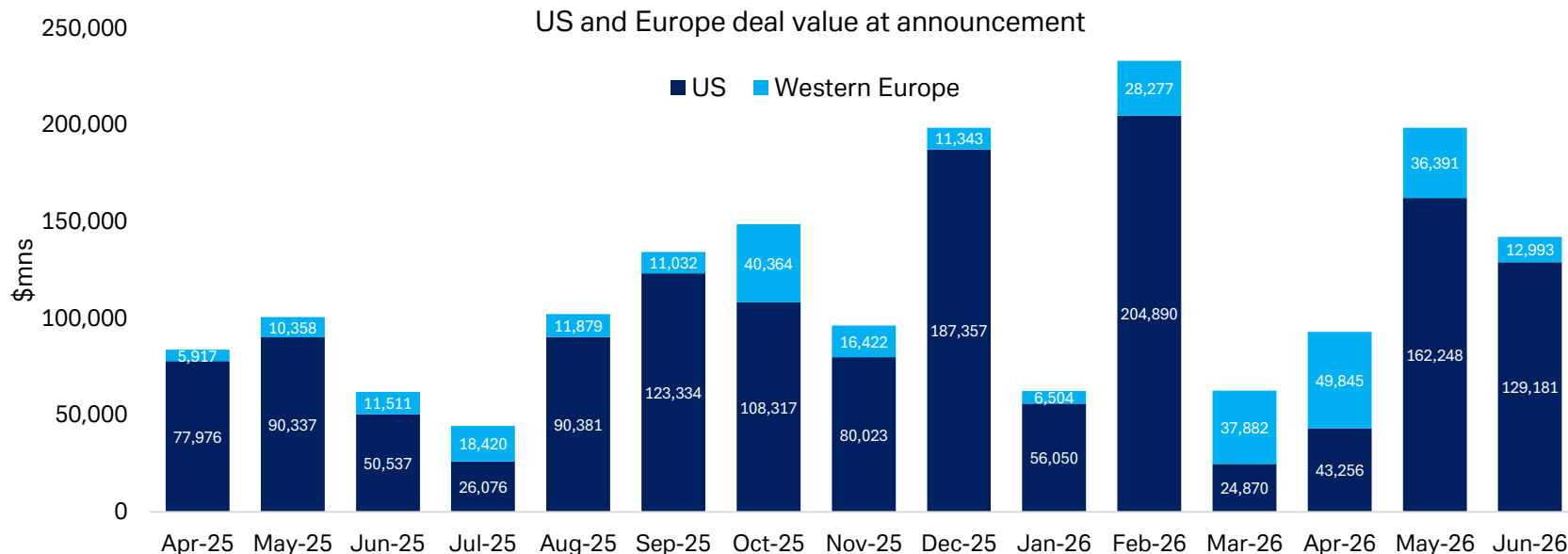


Source: Deutsche Bank Research, Bloomberg Finance LP. Data updated June 30. The following are based on AIQ ETF, BLOK US Equity ETF, and ARKF US Equity ETF.

6. M&A



After a halt in April, Q2 was a blockbuster quarter for mega-cap AI IPOs, with Anthropic (\$65bn) and OpenAI (\$852bn) filings. Beyond IPOs, semiconductors M&A led the way, including Analog Devices' proposal to acquire Empower Semiconductor for \$1.5bn in May and ON Semi's \$6.8bn acquisition of Synaptics Inc, announced on 25 June.



Source: Deutsche Bank Research, Dealogic. Values up to June 29.

DB select forecasts



	Company	Reporting Currency	Listing Currenc	Current Price	Target Price	Rec.	M. Cap	2026E	2027E	2026E	2027E
IDM	NXP	USD	USD	277.0	315.0	Buy	61.4	103x	15.0x	12.6x	10.0x
	Intel	USD	USD	128.3	100.0	Hold	565.9	111.7x	77.7x	34.5x	27.3x
	ST Microelectronics	USD	EUR	62.6	75.0	Buy	57.0	51.1x	24.1x	18.5x	11.8x
	Infineon	EUR	EUR	78.3	90.3	Buy	102.3	40.3x	30.9x	18.6x	14.8x
	Texas Instruments	USD	USD	285.4	240.0	Hold	227.9	35.7x	29.1x	22.2x	20.2x
	Renesas*	JPY	JPY	4,799.0	NA	NR	48.7	18.6x	17.5x	17.6x	15.0x
	Analog Devices	USD	USD	386.9	365.0	Hold	165.4	30.2x	26.5x	25.5x	22.1x
	onsemi	USD	USD	90.7	110.0	Buy	31.2	26.8x	16.8x	18.3x	12.4x
	Microchip Technology*	USD	USD	87.9	NA	NR	41.8	32.0x	23.2x	23.0x	17.1x
	ASML	EUR	EUR	1,578.2	1600.0	Buy	612.6	48.2x	32.6x	37.9x	25.9x
Semi Equip	Applied Materials	USD	USD	626.8	550.0	Buy	436.7	48.2x	35.3x	40.8x	30.1x
	KLA	USD	USD	248.6	1750.0	Hold	285.0	56.3x	46.3x	46.7x	38.6x
	Lam Research	USD	USD	379.1	325.0	Buy	416.0	55.5x	42.5x	45.8x	35.4x
	TEL*	JPY	JPY	74,000	NA	NR	187.7	49.9x	38.6x	36.0x	27.4x
	Screen*	JPY	JPY	16,665	NR	17.2	26.9x	23.2x	18.2x	14.9x	13.2x
	ASM Int'l	EUR	EUR	948	925.0	Buy	46.8	44.5x	36.7x	29.3x	24.0x
	VAT Group	CHF	CHF	669.8	550.0	Hold	21.8	64.8x	46.0x	46.1x	33.7x
	BE Semiconductor	EUR	EUR	282.7	275.0	Buy	22.9	66.4x	45.1x	55.1x	38.4x
	Technoprobe	EUR	EUR	33.9	42.0	Buy	21.6	74.0x	44.4x	45.4x	26.6x
	AMD	USD	USD	521.6	365.0	Hold	746.3	70.1x	43.5x	56.8x	34.9x
Fabless	MediaTek	TWD	TWD	3,910.0	2850.0	Buy	172.5	57.8x	34.7x	44.8x	26.8x
	Nvidia	USD	USD	192.5	255.0	Hold	4,088.5	21.3x	15.8x	17.8x	12.3x
	Marvell	USD	USD	266.8	240.0	Buy	204.8	67.5x	43.8x	50.6x	33.1x
	Broadcom	USD	USD	365.0	315.0	Buy	1,523.9	27.9x	7.0x	27.9x	13.2x
	Qualcomm	USD	USD	189.4	160.0	Hold	175.2	18.0x	17.7x	14.4x	14.4x
	TSMC	TWD	TWD	2,370.0	2500.0	Buy	1,690.5	24.1x	18.8x	15.8x	12.2x
	UMC*	TWD	TWD	164.0	NA	NR	56.7	35.1x	28.4x	15.9x	12.9x
	HKMIC*	USD	USD	82.3	NA	NR	104.4	59.7x	24.6x	20.1x	14.1x
	Tower Semiconductor*	USD	USD	249.9	NA	NR	24.7	70.8x	41.1x	38.4x	25.6x
	arm	USD	USD	334.3	205.0	Hold	313.3	162.2x	123.3x	133.1x	105.0x
IP/EDA Comp	Synopsys	USD	USD	454.3	590.0	Buy	76.3	29.8x	25.2x	17.4x	15.8x
	Cadence	USD	USD	377.3	375.0	Buy	91.3	47.3x	39.1x	36.2x	30.6x
	Logitech	USD	CHF	78.8	80	Hold	13.7	17.3x	15.8x	14.3x	13.5x
	Apple*	USD	USD	283.8	NA	NR	3,657.4	31.6x	28.6x	23.6x	21.0x
	HP Inc*	USD	USD	22.9	NA	NR	18.4	7.6x	7.5x	5.6x	5.5x
	Samsung Electronics*	KRW	KRW	327,900	NR	1,092.6	5.2x	5.2x	4.0x	2.4x	2.4x
	SK Hynix*	KRW	KRW	2,643,000	NA	NR	1,072.8	8.5x	5.8x	6.1x	3.5x
	Micron	USD	USD	1,132.3	1550	Buy	1,122.2	11.2x	6.8x	8.5x	5.0x
	Cisco Systems*	USD	USD	113.8	NA	NR	393.5	25.3x	22.8x	18.8x	17.0x
	Ericsson	SEK	SEK	32.5	14.4x	Hold	14.4x	12.9x	7.5x	6.8x	6.8x
Telecom Equipment	Motorola Solutions*	USD	USD	402.9	NA	NR	58.7	23.7x	21.7x	17.2x	15.7x
	Nokia	EUR	EUR	11.4	13.50	Buy	65.4	35.2x	29.6x	20.5x	17.0x
	Ciena Corp*	USD	USD	479.5	NA	NR	59.6	67.6x	45.7x	46.6x	33.5x
	5 Networks*	USD	USD	394.5	NR	19.5	23.6x	21.8x	16.2x	14.9x	14.9x
	Keysight*	USD	USD	328.7	NA	NR	49.3	31.4x	27.2x	24.0x	21.2x
	ZTE Corp*	CNY	CNY	35.3	NA	NR	20.3	23.5x	20.7x	16.6x	15.3x
	Adobe	USD	USD	202.7	260.0	Hold	77.2	8.2x	7.0x	5.9x	5.1x
	Amadeus	EUR	EUR	58.0	55.0	Buy	15.3x	13.8x	13.4x	8.3x	8.3x
	Autodesk	USD	USD	196.3	330.0	Buy	39.9	15.8x	13.6x	12.1x	10.3x
	Cadence Design	USD	USD	377.3	375.0	Buy	103.1	47.3x	39.1x	34.8x	29.5x
Software Comps	Dassault Systemes	EUR	EUR	17.9	21.0	Hold	26.7	13.7x	13.0x	9.3x	8.3x
	Hexagon	SEK	SEK	81.0	86.0x	Hold	22.2	26.6x	23.5x	13.8x	12.7x
	Intuit	USD	USD	267.7	530.0	Buy	70.5	10.6x	9.2x	7.4x	6.1x
	Microsoft	USD	USD	373.0	550.0	Buy	2,784.2	19.0x	17.2x	11.9x	9.8x
	Nemetschek	EUR	EUR	52.6	NA	NA	7.1	19.6x	16.3x	13.6x	10.9x
	Oracle	USD	USD	148.5	300.0	Buy	432.8	15.3x	15.6x	11.6x	8.6x
	PTC	USD	USD	115.7	180.0	Buy	13.2	14.2x	12.9x	10.2x	9.3x
	Sage	GBP	GBP	817.6	900.0	Hold	10.7	14.4x	12.5x	10.4x	8.8x
	Salesforce	USD	USD	158.4	255.0	Buy	125.2	11.3x	10.3x	13.3x	7.8x
	SAP	EUR	EUR	135.9	200.0	Buy	173.9	7.0x	14.3x	10.2x	9.7x
Services Comps (SBC)	ServiceNow	USD	USD	98.3	135.0	Buy	102.9	24.2x	19.8x	2.8x	10.5x
	Synopsys	USD	USD	454.3	590.0	Buy	84.5	29.8x	25.1x	16.7x	15.1x
	Workday	USD	USD	124.2	180.0	Buy	33.3	11.7x	9.8x	9.6x	7.8x
	Accenture	USD	USD	129.0	140.0	Hold	80.3	7.6x	7.1x	4.9x	4.6x
	Capgemini	EUR	EUR	89.7	125.0	Buy	77.7	7.2x	6.5x	4.8x	4.4x
	CGI	CAD	CAD	91.3	NA	NA	14.2	9.7x	8.9x	6.7x	6.1x
	Cognizant	USD	USD	40.0	70.0	Buy	19.2	6.7x	6.1x	3.8x	3.1x
	HCL Technologies	INR	INR	1,100.7	NA	NA	31.7	15.5x	14.2x	9.1x	8.3x
	IBM	USD	USD	271.6	NA	NA	254.5	17.9x	17.9x	13.8x	12.7x
	Infosys	INR	INR	1041.2	NA	NA	44.4	13.4x	12.6x	7.9x	7.3x
	TCS	INR	INR	2094.7	NA	NA	79.9	13.9x	12.9x	9.6x	9.2x
	Wipro	INR	INR	175.0	NA	NA	19.5	12.7x	12.1x	7.6x	7.1x

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Market cap
in EUR bn

Market cap
in USD bn

Source: Deutsche Bank Research, Bloomberg Finance LP for non-covered companies.
Updated June 29 2026.

Appendix 1



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FIFA World Cup: What economics sees that fans don't

June 30, 2026

This summer, two markets are running simultaneously: one financial, one sporting. While financial markets react daily to macroeconomics and geopolitics, the sporting market — currently represented by the football World Cup — unfolds across 16 venues in the US, Canada and Mexico, with 48 nations competing for a trophy and an expected 6bn global viewers.

These markets are more alike than most observers admit. Both:

- Thrive on scarcity and compelling narrative;
- Follow the same microeconomic laws of supply and demand, driving pricing and profit;
- Reprice in real-time based on new information;
- Exhibit "winner-takes-most" structures, where top performers outstrip the rest exponentially; and
- Exhibit herding behavior - retail investors and football fans alike chasing momentum, overpaying at peaks and panic-selling (or abandoning) at troughs.

This makes the FIFA World Cup an unparalleled “economics laboratory”. Over 39 days, it puts core microeconomics on display — across tickets, host cities, tournament design, player pay, fan behavior, and even the business model of FIFA, football’s international governing body, itself.

Authors

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1. The ticket market: Dynamic pricing meets the Tinkerbell problem

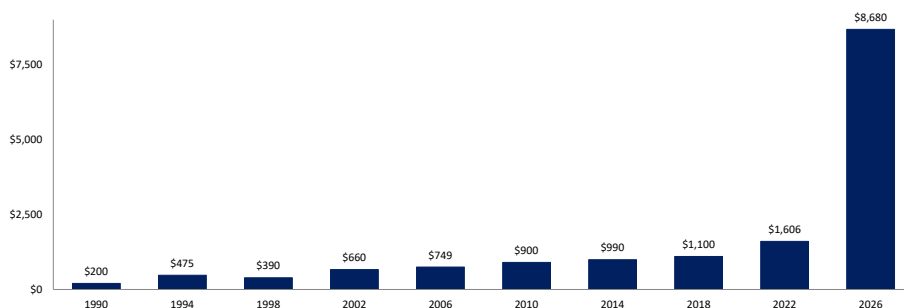
FIFA implemented algorithmic dynamic pricing for 2026 — a first for the World Cup. Think of it like Uber surge pricing: the fare changes depending on how many people want a ride at any given moment. The same mechanism now applies to football tickets, and the effects were immediate.

By February, the median resale ticket on [SeatGeek](#) had reached \$1,291. The US opener against Paraguay saw prices exceed [\\$3,200](#). Premium final seats at MetLife Stadium, with a face value of \$6,730, [were tripled by FIFA to \\$32,970](#) by early May. US President Donald Trump publicly said he would not pay those prices.

By May, demand softened — group-stage prices fell [28%](#) to a median of [\\$928](#). But once results started to matter, prices surged again. After the US beat Paraguay 4–1, tickets for the next US match climbed [65%](#) in 48 hours. By mid-June, [84%](#) of all matches had seen resale prices rise.

This is what we have previously called the “[Tinkerbell Effect](#)”: while the product (a seat) does not change, the belief and narrative surrounding it do and that is enough to move the price. FIFA capitalized on this by creating an [official resale marketplace](#), collecting a 15% fee from both buyer and seller. At the other extreme, Jordan vs Algeria and Curaçao vs Ivory Coast tickets were available for under \$450.

Figure 1: World Cup final ticket prices



Source: [The World Cup Guide](#) Note: Category 1.

2. Host cities: Scarcity priced in, demand missing

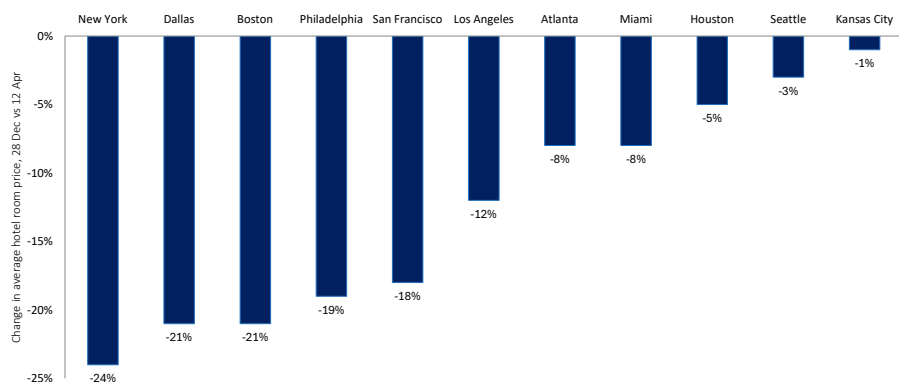
The hotel market illustrates how quickly an anticipated demand shock can reverse. Following the December group-stage draw, room rates in some host cities surged by up to 300% overnight. By April, [80% of US hotel operators were below their initial forecasts](#). New York was down [65%](#) vs expectations; Seattle, [80%](#). Game-day rates had fallen roughly one-third from their December peaks. Mexico was the exception: tighter local supply kept prices elevated.

This shortfall reflects more price-sensitive demand than many expected. And it is not without precedent. The World Cup in France in 1998 offers the closest parallel: organizers predicted 500,000 extra tourists, yet fewer visitors came than in a normal June. Non-football fans simply avoided the country; a pattern economists call the “crowding-out effect.” Overnight hotel stays by non-residents fell [13%](#) year-on-year. The same dynamic is playing out in 2026, compounded by visa and entry frictions that reduced cross-border travel — a factor cited by [70%](#) of hotel operators as a primary drag on international demand. Several high-profile cases reinforced this constraint, including denied entry to officials and ticket restrictions affecting entire national allocations.



The overall macroeconomic impact is limited. The US GDP uplift is estimated at about [0.05%](#) — [negligible relative to the headline projections](#). External assessments consistently find realized gains materially below advertised figures, in line with research showing that [12 of the last 14](#) World Cups produced net economic losses for host regions. With Canada's hosting costs at ~C\$1.07bn (~C\$82mn per match), the event is less about economic stimulus and more about prestige spending.

Figure 2: World Cup hotel rates (%)



Source: Business Traveller, citing Lighthouse Intelligence. Comparison of average hotel room prices booked on 28 December vs. 12 April.

3. Fans: "The ball is round", behavior isn't

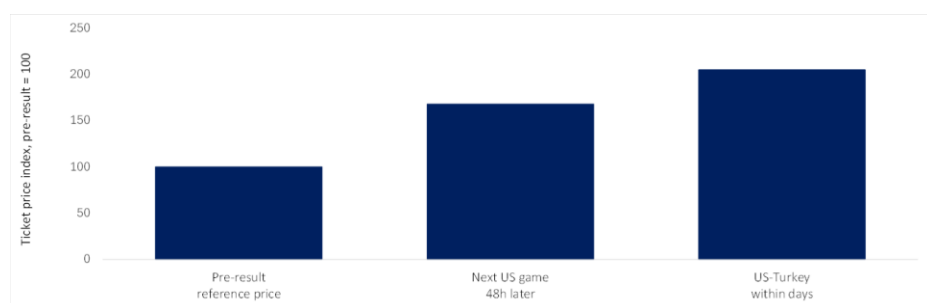
Fans price emotion, not assets. After the US defeated Paraguay 4–1, tickets for the next game surged [68%](#) in 48 hours; US v Turkey prices climbed [105%](#) in days. The product remains unchanged; only the narrative shifted. As Nobel Prize-winning economist Daniel Kahneman showed, people's valuations are reference-dependent and heavily influenced by recent information. In this case, a single victory fundamentally altered fans' willingness to pay.

Fan behavior also reflects Nobel Prize winner Richard Thaler's classic biases. Fans holding tickets for matches their team will no longer play face a sunk-cost dilemma. Many attend rather than sell at a loss, as loss aversion dominates: avoiding a realized loss matters more than optimizing value. What appears to be demand is partly behavioral inertia leading to predictable mispricings. Seats are filled not because their marginal value has increased, but because past spending anchors decisions. Markets clear, but not efficiently.

The contrast is broadcasting. While stadium access is scarce and volatile, global viewership scales at near-zero marginal cost. FIFA's ~\$4.2bn in broadcasting rights captures value from the marginal viewer, not the marginal ticket buyer — a reminder that, in this market, price reflects psychology as much as scarcity.



Figure 3: How winning changes ticket prices



Source: seatgeek.com

4. Tournament design: All about game theory

Rules shape behavior, sometimes more than talent does. Tournament design significantly influences outcomes. The 1982 match between West Germany and Austria is the classic illustration: both teams knew that a 1–0 win for West Germany would send them both through at Algeria's expense, and that's what happened. Given the rules, the outcome was entirely rational. FIFA's response — requiring final group-stage matches to kick off simultaneously — was an institutional fix to a game-theory problem.

The 2026 format partly recreates these conditions. With 12 four-team groups and eight best third-place finishers advancing to the round of 32, there are again situations where teams gain more from managing a result than from maximizing performance. Cape Verde illustrated the new incentive structure: three draws were enough to qualify (the first time since World Cup 1998). The early phase of the tournament rewards not losing as much as it rewards winning.

This played out on the pitch. Against England, Ghana showed little interest in pressing for a win; having already won their earlier match, a draw was the rational outcome to fully secure qualification. England's possession — however dominant — could not change that calculus. Norway, already through and needing only a draw to top the group, fielded a rotated side against France, rationally conserving energy for the knockout rounds.

Yet, 2026 also exposes the limits of purely incentive-based models. Turkey, despite being already eliminated and with nothing at stake, beat the US 3–2 in stoppage time in a match with no bearing on standings. No prize-money model — with payouts ranging from [\\$9mn](#) for a group-stage exit to \$50mn for the champion — would have predicted that result. Pride, national honor, and players auditioning for future contracts generated an intensity that financial incentives alone cannot explain.

Incentives matter, but they do not explain everything.



Figure 4: Ranking of the third best

Ranking	Country	Points	+/-
1	DR Congo	4	+1
2	Sweden	4	0
3	Ghana	4	0
4	Ecuador	4	0
5	Bosnia-Herzegovina	4	-1
6	Algeria	4	-2
7	Paraguay	4	-2
8	Senegal	3	+2
Eliminated			
9	Iran	3	0
10	South Korea	3	-1
11	Scotland	3	-3
12	Uruguay	2	-1

Source: FIFA; <https://www.skysports.com>.

5. Penalties: Behavioral economics under pressure

As tournament rounds advance, quality differentials compress. By the semi-final stage, the marginal gap between sides is small enough that outcome is determined not by aggregate talent but by individual decision-making under extreme pressure. Penalty shootouts are that mechanism institutionalized.

The incidence is significant. Four of the last 10 World Cup champions survived at least one shootout en route to the trophy — in 2022, Argentina survived two against the Netherlands and France. A material share of tournament outcomes are therefore decided not by 90-minute performance differentials but by a discrete, high-variance event with well-documented behavioral distortions.

Shootouts perfectly illustrate loss aversion, mixed-strategy equilibrium, and irreversible decision-making. Kahneman and Tversky's framework explain how players, weighing the asymmetric reputational cost of missing against scoring, systematically favor "safe" shots even when an unpredictable strategy is superior. The fear of missing is not irrational; it is a rational response to asymmetric social payoffs.

The data are unambiguous. World Cup shootout conversion stands at ~70%, against ~80% for penalties in normal and extra time — a 10pp gap attributable to psychological load, not technique or distance. Premier League conversion runs at 80% in lower-stakes contexts. The distance is unchanged; what changes is the psychological weight attached to each kick.

The differential in success rate may be explained by goalkeepers systematizing their preparation for shootouts with more transparent data. Pickford's water bottle at Euro 2024 — carrying printed tendency data on each Swiss taker, direction by direction — captured this precisely: "AKANJI: DIVE LEFT." He dived left and saved it. England used the same approach in the 2018 World Cup round-of-16 against Colombia. Instinct has been replaced by a signal-processing problem. As goalkeepers industrialize their reads, kickers randomize further — producing exactly the mixed-strategy equilibrium game theory predicts, with neither side able to sustain a systematic edge.



The underlying condition remains fixed: a professional athlete, optimally prepared, missing from 12 yards not because of technique but because tournament stakes have recalibrated every cognitive signal. A penalty is ultimately a test of psychological resilience, not technical skill.

Figure 5: 10 last World Cup winners' penalty shoutouts

Winner	Match won on penalties
Argentina 2022	Netherlands (1/4); France (final)
France 2018	—
Germany 2014	—
Spain 2010	—
Italy 2006	France (final)
Brazil 2002	—
France 1998	Italy (1/4)
Brazil 1994	Italy (final)
Germany 1990	England (1/2)
Argentina 1986	—

Source: FIFA; <https://www.skysports.com>.

6. Players: The GOATs take it all

In 1981, economist Sherwin Rosen argued that small differences in talent, amplified by technology that lets the best performers reach unlimited audiences, produce extreme income inequality. The 2026 squad lists are the most striking illustration of that thesis in the sport's history.

Cristiano Ronaldo of Portugal enters his sixth World Cup at the age of 41 with annual earnings of [~\\$300mn](#) — a base salary of ~\$235mn at Al-Nassr (tax-free) plus endorsements including a lifetime Nike deal — making him the first-ever soccer player to cross \$1bn in net worth. Argentina's Lionel Messi earns [~\\$140mn](#) annually; his Inter Miami contract includes a revenue-sharing clause tied to Apple TV's MLS Season Pass subscriptions, giving him equity-like upside in platform growth. France's Kylian Mbappé, at [~\\$100mn](#) per year, is already building a venture portfolio before age 28.

Below this elite tier, the drop-off is steep. Mid-squad players earn 10–20% of their income from endorsements. New Zealand's Tim Payne worked in construction between international caps. If football's income distribution were mapped as a Gini coefficient — the standard measure of inequality used for whole economies — it would likely rank among the most unequal in the world.

The Saudi Arabian influence makes this more extreme still, with sovereign wealth capital deployed as soft power, pricing players at levels no normal commercial model would typically justify. This is structurally identical to US hyperscalers committing over \$700bn to AI infrastructure in 2026: strategic positioning, not near-term return.

Figure 6: World Cup 2026: Top 5 vs. bottom 5 squad market values (€mn)

Top 5	Market value	Bottom 5	Market value
France	€1,520mn	Qatar	€20mn
England	€1,360mn	Jordan	€20mn
Spain	€1,220mn	Iraq	€21mn
Portugal	€1,010mn	Curaçao	€26mn
Germany	€947mn	Iran	€32mn

Source: [Transfermarkt](#); Deutsche Bank Research.



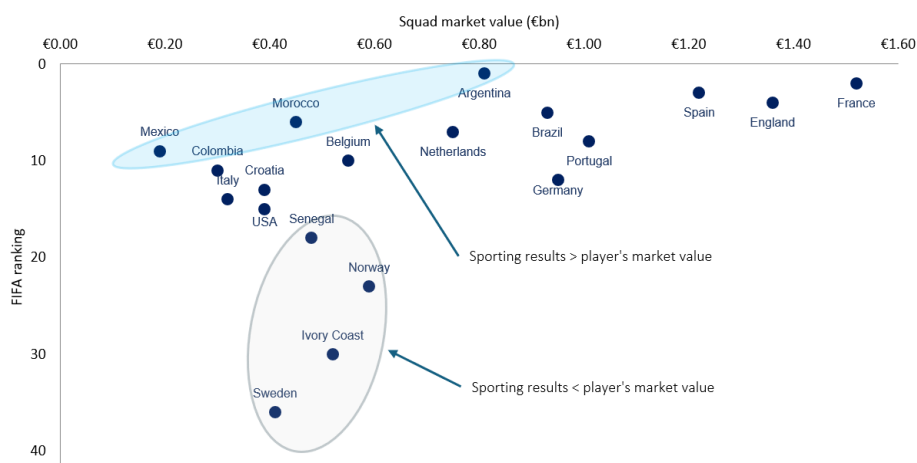
7. Teams: Better than the sum of their GOATs

Moneyball — Michael Lewis's 2003 book about the Oakland Athletics — showed how statistical analysis could greatly assist in building a winning team. It works particularly well in baseball because the sport decomposes into a large number of relatively homogenous individual duels, simplistically: pitcher against batter, each with a measurable contribution that can be compared with many other similar events. The whole approximates the sum of its parts.

Football is more complicated as it has a low number of repeatable scoring 'duels'. Rather, a player's value is frequently derived from hard-to-measure intangibles that necessitates the use of complex 'non-linear dynamic system models'. These are only starting to understand how players interact with each other in a way that can be shaped by training and chemistry built over months. Alchemy matters in a way that spreadsheets find it difficult to capture. The World Cup bears this out. Italy's winning team in 2006 had five players that knew each other well from Juventus (Buffon, Cannavaro, Zambrotta, Camoranesi, and Del Piero); Spain in 2010 had seven from Barcelona (Valdés, Puyol, Piqué, Xavi, Iniesta, Busquets, and Pedro); Germany in 2014 had six from Bayern (Neuer, Lahm, Boateng, Schweinsteiger, Kroos, and Müller). In 2026, the pattern holds among the favorites. Germany is again built around a Bayern spine of six — Neuer, Kimmich, Tah, Musiala, Goretzka, and Pavlović — alongside four from Dortmund and three from Stuttgart. Spain again leans on Barcelona, with Cubarsí, Pedri, Lamine Yamal, and potentially Ferran Torres and Gavi. France has six Paris St Germain players in the mix.

The counterexamples are equally instructive. Cohesion without quality produces coordination without results; the winning formula requires both. Saudi Arabia fielded six or seven Al-Hilal players; Czech Republic 10 from Slavia Prague. Both are eliminated.

Figure 7: Correlation between FIFA ranking and team market value (€ bn)



Source: [Transfermarkt](#), FIFA, Deutsche Bank Research.

8. "Football is a simple game", scaling isn't

FIFA is best understood not as a sports organizer but as a platform — like Apple or Uber — sitting in the middle between two groups (teams on one side; broadcasters, sponsors and fans on the other), capturing value from both. The more users on each side, the more valuable the platform becomes.



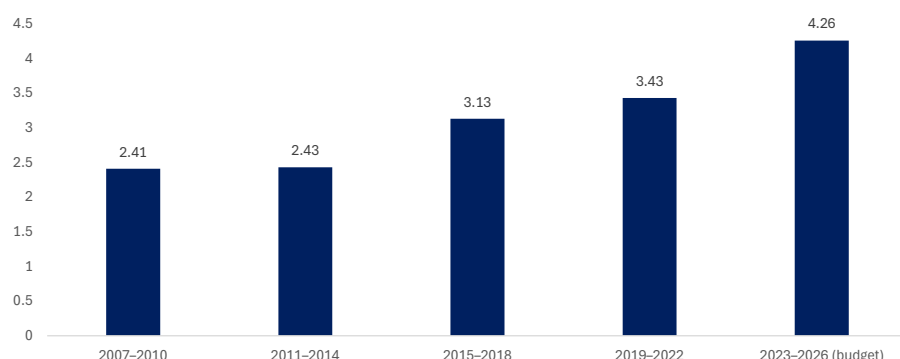
This year's move of expanding the tournament from 32 to 48 teams increased the number of matches from 64 to 104. More matches, combined with two new mandatory 3-minute stoppage breaks per game, mean more advertising slots, more sponsorship activations, and more data generated. The revenue arithmetic is direct: [broadcast revenue above \\$4.2bn](#) (US rights up ~94% vs 2022), [sponsorship at ~\\$2.4bn \(+37%\)](#), and [total cycle revenue approaching \\$13bn](#), up from \$7.6bn in Qatar. Scale, not scarcity, is now the dominant strategy.

FIFA is also moving further into the value chain. Its official resale marketplace mirrors Apple's App Store: a controlled secondary market framed as consumer protection, charging 15% fees on every transaction. This is vertical integration — capturing more of the value the platform creates.

The next frontier is data. With 150mn data points generated per match and 104 matches played, the 2026 tournament represents the largest structured sports dataset ever assembled. Data, not just matches, is becoming a monetizable asset.

The trade-off is familiar to every platform that has ever grown. More content increases reach but risks diluting quality at the margin. Curaçao vs Tunisia in a half-empty stadium is not Brazil vs Argentina, whatever the algorithm prices it at. The balance between scale and quality is now FIFA's central strategic question.

Figure 8: FIFA World Cup broadcast rights revenue has nearly doubled over the past two decades (\$ bn)



Source: FIFA Annual Reports (2010, 2014, 2018, 2022, 2024). Figures refer to four-year commercial cycles; 2023–26 is FIFA's budgeted revenue.

9. Football market as capital markets: A broader ecosystem

The maturation of capital markets produced more than trading profits. It generated an entire industrial infrastructure: data terminals (Bloomberg, Refinitiv), analytics vendors, and retail execution platforms that democratized market access. Football's commercial ecosystem is replicating this structure, driven by the same dynamic: more data, more participants, more liquidity.

The data layer is expanding fastest. FIFA is tracking ~150mn data points per match in 2026, with the match ball's internal sensor logging 500 movements per second. Every player is 3D-scanned and tracked 50 times per second, generating a continuous biometric and performance data stream across all 104 matches. FIFA's AI Pro system, distributed to all 48 teams, provides identical pre- and post-match analytical capabilities via AI agents. The analogy to Bloomberg terminals holds: what those terminals did for price discovery in the 1980s, real-time tracking infrastructure is doing for football performance and wagering today.



The wagering market has scaled accordingly. Global wagers were [\\$35bn](#) at Qatar 2022; 2026 is projected to surge by 43% to exceed \$50bn driven by the expanded format, broader regulatory access, and North American time zones. Prediction markets represent a new layer: Kalshi and Polymarket alone surpassed Super Bowl prediction-market volume within 48 hours of tournament kickoff.

Brazil is the sharpest case study in what happens when deep football fandom meets regulatory liberalization. Online betting was legalized in 2018 but unregulated until January 2025. The results were immediate: licensed operators generated [\\$7bn](#) in gaming revenue in the first year, with [BRL 10bn](#) collected in tax revenue, making Brazil one of the world's largest regulated betting markets in its first year of operation. The driver is simple: roughly one in four of Sofascore's Brazilian users are active bettors, and [18 of 20](#) Série A clubs carried betting sponsors in 2025 — though that figure has since pulled back to [12–13](#) as the market matures. The fandom is the distribution network.

The capital markets analogy holds at the ecosystem level. Retail participation, data infrastructure, and institutional liquidity are scaling simultaneously, each reinforcing the others. The sports analytics market alone — [\\$2.29bn](#) in 2025, forecast at [\\$4.75bn by 2030](#) — is the infrastructure layer that makes the wagering layer possible, precisely as Bloomberg and Refinitiv underpinned the growth of retail trading.

The economic product has shifted in the process: the World Cup used to monetize attention through tickets, sponsorship, and broadcast rights. The same 90 minutes now generate a continuous data inventory — odds, expected goals (xG), corners, cards, player props, micro-markets — and what is being priced is real-time uncertainty, not a match result.

10. Prediction markets as capital markets: Repricing in real time

Financial markets price expectations, not outcomes. When tensions flare around the Strait of Hormuz, oil prices move immediately: the market is revising its probability distribution over future states. Prediction markets apply the same framework to football: each contract is a continuously updated implied probability, and every red card, substitution, or missed penalty is a signal that triggers instant repricing.

The market is no longer marginal. Peak betting traffic is expected to average [100,000 wagers per minute](#) around key matches in the 2026 World Cup. US sportsbooks (FanDuel, DraftKings, BetMGM) run full tournament slates. But the more structurally interesting development is the rise of prediction markets proper: Kalshi and Polymarket together went from [\\$2.2bn to \\$4.8bn](#) in trading volume on the opening day alone, surpassing Super Bowl prediction-market volume within 48 hours. Unlike sportsbooks, these platforms allow participants to enter and exit positions continuously, acting as liquid secondary markets in probabilistic outcomes.

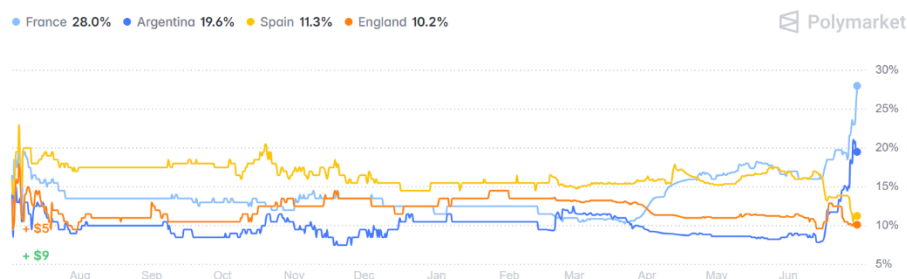
Argentina's tournament offers the cleanest illustration. The defending champions entered at [~9%](#) implied probability — unusually low for a title holder — reflecting market uncertainty about Messi at the age of 38 and a congested field. A perfect group stage (three wins, no goals conceded) pushed that to [~13%](#). Then came the more revealing repricing: after the group stage draw, Argentina's probability surged to [~21%](#) without kicking another ball. Portugal's failure to top its group had handed Argentina an easier path to the semi-finals with no heavyweight opponent until that stage. The market priced the bracket, not a



result. No new information about Argentina's squad quality had changed; what changed was the probabilistic map of their path to the trophy.

That is the efficient market hypothesis operating in compressed time. The tournament is, in this reading, a six-week price discovery mechanism: 104 matches, \$50bn in open interest, and a continuous information flow that no participant can fully anticipate.

Figure 9: World Cup Winner



Source: [Polymarket.com](https://polymarket.com).

Conclusion: Can America make football great again?

The 2026 World Cup has been a live economics seminar mostly confirming the theory. Dynamic pricing has extracted value but triggered a backlash. The anticipated demand boom in host cities has softened under geopolitical pressure. Game-theory distortions have emerged on cue. Superstar pay has widened income disparity further. Fans have overpaid at peak and held tickets at a loss. And FIFA, like any dominant platform, continues to capture a larger share of the value it enables.

The final whistle blows on July 19 at MetLife Stadium in New Jersey. Regardless of the score, the economic outcome is already visible: higher prices, strong revenues, and a tournament that has solidified football's role as both a global spectacle and a significant business engine. Andrew Giuliani — executive director of the White House's World Cup task force — has already suggested expanding the tournament to 64 teams and proposed a US bid for 2038. The arithmetic is simple: more matches, more markets, more revenue.



Appendix 1

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Washington's Post-Iran Playbook: Risks, Deals and Pressure Points

June 29, 2026

Executive Summary

- Markets should expect more policy volatility, not less, after Iran. Washington's willingness to consider using force means geopolitical tail risks will likely remain difficult to price.
- The midterms may cap some near-term volatility but increase incentives for visible foreign policy wins. President Trump has reason to avoid market-disruptive escalation before November, but pressure for quick wins could accelerate action on Cuba and Europe.
- After November, foreign policy could become an even bigger focus. If domestic legislating becomes harder, the administration is likely to lean more heavily on executive tools, sanctions, tariffs, security deals and international resource coalitions.

Context

- The first half of 2026 has demonstrated that the Trump administration has a higher tolerance for market volatility than many anticipated, including decisions that risk disrupting trade, commodity flows and supply chains.
- Markets will remain sensitive to signalling from the Trump administration on international security. Trump's willingness to threaten force to advance talks drove repeated bouts of volatility in H1 – but the signals will remain hard to price as the continued ambiguity may lead markets to misinterpret geopolitically driven tail risks over the next two and a half years.
- The Iran ceasefire partially frees Trump to refocus on longstanding policy priorities in the second half of 2026. But domestic political pressures will likely weigh more heavily on the Trump administration's foreign policy post-Iran. Public support for further military campaigns – especially lengthy ones or those involving ground forces – is low. But investors should not overestimate how far this constraint will shape Trump's international strategy over the coming two and a half years.
- The administration has shown it is willing to use, or threaten to use, air and naval forces to resolve international disputes where negotiations stall. The interventionism of H1 2026 defied almost every expectation set in 2025 for Trump's second term. While it is not our base case, investors should consider tail-risk scenarios that hit markets through energy supply, freedom of navigation in key waterways, or spillover into tariffs and export controls.

From ceasefire to stress test

- Even after the Memorandum of Understanding with Iran, the US faces serious diplomatic and security challenges in the Middle East. The Phase 2 negotiations with Iran that began on 21 June (focusing on the nuclear programme and deal implementation) will be complex and almost certainly run past the initial 60 days. Reopening the Strait of Hormuz to full pre-war

Authors

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traffic could take months and require a major military commitment to fully de-mine its central corridor and reassure shipping firms.

- Washington is consequently pushing European allies and other regional actors to shoulder more of the burden after the initial re-opening. Recent strains in US-Israel relations could complicate the administration's efforts to maintain delicate ceasefires in Lebanon and Gaza, and have already affected the truce with Iran and the navigability of the Strait of Hormuz. Alongside US efforts to manage post-war security assurances for Gulf Arab countries ([which we discussed in our recent note](#)), the Middle East will likely remain a focus for the rest of Trump's term.

NATO's burden-sharing bill comes due

- We expect pressure on European allies to carry more of the security burden will intensify as US stockpile and resource constraints bite after the Iran war. Trump's demands will likely headline the NATO Leaders Summit in Ankara on 7-8 July. It is slated to assess the credibility of national defence spending plans and advance the emerging "NATO 3.0" agenda under which European NATO members take on more responsibility for their own conventional defence.
- Ahead of the meeting, the US has announced some military pullbacks from Europe and reportedly notified allies of further planned reductions to its NATO Force Model contributions. To slow the withdrawal of hard-to-replace US capabilities (such as air defence, long-range weaponry and strategic airlift), European leaders will likely stress their commitment to NATO's new 3.5%-of-GDP target for "core" defence spending and their readiness to help clear the Strait of Hormuz.

Tariff pressure remains a dealmaking tool

- The July expiry of Section 122 tariffs and their replacement by Section 301 levies should keep pressure on countries to strike bilateral deals favourable to the US, spanning trade, investment, industrial and technology concessions. While EU-US trade relations are more stable now than in early 2025, there remains a tail risk that a transatlantic security or sovereignty dispute (renewed American claims on Greenland, for instance) spills over into the economic relationship.

Ukraine diplomacy sees renewed momentum

- Trump will likely renew his push for a ceasefire in the Russia-Ukraine war. The G7 Leaders' 16 June commitment to boost support for Ukraine and increase pressure on Russia struck European capitals as an encouraging sign that the US was converging on their approach to the conflict. After meeting Volodymyr Zelensky the same day, Trump signalled he could reinstate Russian oil export sanctions to press Moscow towards a settlement. On 17 June, the US Treasury's most recent waiver of Russian oil sanctions expired. But US-Europe rhetorical alignment could reverse quickly – for instance if US-Russia-Ukraine negotiations resume in a format that largely excludes the EU/UK. While a ceasefire this year is far from guaranteed, any deal brokered by the Trump administration could offer Russia sanctions relief and access to frozen assets.



Asia policy balances deterrence with détente

- The US strategy for the Asia Pacific is crystallising around pragmatic balancing with China and more burden-sharing with allies. In a 30 May speech to the Shangri-la Dialogue, Defence Secretary Pete Hegseth reaffirmed the 2026 US National Defence Strategy's aim of building up forces to strengthen deterrence and denial along the First Island Chain and in the Taiwan Strait. But with resources stretched and stockpiles to rebuild post-Iran, Washington will likely pair this build-up with mounting demands that allies spend more on their own defence, using incentives such as favourable access to weapons programmes, intelligence sharing and industrial collaboration.
- We expect the administration to prioritise stabilising relations with Beijing in the lead-up to the Trump-Xi summit in Washington, D.C. on 24 September, where the 2025 Busan trade and critical minerals truce may be extended. Even so, a steadier US-China relationship will not undo deepening long-term competition on military, economic and technology issues. Strong bipartisan alignment in Congress on a more hawkish China policy should also keep pressure on the administration to limit any rapprochement with Beijing.

The Americas become a legacy play

- Extending the “Donroe Doctrine” in the Americas should offer Trump a legacy-building opportunity. The US stands to gain from a rightward shift in regional governments (now including Colombia and Peru), letting the administration expand its Shield of the Americas initiative to counter drug trafficking and curb Chinese involvement in critical infrastructure. US pressure on Cuba is mounting, which may lead to more brinksmanship or a partial negotiated solution with Havana in the next few months.
- We expect Trump to also push to widen American involvement in anti-cartel efforts in Mexico, which the Sheinbaum government will likely accommodate through already-close US-Mexico security cooperation. The US will likely also keep leading the rapid re-development of Venezuela's oil and minerals sector and collaborating with the Rodriguez government on anti-gang measures. The drive for a democratic transition in Caracas is expected to persist but may take longer to materialise as the administration focuses on near-term stability.

The impact of midterm elections

- The 3 November midterms will likely bring domestic politics to the fore and shape the Trump administration's international agenda. The thesis that the midterms will steady US policy is partly credible: Trump has a strong incentive not to shock equity and bond markets or stoke inflation ahead of the vote. But the midterms will cut both ways. Some foreign policy issues should benefit from more stability before November – namely US-China ties and the security situation in the Gulf. Yet an administration looking to demonstrate strength through quick wins may pursue other issues more aggressively. Investors should watch rising US pressure on Cuba and the political relationship with Europe as likelier sources of nearer-term movement.
- What happens after the midterms actually matters more. Recent polling and historical precedent – governing parties usually shed seats at midterms – point to Republicans losing control of at least the House of Representatives. Even so, this would not decisively restrict the administration's foreign policy in its final two years: the US executive holds broad foreign affairs and military powers. While a Democratic-controlled House may attempt to constrain



certain Trump initiatives through spending and oversight powers, the administration has already proved willing to work around congressional limitations.

- With little room to pass major domestic legislation, Trump is likely to become a “foreign policy president”, focused on cementing his legacy. This should include: embedding existing tariff and investment deals with key US partners; deepening US leadership in the Americas; creating the conditions for a pull-back of American resources from Europe (including brokering a Russia-Ukraine ceasefire); and broadening US efforts to reduce rare earths and critical minerals dependence on China, including through international coalitions such as the US-led Pax Silica initiative.



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Deutsche Bank
Research

Q2 2026 Global Markets Survey

June 2026

Jim Reid | Camilla Siazon

With special thanks to Anthony Chaimowitz from the dbDataInsights team



Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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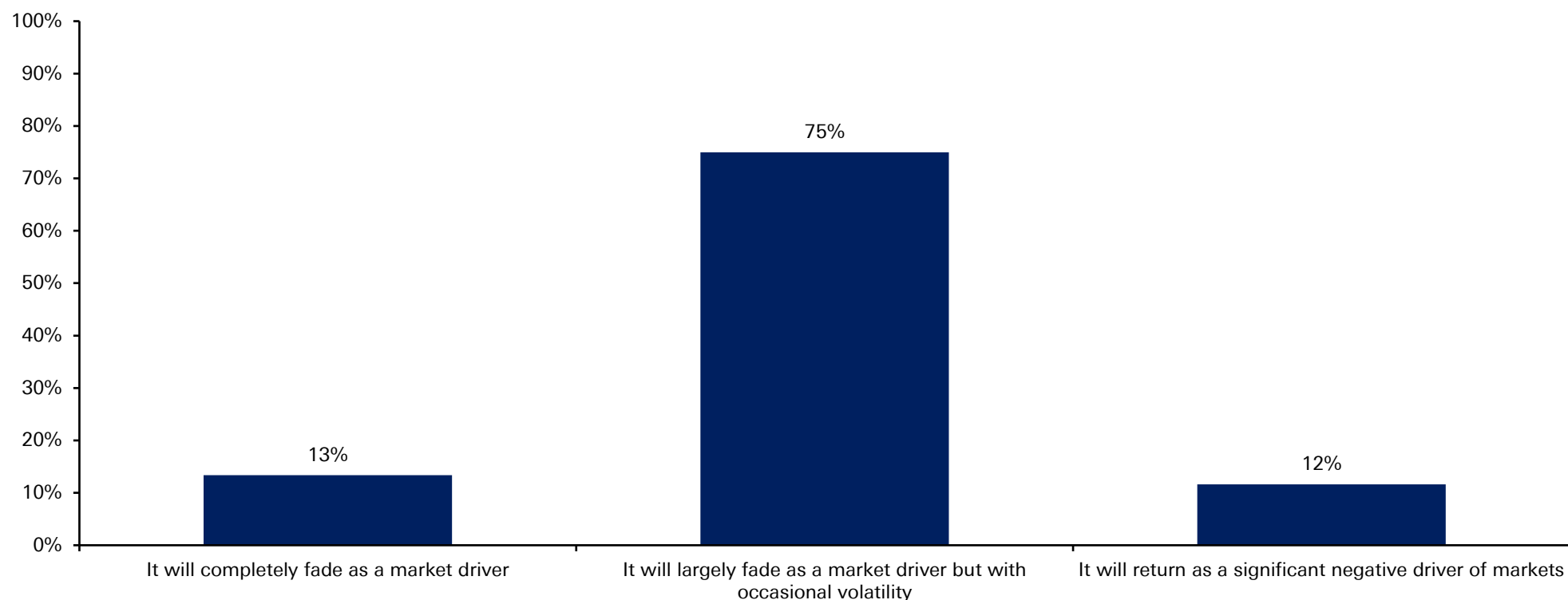


Our Q2 2026 global financial market survey had 275 responses from around the world, conducted between June 22nd and June 26th.

An overwhelming 75% believe the Iran conflict will now largely fade as a market driver into year-end, albeit with occasional volatility. With the 13% who think it will completely fade as a factor that makes 88% who are reasonably sanguine about events in the Middle East. Just 12% think it will return as a big negative driver.



When thinking about the Iran conflict, which of the following best reflects your view of its impact on financial markets for the rest of 2026?

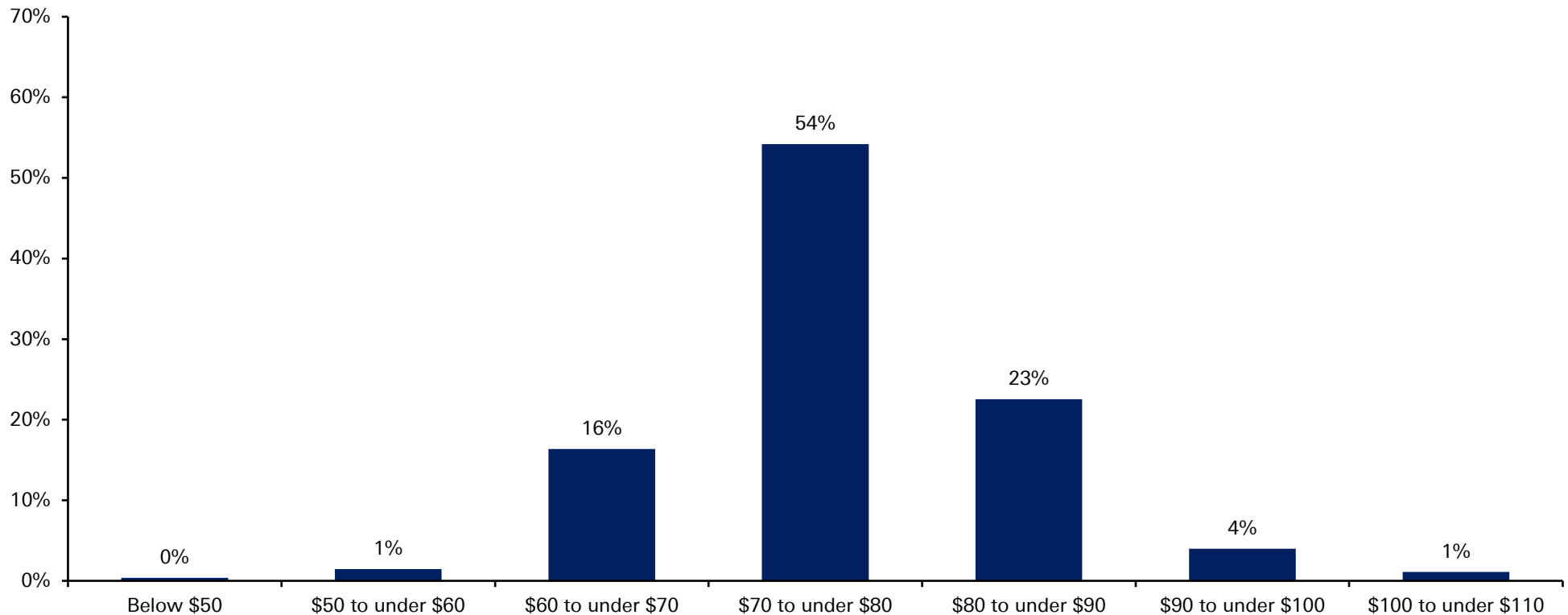


Source: dbDataInsights Survey, Deutsche Bank Research

The more sanguine view on Iran has also led investors to dial down expectations of higher oil prices. A majority of respondents now think Brent will stay between \$70-80/bbl for the remainder of the year. Compare that to April, when Brent peaked at ~\$118/bbl.



Brent crude oil prices started the year at around \$60, rose to approximately \$70 prior to the Iran conflict, peaked near \$120, and have since declined to around \$80. With this in mind, where do you expect Brent crude prices to average for the remainder of the year?

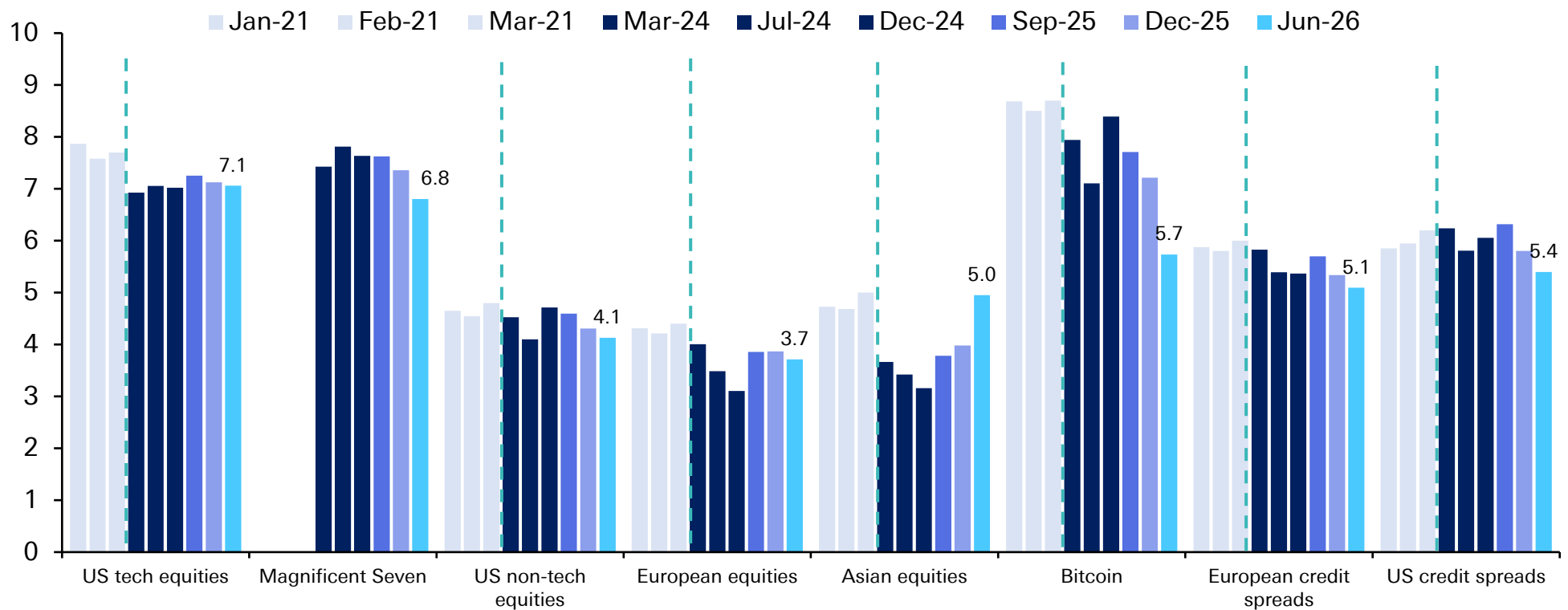


Source: dbDataInsights Survey, Deutsche Bank Research

In our regular bubble question that we first asked in 2021, and then after a gap, in most quarters since 2024, we've seen the lowest perceived bubble risk for the Mag-7. Both US tech and Mag-7 still score around 7 out of 10 on our bubble risk scale, though. Note that Bitcoin bubble risk has collapsed to 5.7 after the near halving since last October.



Looking at the following assets and using a scale of 0 to 10, where 0 means 'No Bubble' and 10 means 'Extreme Bubble', please tell us where you think these assets currently are.

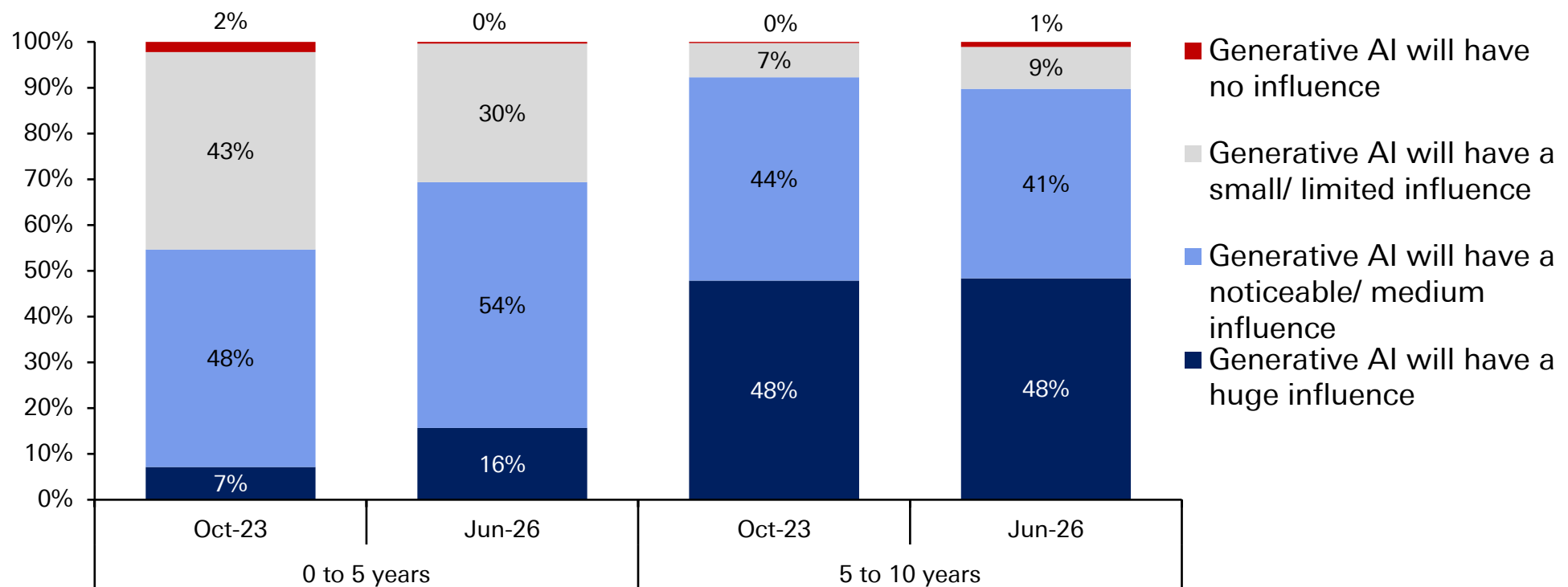


Source: dbDataInsights Survey, Deutsche Bank Research

We last asked about Gen AI's potential for productivity growth back in October 2023. Its expected near-term positive impact has increased since then – although some of that likely reflects the nearly 3 years of extra time since we last asked. Surprisingly, the anticipated 5- to 10-year impact has hardly changed, even if just under half still think it will have a huge impact on productivity.



What influence, if any, do you think future iterations of Generative AI will have on productivity growth within the following time frames?

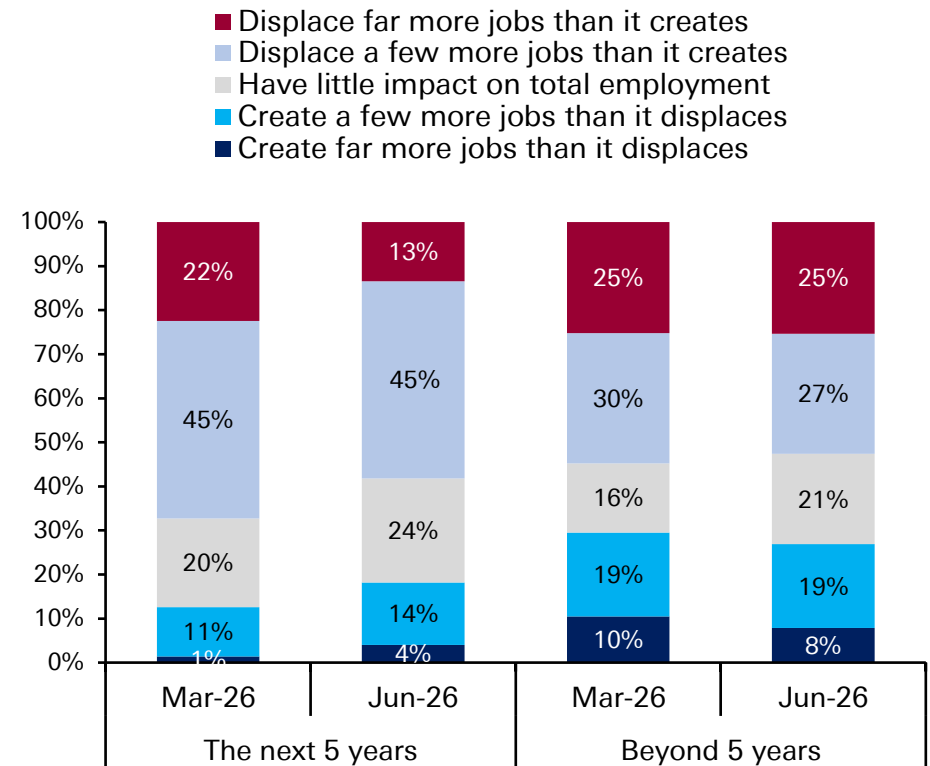
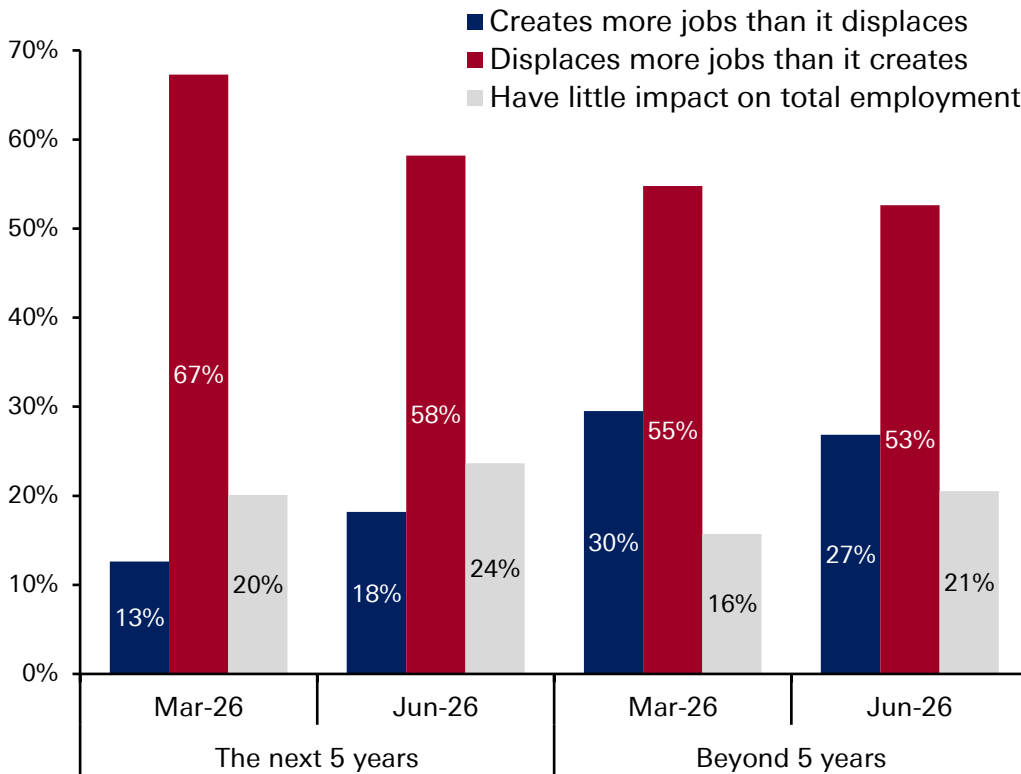


Source: dbDataInsights Survey, Deutsche Bank Research

Market sentiment on AI's impact on jobs remains pessimistic, but we've seen a small improvement since we asked in March. The first chart aggregates the data into binary positive or negative, and the second breaks it down as to how positive or negative.



For the following time horizons, which best describes your expectation for AI's impact on employment? Assume a scenario where AI adoption stays on its current trajectory.

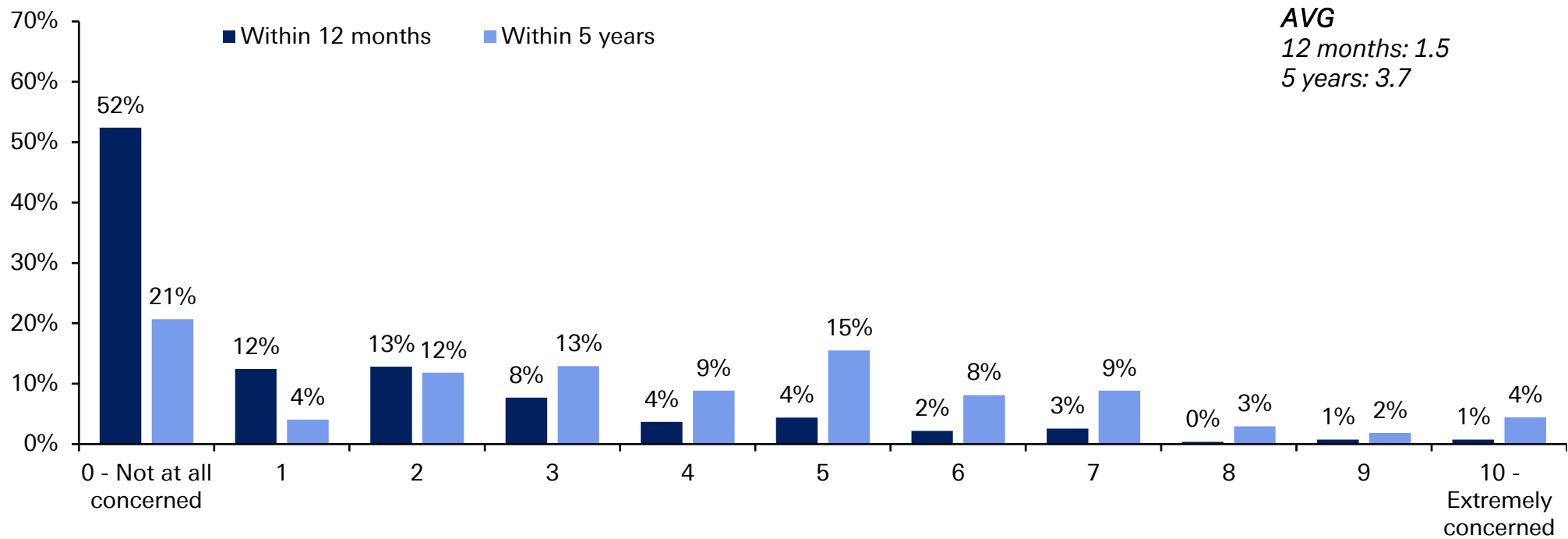


Source: dbDataInsights Survey, Deutsche Bank Research

52% say they are not concerned at all about their own jobs because of AI within the next 12 months. This drops to 21% within the next 5 years. The average scores (scale of 1-10) for these periods are 1.5 and 3.7, respectively. So, combining the last two charts suggests that people are more concerned about other people’s jobs than their own due to AI.



Using a scale of 0 to 10, where 0 is ‘Not at all concerned’ and 10 is ‘Extremely concerned’, how concerned are you that you will need to look for a new job within the following time frames because of AI?

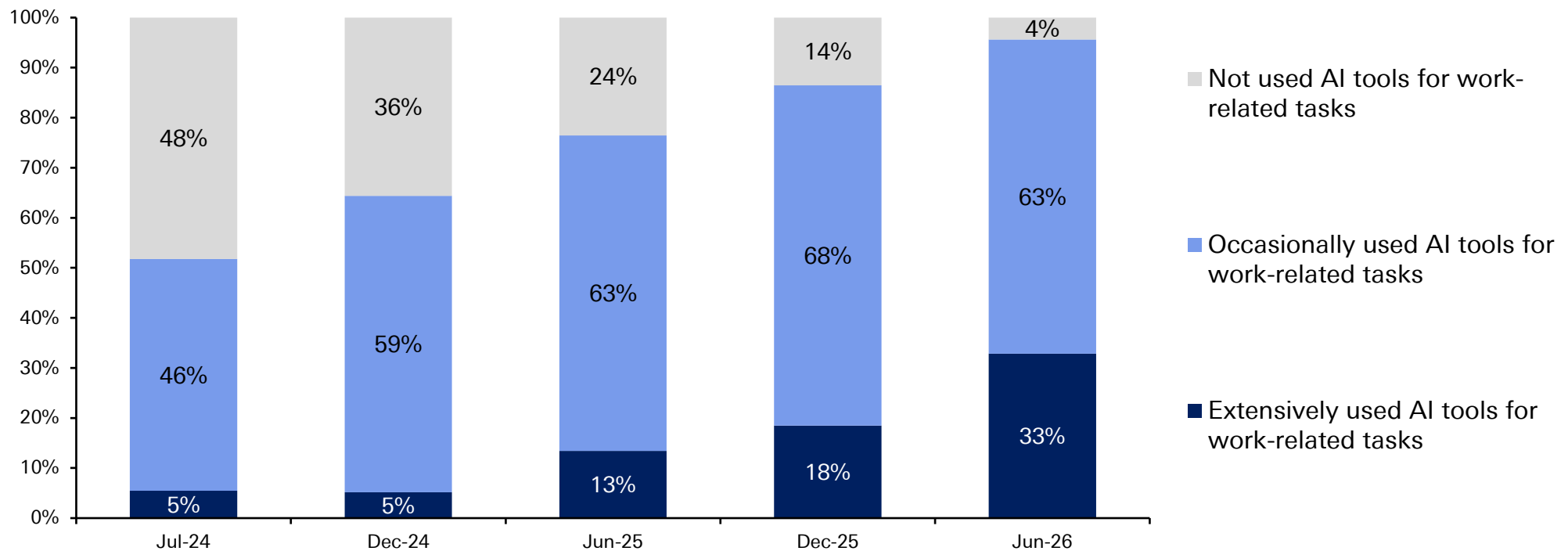


Source: dbDataInsights Survey, Deutsche Bank Research



Since March, AI adoption at work continues to accelerate as it has done steadily since we first asked 2 years ago. Now only 4% have not used AI for work, compared to 14% last December and 48% 2 years ago. “Extensively used” has also risen from 18% to 33% now since December – a pretty big increase.

Which of the following best describes your use of AI tools for work-related tasks within the last three months?

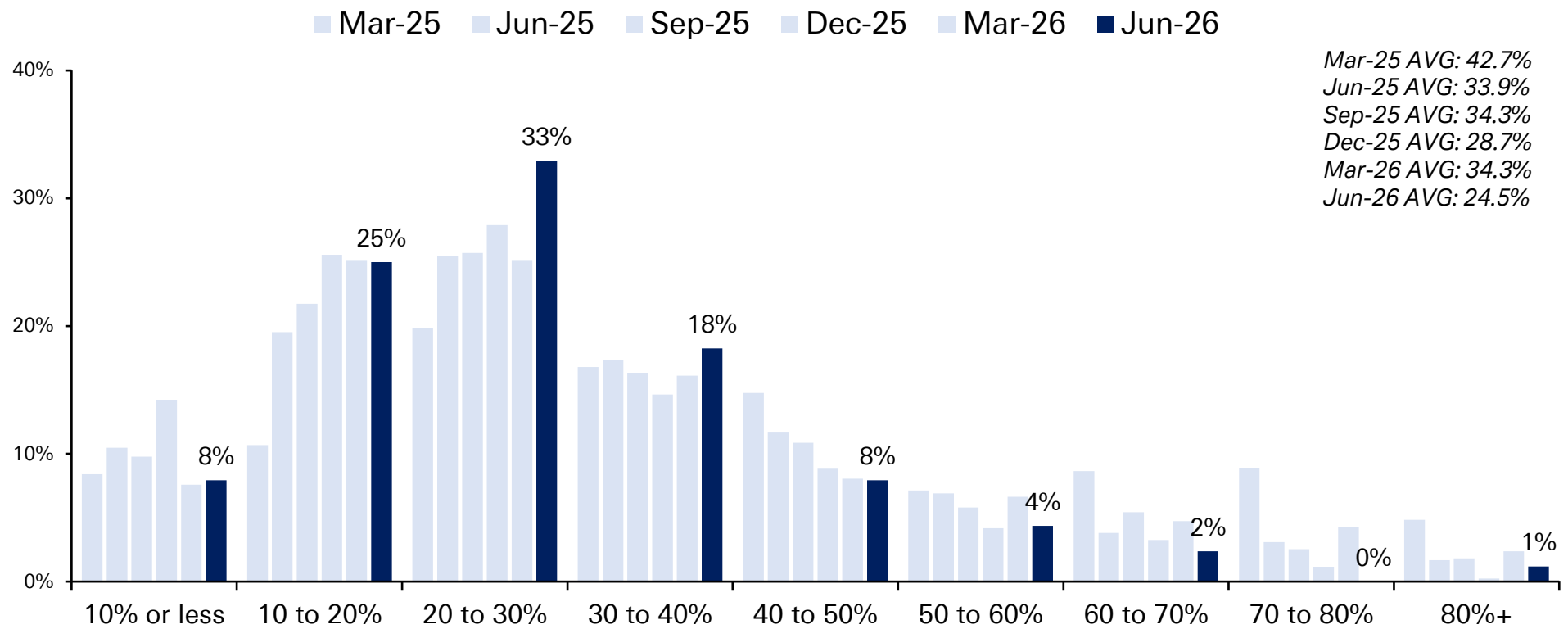


Source: dbDataInsights Survey, Deutsche Bank Research

Average perception of 12-month US recession risk is back down to 24.5%, its lowest score since we asked the question in this format last March. Back then it was 42.7% – and even this March, when the Iran War was in its early stages, it was 34.3%.



What do you think the probability of a US recession in the next 12 months is?

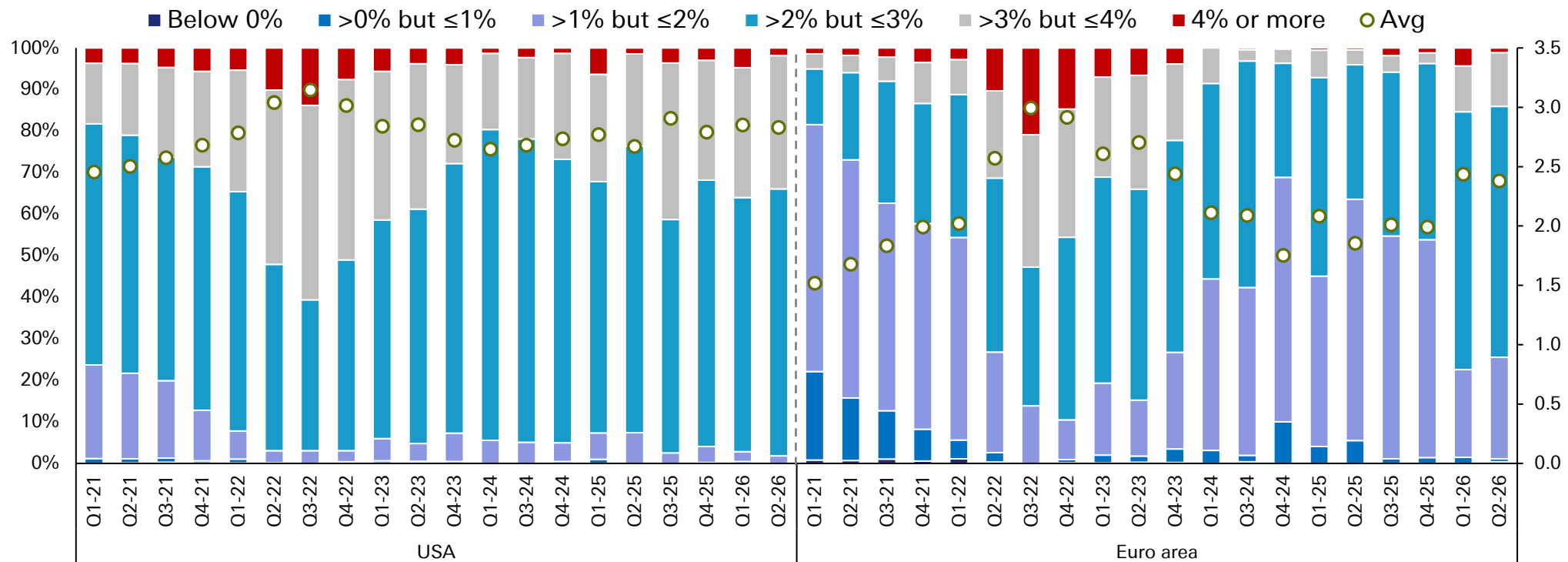


Source: dbDataInsights Survey, Deutsche Bank Research

Europe inflation expectations over the next 5 years are unchanged from March, at 2.4%. 5-year US inflation expectations have edged back down to 2.8% from 2.9% in March.



Looking at the below, what do you think inflation will average in the following regions over the next 5 years?

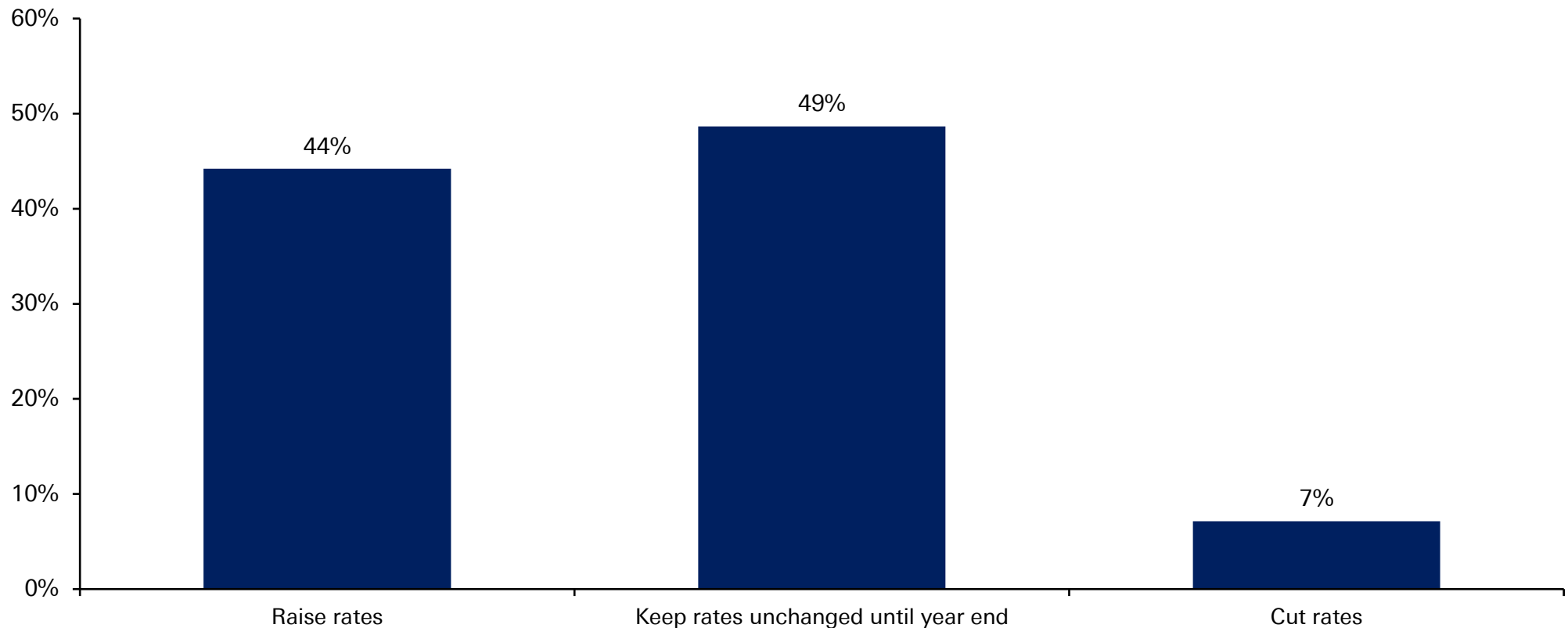


Source: dbDataInsights Survey, Deutsche Bank Research

Markets are pricing in almost one-and-a-half hikes by December. Yet perhaps the receding threat of an oil-driven shock has made our respondents lean slightly more dovish, with 49% thinking the Fed will stay on hold this year and 7% thinking cuts are still on the agenda. 44% think a hike is coming.



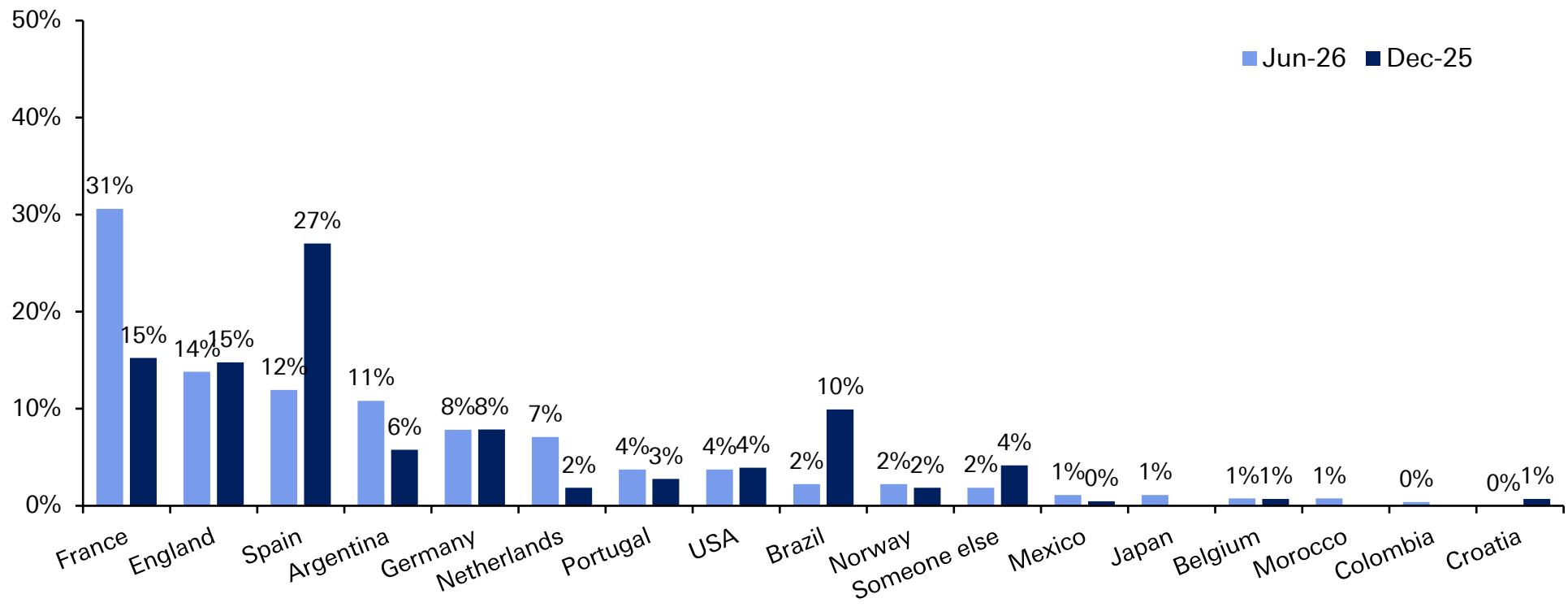
What do you think the Fed's next move will be in 2026?



And finally, a majority (31%) think France will win the World Cup this year, followed by England (14%). Spain and Brazil have seen the biggest shifts in opinion since December.



Who do you think will win the 2026 football World Cup?





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US at 250 – Why has the US been so successful and can it continue?

June 24, 2026

US at 250

To mark the 250 year anniversary of the US Declaration of Independence, this paper looks at how the US emerged as a global superpower, and why it's likely to remain one despite new challenges.

First, we consider how the US went from being a comparatively small country to the world's pre-eminent global power. These reasons range from the US' natural advantages, like favourable geography, to factors like its institutional stability and risk-tolerant capital markets.

We then consider the challenges that threaten US outperformance. China's rapid growth means the US faces its biggest rival since its own emergence as the world's dominant power. China's rise also coincides with several other issues: the rules-based international system that the US helped create is under immense strain; the reserve currency status of the US Dollar is under pressure; and the country's public debt-to-GDP ratio is set to hit new records in the coming years.

Finally, we look at why the US is likely to sustain its outperformance, thanks to a collection of reinforcing advantages. In fact, the US has consistently emerged from challenging periods successfully, be that after the Great Depression, the malaise of the 1970s, and again around the GFC and its aftermath. We conclude with a few lessons for investors and for the rest of the world.

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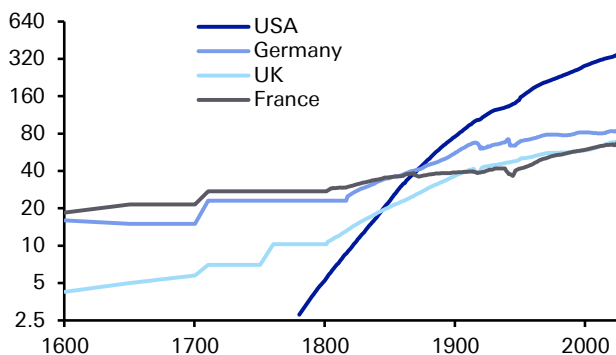
Jim Reid
Global Head of Macro and Thematic Research



1. The Lessons from History: What underpins America's Success?

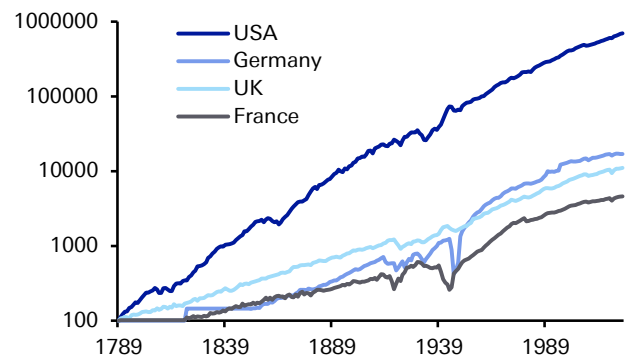
Before delving into the drivers of US success, let us bring with a few highlights of its historical outperformance. Since the United States' founding, the country has achieved remarkable economic success on a whole range of metrics. For one, the US saw rapid growth in population. It started as a comparatively small entity at its founding, but it quickly surged to overtake the large European countries in the mid-19th century and has continued to outpace them since. And with more people, it was little surprise that US GDP rapidly outpaced other countries too.

Figure 1: Population on a log scale (millions)



Source : Finaeon, Deutsche Bank

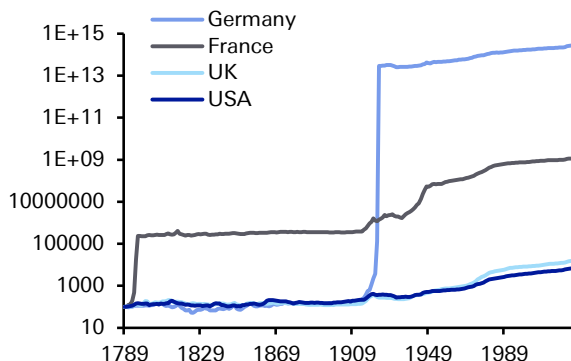
Figure 2: Real GDP on a log scale (1789 = 100)



Source : Finaeon, Deutsche Bank

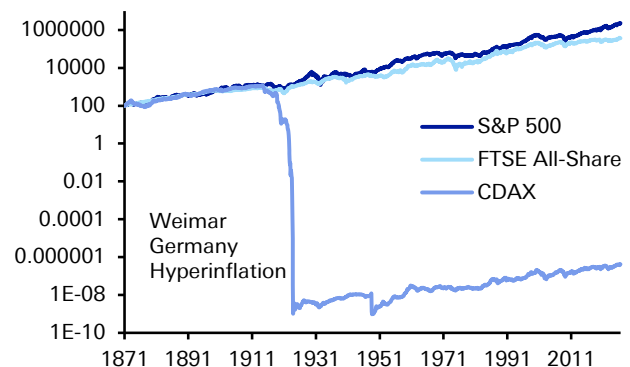
This success extends beyond GDP growth. Unlike France or Germany, the US has never had a bout of hyperinflation in its history, with its currency maintaining its value comparatively well. Meanwhile, the US' long-term equity performance has also been very strong. In the time since equity data is available from the late-19th century, real returns have outpaced the UK and Germany by substantial margins.

Figure 3: Consumer Price Index on a log scale (1789 = 100)



Source : Finaeon, Deutsche Bank

Figure 4: Real equity returns on a log scale (Jan 1871 = 100)



Source : Finaeon, Deutsche Bank



So what has caused this incredible success?

The first reason, and one of the most critical historically, has been the US' political and institutional stability. Clearly, there have been moments of intense turmoil, most notably with the Civil War in the 1860s. Its historical path also should not be over-romanticised, with US territorial expansion during the 19th century having many similarities to that of the Old World colonial empires. Nonetheless, the US is incredibly rare in that its political system is recognisably the same over the last 200 years. This relative stability and early development of property rights provided a fertile environment for long-term investments, which in turn has aided economic growth over the centuries.

The second reason is its geographic advantages. The US possesses vast arable land, navigable rivers, large coastlines, and access to two oceans. It therefore found itself more insulated from the destructive effects of the world wars, with productive capacity not impacted in the same way. Moreover, the country borders Mexico and Canada, who in both population and GDP terms are much smaller than the United States, meaning it didn't face the security risks that many European powers faced over the 19th and early-20th centuries.

The third reason has been its abundance of energy resources, particularly relative to Europe. In large part a corollary of its geographic strengths, this energy abundance has given the US several advantages. First of all, lowering costs for households and industry, which has helped the economy be more resilient against geopolitical shocks. Moreover, with the US becoming a net energy exporter in recent years, it also strengthens the external position.

The fourth reason for the US' relative success was that its main competitors in Europe were deeply affected by the world wars and wider political turmoil. Their productive and financial capacity was severely degraded, with the destruction causing huge loss of life as well. Even among those who survived, many scientists, engineers and entrepreneurs left for the US in the first half of the 20th century. Whilst not a US "success" as such, this meant that on a relative basis, the US' divergence widened considerably. Indeed, in the first half of the 20th century, the US economy grew by 5.7 times, Germany by 3.4 times, the UK by 2.0 times, and France by 1.8 times.

The fifth reason for the US' outperformance is scale. It has a large domestic market of over 300m people, with high average incomes, a common language, and low internal barriers to trade. This has given firms the ability to reach a large-scale domestically before expanding abroad. Indeed, it is notable that 8 of the world's 10 largest firms are based in the US, whereas none are in Europe.

The sixth reason is the structural advantage of the US Dollar, which remains dominant in global trade and FX reserves. This dollar demand matters because it lowers borrowing costs and raises demand for US Treasuries. In turn, it leaves the US with an exceptional capacity to run bigger fiscal and external deficits without facing a funding crisis, and expands the geopolitical leverage that the US possesses. This has been dubbed the "exorbitant privilege", and has been a major advantage in recent decades.

The seventh reason is financial depth. The US has a large banking system, but also a wide range of non-bank financing, which means startups have access to other sources of capital. In fact, in the decade from 2013 to 2023, annual venture capital



financing was 0.7% of GDP in the US, compared to just 0.2% in the EU. This financial depth is important because innovative firms can often be loss-making for lengthy periods, so countries with bigger pools of patient risk capital are better placed to commercialise new technologies.

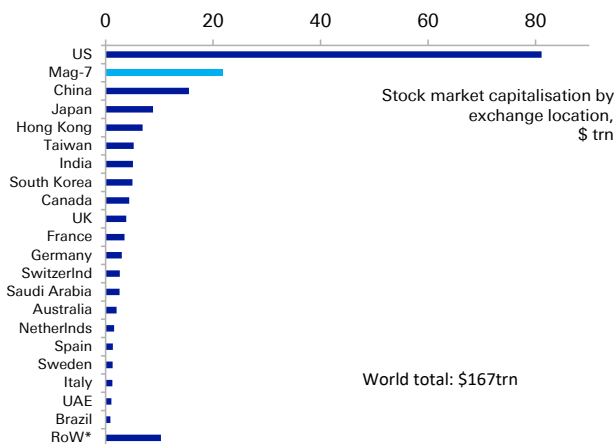
The eighth reason is its self-compounding advantages in education and research. The US has many of the world's strongest research universities, including 7 of the top 10 in the Times Higher Education World University Rankings for 2026. Moreover, this research also has an economic angle, as scientific discoveries feed into startups, workforce training and industrial scale-ups. In turn, this becomes self-compounding, because top global talent is attracted to the US, and ensures it remains at the forefront of new sectors such as artificial intelligence.

The ninth reason is its pro-business architecture, including a greater tolerance of business failure than many European systems. For instance, Chapter 11 reorganisation is designed to preserve and restructure viable firms rather than liquidate them. This more positive approach to business failure ties into the empirical literature, which suggests that more debtor-friendly and efficient insolvency frameworks are associated with greater entrepreneurship and innovation.

The tenth reason is adaptability. The cultural acceptance of failure and capacity to reinvent itself have boosted US ability to adapt to the changing world. This has allowed it to navigate repeat boom-and-bust cycles, as well swings of the policy pendulum – between openness and isolationism, between protectionism and free trade – without threatening overall the institutional stability we highlighted above. And while capitalism has been a key underlying driving force, it is pragmatism rather than ideology that have driven continued success over time.

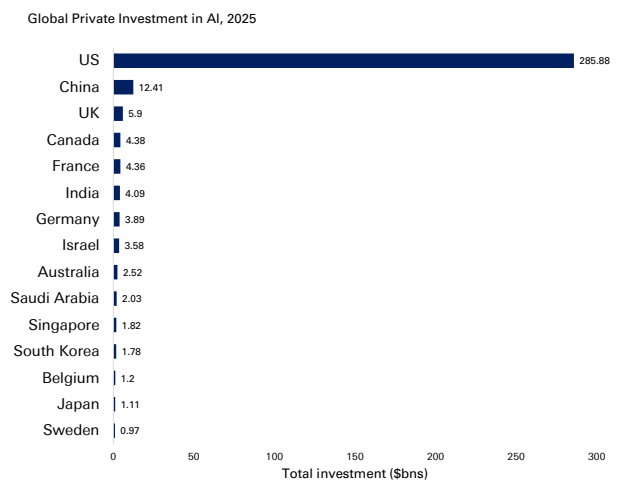
It is also important to note that these advantages do not play out individually, but are mutually reinforcing, with the interaction between them allowing the US to benefit from network and externality effects that few if any countries can match.

Figure 5: The US accounts for nearly half of the world's public stock market capitalisation



Source : Bloomberg Finance LP, Deutsche Bank Research; Data as of 22 June 2026
* Rest of World comprises of a further 62 country exchanges.

Figure 6: US private AI investment reached \$285bn in 2025, more 20x as much as in China



Source : Stanford University 2026 Artificial Intelligence Index Report, Deutsche Bank Research



US history has been marked by challenges and recoveries

That said, this success story was far from a straight line. **US outperformance has frequently been questioned, yet it has repeatedly defied the sceptics, including at numerous points in the last century.**

Right from the country's founding, questions were raised about its long-term potential, and in the Civil War of the 1860s, the nation's survival was at stake after less than a century.

Even over the past 100 years, which has ostensibly been a period of US pre-eminence, there have frequently been doubts. For example, after the Wall Street Crash of 1929, the **Great Depression** saw unemployment peak above 25% and remain above 10% for an entire decade. The associated stock market collapse saw the S&P 500 fall by -86% from peak-to-trough, not reaching its 1929 peak again until 1954. This was a global shock, but it undermined faith in the US system of free-market capitalism.

However, the country eventually pulled out of the slump. That started with Franklin Roosevelt's New Deal program, which helped to restore confidence and economic activity. Then the start of WWII saw economic activity surge even further, particularly as there wasn't fighting on US soil. By the end of WWII, the US had never been in a stronger position relative to its peers, not least with Europe having to rebuild after the war.

Similar doubts were clear as the US grappled with the various **crises of the 1960s and 70s**. This period saw major political turmoil and multiple assassinations, including President Kennedy in 1963, his brother and presidential candidate Robert Kennedy in 1968, and civil rights leader Martin Luther King Jr. in 1968. A few years later, Richard Nixon became the first president to resign from office amidst the Watergate scandal. US military power was also facing questions at this time, as the Vietnam War turned into a protracted conflict that also faced substantial domestic opposition.

Then on the economic front, inflation started to gather pace from the late-1960s, surging further after the first oil shock of 1973, leading to a 16-month recession. Measures were imposed to conserve energy, including the National Maximum Speed Limit, which wasn't repealed until 1995. Then in 1979, a second oil shock drove inflation up again.

This turmoil raised questions about whether policymakers could effectively tackle the crises of the day. In 1979, President Jimmy Carter delivered what was widely referred to as the "malaise" speech, although it was actually called "A crisis of confidence". He said there was "growing doubt about the meaning of our own lives and in the loss of a unity of purpose for our nation." He spoke of a "growing disrespect" for institutions, and said how the public often "see paralysis and stagnation and drift." Even the US president was acknowledging that the country was facing huge issues.

Nevertheless, the US then recovered strongly into the 1980s and 90s. Inflation was tamed, albeit with drastic monetary tightening, and the economy saw rapid growth once it recovered from the early 1980s recession. In 1991, the dissolution of the Soviet Union marked an end to the Cold War, and the 1990s were widely considered a unipolar moment where the US was left as the unmatched global superpower. So once again, the US came through a period of doubt over its success.

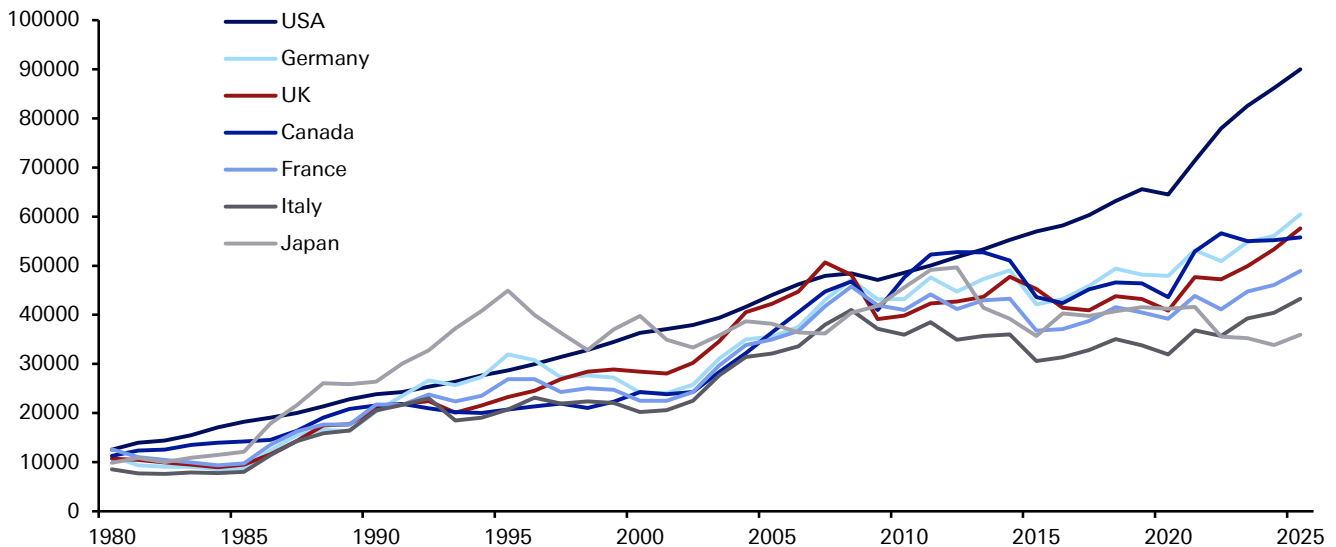


But the unipolar moment proved brief, as the **global financial crisis** cast fresh doubt over US prestige. Unemployment surged above 10%, and the economy entered an 18-month recession, the longest since the Great Depression. Moreover, the subsequent recovery was very sluggish, with growth rates well below those of previous decades. Economists resurrected the idea of “secular stagnation”, originally put forward by Alvin Hansen in the 1930s, suggesting the US faced an era of permanently slower growth.

These questions about US pre-eminence came as China was growing rapidly at the same time. In 2005, China’s GDP (in USD terms) was still only 18% of the US total, but that surged to 62% by 2015. That was an incredible shift in relative economic weight in the space of a decade.

But for all the doubts of the 2010s, the secular stagnation narrative has since disappeared in the 2020s. The US economy bounced back remarkably quickly from the pandemic, aided by huge quantities of stimulus that were far larger than 2008. Moreover, recent years have seen the US at the forefront of AI developments, raising hopes of a new productivity surge, with its lead extending over the other advanced economies. So yet again today, the US has emerged from a rockier period around the financial crisis and its aftermath, into a relatively strong position that includes the highest GDP per capita of any G7 country.

Figure 7: GDP per capita (USD terms)



Source : Haver Analytics, Deutsche Bank

2. The Challenges Ahead: Why the US now faces a crucial decade

The above discussion highlights how the US has repeatedly overcome periods of doubt, with predictions of American decline proving premature each time. However, in several respects today’s challenges feel more acute. The US is not as dominant a geopolitical power as it used to be, given China’s emergence as a serious rival. Moreover, its debt-to-GDP ratio is on the verge of new records, meaning its fiscal space is far more constrained to deal with new challenges. As such, the next decade will offer a further test of US adaptability, and whether the political system can respond to these interacting pressures.



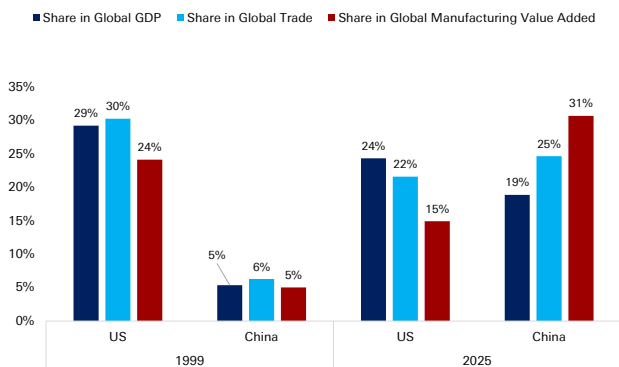
What are the biggest near-term challenges for the United States?

Since the end of the Cold War, the most significant adjustment has been the end of the unipolar moment and China's emergence as a genuine peer competitor, which is different in scale and capability from any rival the US has faced since the late-19th century. China has already overtaken the US on manufacturing output, merchandise trade, and GDP (on a PPP basis), and is rapidly closing the gap in advanced technologies and semiconductors. The US retains decisive leads in nominal GDP, capital markets depth, the dollar system, and frontier innovation, but the gap has narrowed substantially.

The military balance tells a similar story. US defence spending remains roughly three times China's at market exchange rates, but the PPP-adjusted gap is far smaller. More broadly, the gap between US military spending and other major economies has shrunk over the past 15 years. That comes as the proliferation of low-cost precision and autonomous systems is eroding the historic advantages of expeditionary power projection, raising the cost of underwriting the global security order the US has provided since 1945.

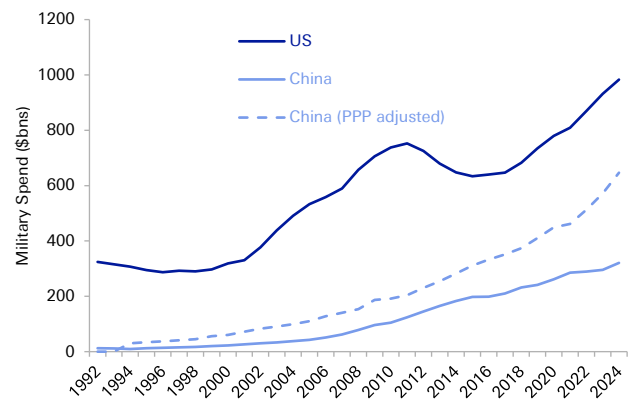
We can also see the emergence of Chinese leadership in a whole range of areas. Among others, China accounts for a third of all global manufacturing activity, with a sector as large as the US, Japan, Germany and South Korea combined. The cost of electricity there is a fraction of what people pay in Europe or the US. And just as the US dollar's role has come under pressure, China is strengthening the global influence of the RMB, in part because of its vast domestic savings surplus.

Figure 8: China has overtaken the US as the world's largest trade and manufacturing hub, though the US is still ahead on nominal GDP



Source : Deutsche Bank Research, UN Industrial Development Organization, Haver. All in Constant 2020 USD, Data for 2025

Figure 9: US lead on military spending has narrowed over the past 15 years



Source : World Development Indicators, World Bank Data, Deutsche Bank Research

The emergence of China comes as the rules-based international system designed and led by the US for eight decades is under strain from multiple directions. Admittedly, these strains have not just emerged. Even before the Trump administration, emerging markets had grown in relative size and become less willing to accept Western-designed institutions. Meanwhile, the political support for globalisation has frayed across advanced economies.

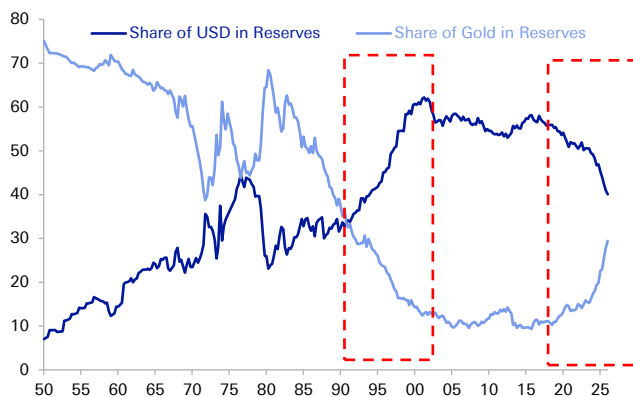


But recent US policy choices have raised profound questions about the future global order. In 2025, the US effective tariff rate rose to its highest since the 1930s, representing the largest peacetime trade barrier increase in a century. Strategically, the US alliance network has been an enormous economic asset, underwriting dollar dominance, securing trade routes, and providing the soft-power foundation for global rule-setting. Yet US membership of NATO has become an openly debated question, and tariffs have been raised on US allies as well as rivals.

This uncertainty has contributed to growing pressure on the US' status as the global reserve currency. The dollar's share of global reserves has fallen from roughly 72% to 58% over two decades — a gradual decline rather than a collapse, but a clear trend nevertheless. The reduction has accrued to a basket of "non-traditional" reserves and to gold, which has seen the largest central bank buying programme in over half a century. In addition to the overall transition away from a 1990s unipolar world, specific factors have added to the challenges that dollar dominance faces. This includes the weaponisation of the dollar through sanctions — notably the freezing of Russian central bank reserves in 2022, which gave non-Western-aligned countries a strategic incentive to develop alternatives even at material cost — as well as the structural [decline of the petrodollar arrangement](#) as energy trade has fragmented and the US has become a net oil exporter.

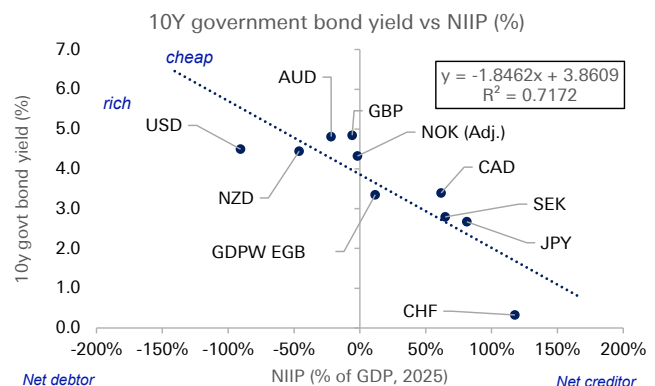
Irrespective of the cause, were the US "exorbitant privilege" to erode further, this would bring major costs to the US and allow other currencies to benefit (see also the DBRI note last year on the [potential benefits for the euro](#)). As mentioned earlier, the structural subsidy that the US receives from its reserve currency status includes lower borrowing costs, the ability to run persistent external deficits without the discipline imposed on other economies and the advantages of the dollar being the unit of account for global trade, commodities and cross-border lending.

Figure 10: The "end of history" in the 1990s saw a sharp rise in the USD and decline in gold's share of reserves



Source : Deutsche Bank, Bloomberg Finance LP, Eichengreen, Chitu and Mehl (2014), IMF International Liquidity.

Figure 11: 10yr yields vs Net international investment Position – An illustration of US "exorbitant privilege"



Source : Deutsche Bank, Haver Analytics, Bloomberg Finance LP, IMF Updated 22 June 2026

None of this implies imminent displacement. No alternative is remotely ready to assume the dollar's role. Network effects and inertia in trade invoicing and central bank balance sheets all favour continued dollar dominance. Indeed, the pound sterling held a prominent role for many decades after the US had emerged as the world's largest economy. So the realistic risk for the next decade is not collapse but gradual erosion, with a slow loss of the privilege margin that has boosted US economic performance.

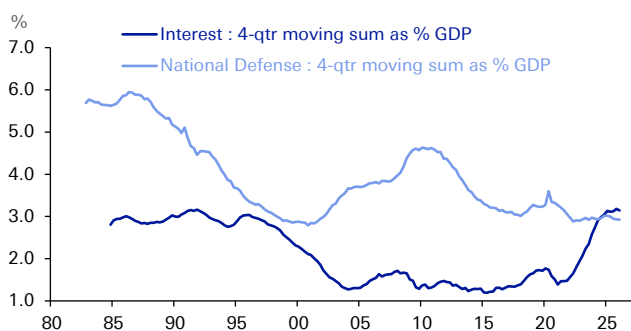


The US fiscal trajectory is the most plausible catalyst to accelerate that erosion, and for institutional investors is the single most concrete macroeconomic risk facing the United States. Over the last four years since 2022, the federal deficit has been consistently running at around 5-6% of GDP. Those are the highest peacetime deficits in US history outside of a major recession, and this is happening in a full-employment economy. Debt held by the public is set to surpass 100% of GDP this year, and the CBO projections point to an unsustainable trajectory. Interest payments on the federal debt now exceed defence spending and are the fastest-growing line item in the budget.

Most significantly, the binding constraints are now set to arrive within the next decade. The latest [estimates](#) project that the Social Security trust fund will be depleted by late 2032, which would trigger an automatic benefit cut absent legislative action. The Medicare hospital insurance trust fund faces a similar reckoning shortly after. So while the unsustainability of the US public debt trajectory has been in the headlines for many years, these events are now more imminent – ones that the next US administration that takes office after the 2028 election will have to face.

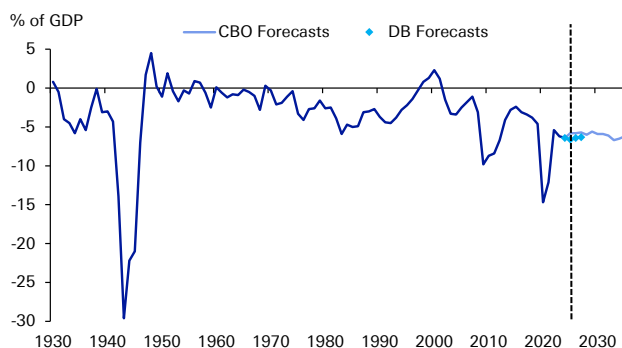
To be fair, the fiscal challenge is far from arithmetically catastrophic – a combination of AI-driven productivity growth and moderate fiscal consolidation could lead to significant improvements in the long-term debt profile. The difficulty is whether the political system can deliver such an outcome. Past fiscal consolidations required bipartisan compromise. Yet the current political environment offers little prospect of similar cooperation, with polarisation a mounting issue. Attempts in recent decades (like the Simpson-Bowles Commission in 2010) have failed to gain sufficient support.

Figure 12: The Crossover: Rising Federal Interest Payments have Overtaken Defense Spending



Source : CBO, OMB, Haver Analytics, Deutsche Bank Research

Figure 13: A Century of Deficits: US Federal Budget Balance from the Great Depression to the 2030s



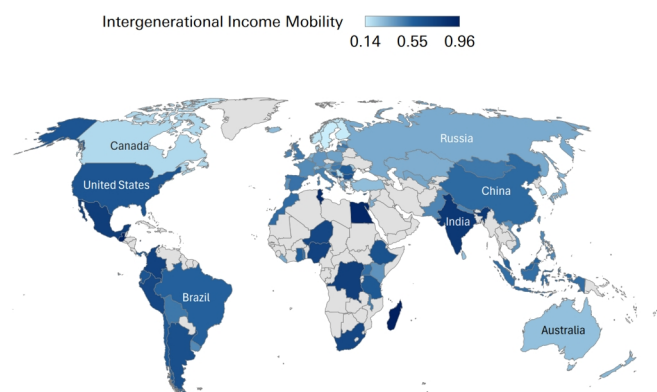
Source : CBO, OMB, Haver Analytics, Deutsche Bank Research

In the meantime, demographic trends are compounding both the fiscal and innovation challenges. The US fertility rate has fallen to 1.6 and the dependency ratio is set to deteriorate sharply as the baby boomer cohort fully enters retirement. Even though the population is comparatively younger than many other developed countries, it is still ageing. For example, the UN projects that the share of the US population aged 65 and over will rise from 19% in 2026 to 22% by 2036, rising further to 28% by the end of the century.



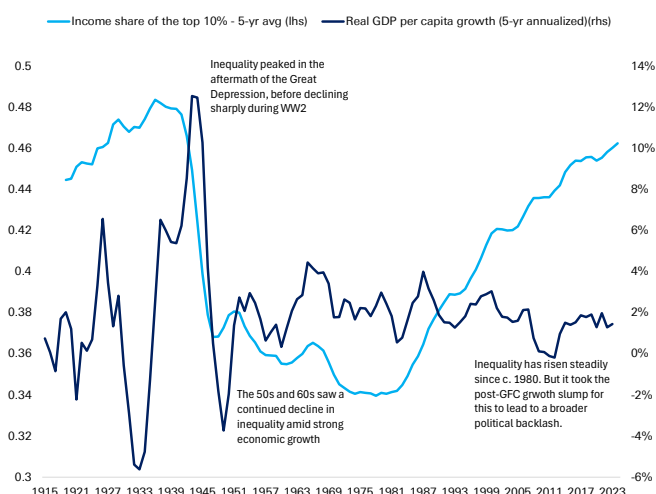
An important connecting driver across fiscal, political and demographic challenges has been a rise in inequality. Its steady rise since the 1980s has gone hand-in-hand with a rise in political polarisation, especially once combined with the economic shock that followed the GFC and the sluggish recovery in its aftermath. A corollary of this is that the traditional “American dream” of social mobility has found itself under mounting pressure. The US now has lower intergenerational income mobility than almost all DM economies, which risks corroding the social contract on which open markets and creative destruction depend. It has also led to negative social effects, notably a widening of the life expectancy gap between Europe and the US since the 1980s.

Figure 14: The US has lower intergenerational mobility than most DM economies (lower score = higher mobility)



Source : World Bank Data, Deutsche Bank Research

Figure 15: The rise in inequality since 1980 added to the political challenges facing the US, especially after growth slowed post-GFC



Source : Finaeon, World Inequality Report, Deutsche Bank Research

3. Why the US Should Remain a Long-Term Success Story

The challenges facing the US are real, but the weight of evidence still suggests it will remain the world's leading economy for the foreseeable future. Its collective structural advantages remain difficult to replicate, and it retains the same adaptability that's helped it recover from previous difficulties.

The starting point is that the US should remain the world's largest economy well beyond the critical decade coming up. A global demographic slowdown means that previous assumptions of rapid population and economic growth in emerging markets no longer look inevitable. Conversely, US economic growth has generally surprised on the upside since the pandemic, with the “secular stagnation” narrative of the 2010s quietly fading. Indeed, the US economy has successfully weathered multiple shocks in recent years, ranging from rapid Fed rate hikes, to tariffs, to the Iran conflict. Moreover, there is further upside potential to productivity growth (and hence economic growth) from the US' significant lead on AI.

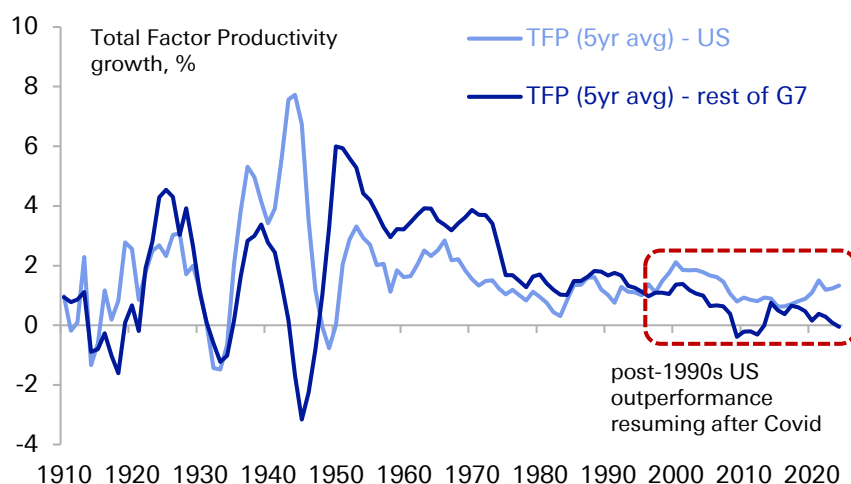


This AI boom is the latest expression of the US capacity for reinvention that stretches back through its history: a willingness to embrace disruptive, capital-intensive technology and absorb the dislocation it brings. AI is one of many new technologies that have often led to economic volatility (and boom-bust cycles on occasion), but whose benefits have delivered broad, economy-wide productivity gains over time. That was clear historically with canals, railroads, and electrification, and also with more recent innovations like the internet.

Thus, the US is well-positioned to lead the AI cycle for several mutually reinforcing reasons: an innovation-friendly ecosystem of research universities and venture capital; a cultural and institutional acceptance of the boom-bust investment cycles that frontier technology requires; the deepest capital markets in the world to fund it; and abundant domestic energy. The shale revolution has transformed the US into a net energy exporter, and cheap, plentiful power is now a decisive input advantage in the race to build and run data centres. Few competitors can match this combination.

The AI boost comes at a time when the US has already re-established its productivity outperformance in recent years. Since the pandemic, US productivity growth has risen above its post-1970 average, which in turn is boosting GDP and the revenue base. The potential fiscal arithmetic is powerful. According to CBO estimates, if productivity growth were just half a point above their baseline in the coming decade, then debt held by the public would only rise to 110% rather than 120% by 2036 – see our colleagues' note on the [US debt outlook](#) last autumn for more. However, productivity gains alone are unlikely to resolve the fiscal issue, in part because the political incentive is to spend the fiscal dividend rather than save it and in part given the uncertainty of how the gains would be distributed — a theme we return to below.

Figure 16: US has resumed its productivity outperformance in recent years

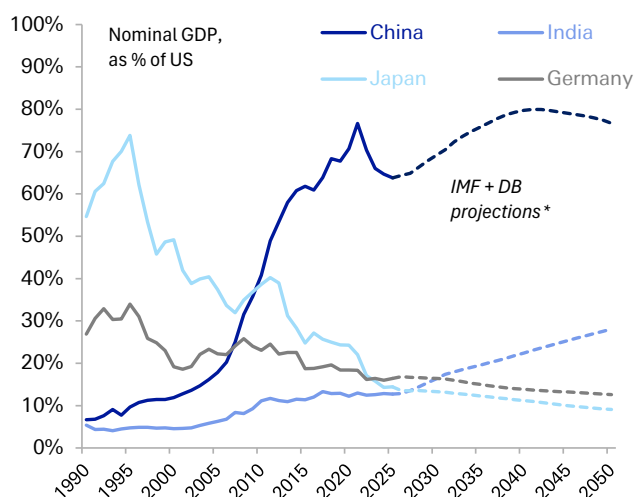


Source : Deutsche Bank Research, www.longtermproductivity.com
Rest of G7 is the simple average of Canada, France, Germany, Italy, Japan and the UK.



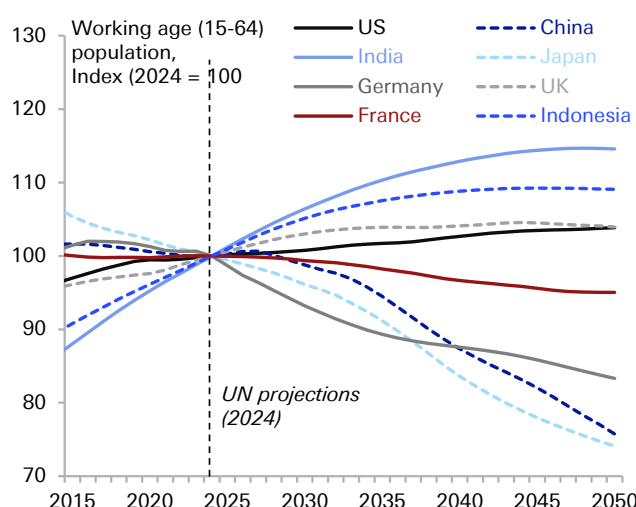
On demographics, the US also enjoys a clear advantage over its principal rivals. China's working-age population will begin a pronounced contraction in the 2030s, with Europe in aggregate also seeing a decline, whereas the US will see moderate growth under a normalised migration trend¹. With a growing working-age population, the US would avoid this drag from a shrinking labour force. Moreover, the US has had a consistent capacity to absorb immigration at levels few other developed economies have managed. And with the demographic slowdown now reaching emerging markets, few major economies now have clearly better long-term demographic potential than the US. For instance, while India will maintain positive demographic momentum over the next couple of decades, with its fertility rate now below replacement rate, its advantage will shrink by the middle of the century.

Figure 17: US likely to remain the world's largest economy for the foreseeable future



Source : IMF World Economic Outlook, Deutsche Bank Research
* IMF projections to 2031; DB estimate to 2050 based on (i) UN population projections (2024), (ii) stable productivity growth among DM economies, (iii) gradually slowing labour productivity growth for China (to 2.0% by 2050) and India (3.5% by 2050), (iv) stable REER exchange rates.

Figure 18: The US has better demographics than most DM countries and China, while India's advantage is set to peter out over time



Source : UN Population Database (2024), Deutsche Bank Research
Recent research suggests the 2024 UN demographic projections are likely too optimistic (e.g. www.mercatus.org/macro-musings/jesus-fernandez-villaverde-quandary-global-demographic-decline). However, this downside is largely global and unlikely to see relative US underperformance under normalised migrations trends.

Even on the financial side, whilst the dollar's role has come under pressure, it still retains a dominant and unrivalled position. The network effects that sustain it are exceptionally strong, and there is no obvious successor in the wings, unlike when the dollar overtook sterling in the interwar period. The depth and liquidity of US capital markets, inseparable from the dollar's role, are themselves a structural draw that even US rivals rely upon. Indeed, it is notable that the asset seeing a major increase in official reserves is gold, rather than another currency. And while new payment technologies could bring new risks to dollar dominance, US dominance in the [stablecoin space](#) will help limit these risks for the foreseeable future.

1 The UN population projections assume net migration into the US of c. 1.2m annually in the coming years. This would be roughly in line with the average migration rate since 1960 (0.3pp of population annually), while being well below the c.2.5m peak reached in 2023 but well above the c.300k latest [Census projection](#) for 2026.



The Conditions for Continued Success: Pitfalls to Avoid

Yet much as there are credible reasons for US outperformance to continue, this cannot be taken for granted.

The first key challenge is to ensure that the gains from future technological growth are broad-based, as excessive concentration risks adding to political and institutional dysfunction. We know from history that technological progress does not automatically deliver broadly shared prosperity. For instance, during the early Industrial Revolution, aggregate output rose for decades as wages for ordinary workers stagnated, a period that has been dubbed “Engels’ pause”. This offers a cautionary lesson on AI, and with the [labour share of income](#) in the US already near historic lows today, a further deterioration risks intensifying the social and political backlash building against the technology. It’s important to avoid that, as a backlash against technological advances would in turn negate its gains. A related risk is that if AI gains accrue to capital through profit margins while labour is shed, this would negate the fiscal benefits from higher productivity given the focus of the US tax system on labour taxation.

This brings us to the second challenge – managing the fiscal adjustment and the related timing risk. The US is currently running budget deficits at levels previously unprecedented outside of a major recession or war. The ideal scenario would be some sort of bipartisan consensus, as in the early 1990s, to deal with the fiscal challenges ahead in a pre-emptive manner. However, political incentives to reduce deficits appear limited, raising the prospect that it will be up to markets to ultimately force consolidation. The risk is that this could come at a moment of heightened vulnerability, such as a wider economic crisis or recession, resulting in a harsher correction. A degree of financial repression, especially if combined with genuine consolidation, could end up playing a role in managing the debt burden. But a full embrace of that route — in an extreme case, via debt monetisation by the Fed — would merely accelerate the erosion of the dollar's special status.

The third challenge is protecting the US’ institutional foundations. Institutional stability (and the consistent rule of law and secure property rights underpinning it) have been the bedrock on which the capital-markets and dollar advantages ultimately rest. If that eroded over the coming decades, it would create a more uncertain climate for long-term investments and dampen the international appeal that has attracted top talent and financial capital to the United States.

The fourth challenge is a global one, in that the US has benefited immensely from the post-war system of US leadership. However, that settlement has come under increasing strain in recent years, particularly as emerging markets that have grown in size seek greater influence in global affairs. After all, the US only makes up around 26% of global GDP in USD terms. But with its allies, the influence of the US increases substantially. For instance, the G7 as a whole makes up 44% of global GDP, and if you include further US allies such as Australia and South Korea, its effective alliance network still comprises more than 50% of global GDP in dollar terms. With the US standing as the clear and unsurpassed leader of this group, maintaining this network has enormous benefits.



What will the US' ongoing success imply, and what are the key takeaways?

To wrap up, let us briefly consider two concluding questions. First, what does the above analysis mean for investors looking at the US over the long term. And second, what lessons can the rest of the world learn from the success of the US model over time.

For those allocating capital, the case for the US as the world's pre-eminent investment destination remains intact. An investor into the US buys a bundle that is hard to replicate – productivity leadership, the deepest and most liquid capital markets in the world, the rule of law that underpins them, and the flexibility conferred by the dollar's reserve role. That combination should persist. However, the risk premium around the US has risen, as fiscal strains and a wider geopolitical reshuffling point to a wider distribution of outcomes and bigger tail risks.

In practice, this could play out in several ways. First, the dollar's "exorbitant privilege" is unlikely to disappear within the next decade, but gradual diversification of both reserves and payments is real and persistent. Already, the dollar's global importance has ebbed in recent years. These growing question marks around the dollar's status are the first risk. Second, US equity leadership is now heavily concentrated in a handful of AI-exposed names, so the bull case increasingly rests on a small group of companies, again widening the tail risk outcomes. And third, there is a tail risk of a correlated scenario where a fiscal reckoning coincides with a correction in the AI investment cycle, pressuring equities, duration and the dollar simultaneously, and undermining the diversification that normally cushions portfolios. This is a scenario worth planning for, even if it is not the central case.

What can the rest of the world learn from the US' success? In part, it is simply down to luck, including geographic advantages and an abundance of resources, such as energy, that cannot be replicated elsewhere. However, there are several features that could be reproduced elsewhere.

The first is its resilient institutional architecture, with a legal system and property rights that have provided a favourable environment for long-term investment. Although the US has faced severe crises in the last 250 years, including a civil war, its system of government has been recognisably the same throughout that time.

Second, it has a well developed financial system, and a tolerance for risk that's enabled the US to embrace new technologies and emerge quicker from downturns, with resources shifting quickly to new and productive sectors. US history demonstrates how suppressing volatility and preventing resource re-allocation can be counterproductive over the long term.

Third, a key component of US success has been its adaptability to different contexts, and its ability to leverage existing strengths when under pressure. These range across its energy resources, capital markets, universities, military power, reserve currency status, and technological leadership. Even when one point has appeared under threat, others have been able to compensate.



Appendix 1

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Remembering Alan Greenspan

June 22, 2026

First Impressions

Alan Greenspan, who passed away today at the age of 100, was one of the most consequential figures in the history of American economic policy—and someone I had the privilege of working with for more than two decades, first as a colleague at the Federal Reserve and later as a friend and adviser at Deutsche Bank. His passion for economic analysis, his respect for data, and his willingness to challenge conventional wisdom profoundly influenced my own approach to economics and policy.

I joined the Federal Reserve staff in September 1973 as a newly minted PhD in Economics from the University of Michigan—four years after my new Fed colleague Don Kohn—and served through the chairmanships of Arthur Burns, G. William Miller, and Paul Volcker. When Greenspan arrived in August 1987, I was in mid-level management in the Board's International Finance economic research staff. I still remember the day he arrived. He toured the building with my colleague Ted Truman, meeting many members of the staff, and I had the opportunity to shake his hand and welcome him personally. It was a small gesture, but a characteristic one: Greenspan appreciated the value of his staff and moved quickly to engage them.

A Remarkable User of Expertise

That engagement was immediately substantive. Greenspan had ambitious ideas about how to use the Fed's deep reservoir of talent. He was fascinated by data—not only the headline numbers but obscure series and unusual relationships that others overlooked.

Much of my early Fed career predated the Internet. Yet the institution possessed an extraordinary asset: specialists in virtually every corner of the economy. One could simply pick up the phone and speak with an expert on housing, productivity, corporate profits, or almost any topic imaginable. It was, in effect, a pre-Internet web of distributed expertise. Greenspan understood this instinctively and made exceptional use of it.

Don Kohn, in his remembrance published today at Brookings, recalled that Greenspan arrived “like a kid in a candy store,” eager to tap the knowledge embedded throughout the organization. That description captured him perfectly.

Models, Markets, and Instinct

Greenspan brought macroeconomic instincts somewhat different from those of most Fed staff. I had been trained in the neo-Keynesian tradition, while Alan placed greater faith in markets and the informational efficiency of prices.

Having spent much of my own career modeling trade flows and global economic interactions, I came to appreciate both the power and limitations of formal econometric models. Greenspan shared my strong support for free trade, but he was less inclined than many Fed economists to rely heavily on model-based

Authors

Peter Hooper

Vice-Chair of Research



forecasts. Instead, he searched for revealing relationships in data and drew heavily on information gathered from his extensive network of business leaders and industry contacts.

Over the years I occasionally worked on speeches for him, mostly in an editorial capacity, since he inevitably rewrote drafts into precisely what he wanted to say.

“Irrational Exuberance”

One memorable episode involved the now-famous phrase “irrational exuberance” in a speech prepared for the American Enterprise Institute in December 1996. Several staff members, myself included, were skeptical. Greenspan insisted. History vindicated him.

Years later I asked him the secret of his remarkable longevity and productivity. His answer surprised me: soaking in a hot tub. The same ritual that, as he later recalled, had inspired the concept of “irrational exuberance” sustained him through his tenth decade.

The Productivity Revolution

The 1990s were, in many ways, a magical time for the Federal Reserve. Greenspan earned his reputation as the “Maestro”—the title Bob Woodward gave him in his 2000 book—in large part because of his success in steering the economy through the long expansion of that decade.

By the mid-1990s, many policymakers worried that growth was running beyond the economy’s capacity and that inflation pressures were building. Greenspan disagreed. He became convinced that information technology and the computerization of business processes were producing a genuine acceleration in productivity growth, allowing the economy to sustain faster expansion without inflation.

Greenspan’s recognition of the productivity surge before most others demonstrated his ability to identify anomalies that did not fit conventional wisdom and pursue them to their logical conclusion. He was right. The economy grew faster, and inflation remained subdued. My good friend Steve Landefeld, former Director of the Bureau of Economic Analysis, well remembers how Alan, one of their most ardent supporters, frequently challenged him to expand their measures of sustainable GDP growth.

Communication and Transparency

Greenspan believed that some ambiguity served monetary policy well. He thought the central bank retained greater flexibility when markets focused on incoming economic data rather than endlessly parsing every word uttered by policymakers.

Nevertheless, under his leadership the Fed began its gradual move toward greater transparency, introducing post-meeting statements and the policy bias. Ben Bernanke later accelerated that transition through press conferences, forward guidance, and eventually the dot plot.

Bringing Greenspan to Deutsche Bank

Greenspan was more than a mentor to me; he was an inspiration who demonstrated the importance of rigorous analysis and intellectual curiosity.

He was gracious enough to acknowledge our relationship in his own way. When he inscribed a copy of his memoir, *The Age of Turbulence*, he referred to me as a mentor—a generous inversion that I treasure, even though I would reverse the description.



After leaving the Fed in 1999 and joining Deutsche Bank, I became convinced that Alan would make a valuable adviser to the firm. With David Folkerts-Landau's blessing, I approached him after his retirement in January 2006, but he wanted first to finish his book. Months later, after learning he had accepted a position with PIMCO, I reminded him of our earlier conversation. He agreed to join Deutsche Bank as a senior adviser for two years, beginning in 2007.

Around the World with the Maestro

Alan saw clients globally for Deutsche Bank, and the attraction was exactly what the firm had hoped when he joined as senior adviser: access to his strategic insight and his extraordinary perspective on markets, monetary policy, and risk. Clients everywhere relished hearing the Maestro's latest thinking—not simply because he had once chaired the Federal Reserve, but because he could translate complex economic developments into practical judgments that sophisticated market participants could use.

Intellectual Honesty

Alan inspired me in many ways, though we had our differences. I was uncomfortable with his early confidence that private institutions could regulate themselves more effectively than government authorities—a view that events eventually proved overly optimistic.

Yet those disagreements never diminished my admiration for him. What mattered most was that Alan continued to interrogate his own views when events challenged them. Testifying before Congress in October 2008, Greenspan acknowledged that he had “made a mistake” in presuming that the self-interest of banks and other institutions would be sufficient to protect their shareholders. That willingness to confront error openly, rather than evade it or rationalize it away, was one of the clearest expressions of his intellectual honesty—and one of the qualities I most respected in him.

His Living Legacy

What living legacy does Greenspan leave at the Federal Reserve?

Part of the answer may be found in Kevin Warsh, who was sworn in as Fed chairman just one month ago. At his White House ceremony, Warsh paid explicit tribute: “Chairman Greenspan was the first to tell me and show me what this role demands.”

Warsh's instincts resemble Greenspan's in two important respects. First, just as Greenspan resisted premature tightening in the 1990s because he believed information technology was driving a productivity revolution, Warsh has argued that artificial intelligence could prove a significant disinflationary force. In both cases, the judgment rested on perceiving structural change before the data fully confirmed it.

Second, Greenspan's preference for brevity when discussing the outlook for monetary policy appears to have found an echo in Warsh's early approach. In his first press conference, he reduced the Fed's post-meeting statement from 341 words to 132. As my Deutsche Bank colleague Matthew Luzzetti observed, that move reverses a decades-long trend toward ever greater transparency.

Whether Warsh's views on AI prove as well founded as Greenspan's insights about information technology will be closely watched. Inflation today is stickier, geopolitical risks are greater, and the enormous investment required to build AI infrastructure may itself prove inflationary in the near term.



“We Had Fun”

Don Kohn concluded his remembrance by recalling Greenspan summing up their years together with a simple observation: “We had fun, didn’t we, Don?”

Yes, Alan—we had fun.

And that spirit of curiosity, intellectual excitement, and joy in understanding how the world works enriched economic policymaking in the United States and around the globe. It certainly enriched my own life and career, and strengthened my desire to remain active in professional economics. For that, and for much else, I will always be grateful.



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Brexit 10 Years On: what's worked, what hasn't, what's next?

June 22, 2026

Ten years ago, the UK voted to leave the European Union with a small majority. This marked the biggest self-imposed trade disruption in history (at the time). Much ink has been spilt on understanding the effects of Brexit, and much uncertainty still remains.

In this special report as part of the Deutsche Bank Research Institute, we revisit the Brexit impact 10 years on. We look at what's worked, what hasn't, and what's next?

To be sure, the effects of Brexit are still felt today. The UK economy continues to suffer from tight trade restrictions via non-tariff barriers. And we estimate some meaningful costs to GDP and employment as a result of the UK's departure from the EU. The cost of living has also been modestly higher on the back of Brexit.

It's not all doom and gloom, however. While Brexit may not have been the success some claimed it would be, it hasn't unfolded quite as badly as others thought either. While our Brexit Scorecard shows an economic cost to the UK's exit, our Brexit Balance Sheet brings together things that have been more positive for the UK as well as things which have not, and where we remain unsure.

We conclude on where we think the UK should focus its efforts in the near term while asking the big medium-term question: is there a path back to the EU? No matter how ambitious the UK's reset with the EU may be in the coming years, the UK will need to garner full support from both the electorate and the EU to come to terms on a deal fit for the British economy of the future. The bottom line: while the cost of Brexit cannot be ignored, there may be political limitations on how far the UK can go in re-integrating with the EU.

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Table Of Contents

Executive Summary.....	3
Introduction: Brexit Impact – 10 Years On	4
The Long Goodbye - How The UK Delivered Brexit.....	5
The UK Scorecard: economic performance relative to the G7	7
UK Scorecards in Charts	14
Marking to market the Brexit impact: a doppelganger analysis.....	16
The Brexit Balance Sheet: what's worked and what hasn't.....	20
What's worked?	20
What's not worked?	21
The grey zone: neither worked nor didn't work	22
The Look Ahead: What's Next?.....	25
The near-term questions: how can the UK improve on the TCA?	25
The big medium-term questions: is there a path back?	27
A hypothetical path ahead	31
Appendix.....	32
A. The Synthetic Control Method	32
B. Detailed workings on TCA adjustments	34



Executive Summary

On the 23rd of June 2016, the UK economy and political ecosystem changed forever with the public voting to leave the European Union (EU). Fragmented politics gave way to a fragmented economy with the UK the first country to break from the European Union – marking, at the time, the biggest unilateral trade disruption in history.

The effects of Brexit are still felt today. The UK economy continues to suffer from tight trade restrictions via non-tariff barriers. The changing tide of immigration has also impacted businesses, from agriculture to hospitality and retail. To be sure, the costs of Brexit – though perhaps slightly exaggerated – are real. The economy is smaller as a result of its departure from the EU. The labour market is also smaller and more dislocated. And prices are running higher, given increased cost of EU imports.

Still, the path ahead – though uncertain – is filled with opportunity. The UK's Brexit freedoms have opened a path to better trade deals with other nations. Regulatory autonomy has positioned the UK to stay at the frontier of AI – the next biggest technological revolution. The speed of deregulation has also allowed the UK to benefit from stronger investment growth. And outside the EU, the UK has been able to embark on a more aggressive form of industrialisation.

In this report, we look at how the economy has taken shape a decade since the Brexit referendum. First, we present a scorecard of economic and market measures over the last decade, comparing the UK to its G7 peers (where compatible). We also set out a doppelganger analysis to estimate the running costs of Brexit. Second, we set out a balance sheet approach looking at what's worked and what's not worked. And third, we conclude with the path ahead. Low hanging fruit to improve the UK-EU Trade & Cooperation Agreement (TCA) remains ripe for picking. At the same time, however, big questions around the UK's future with the EU remain unanswered.

Our key conclusions:

- The cost of Brexit, we estimate, is a smaller economy (-4%), lower employment (-2%) and higher prices (+0.7%). While meaningful, these are slightly better readings than the overarching consensus.
- From a market perspective, investors have taken a dim view of Brexit and its consequences. While the currency has been broadly stable for much of the last decade, the disruption to the supply side of the economy has seen the cost of borrowing rise more sharply and seen equities underperform. From a current account standpoint, while income and trade balances have improved, the external economy is more unbalanced between goods and services than ever before.
- The good news is that the options to improve the TCA – without taking any big decisions on deep integration - are meaningful. We estimate that such improvements to the TCA can lift UK GDP by 0.4-0.8%.
- It takes two to tango. No matter how ambitious the UK's reset with the EU may be, the UK will need full support from both the electorate (to avoid any political paralysis) and the EU to come to terms for a better deal.



Introduction: Brexit Impact – 10 Years On

Ten years on from the June 2016 EU referendum, the United Kingdom's economic journey since Brexit remains one of the most contested empirical questions in modern economic history.

Isolating the structural Brexit impact (higher trade friction, reduced labour mobility, regulatory divergence) however, remains fraught with complexity. The overlapping shocks of a global pandemic, a historic energy shock, a global trade war, and a pronounced monetary tightening cycle affected virtually every advanced economy.

In this report we take a multi-step approach to disentangle some of the effects of Brexit and look ahead to what's next.

We start by providing a **brief overview** on how Brexit was delivered, highlighting the two phases that dominated markets and the economy: the initial referendum shock and the constitutional crisis arising as a result of delivering Brexit, and secondly the post-pandemic shock in which the final Trade & Cooperation Agreement (TCA) was both negotiated and implemented.

We then present a **UK scorecard** looking at how the economy fared during the Brexit transition – comparing it to other G7 economies. This includes GDP growth, the labour market, inflation, trade and market performance. The UK – despite Brexit – has outperformed many of its peers in some areas, underscoring its importance on the world stage. Strikingly though, multiple areas and asset classes have still seen material underperformance. Domestic-focused stocks have delivered one-third of the return of peers since the UK's departure from the EU.

As part of our scorecard, we run an empirical estimation exploring the cost of Brexit (as things stand) using a synthetic control method (i.e. the doppelganger method). Here, we cover GDP, employment, and prices (CPI). Under this framework, UK GDP sits nearly 4% smaller. Employment is 2% lower. And prices are 0.7% higher.

The next section presents our **Brexit Balance Sheet**: a non-exhaustive list of what we think has and hasn't worked, and where the jury is still out. To be sure, while the negatives of Brexit are largely well-explored, we also set out some of the positives that position the UK into a more fragmented world.

Turning to the **future**, we estimate the impact on the UK economy of potential deals with the EU. There are plenty of low-hanging fruit to improve the TCA. We set these out in detail. In short, we think there may be something like 0.4-0.8% upside to GDP growth if the UK and EU worked together to reduce some of the non-tariff barriers.

We conclude with some bigger picture thoughts on the longer-term future of the UK's relationship with the EU, and our view on the optimal path forward. To be sure, no matter how ambitious the UK's reset with the EU may be, the UK will need full support from both the electorate and the EU to come to terms on a better deal. Bottom line, while the cost of Brexit cannot be ignored, there may be political limitations on how far the UK can go in re-integrating with the EU.



The Long Goodbye - How The UK Delivered Brexit

The UK's exit from the European Union was anything but swift. While a slim majority of the British public voted to leave the European Union on 23 June 2016, the path to the eventual Brexit Day on 31 January 2020 was a near four-year journey. We start by walking down history lane and reviewing how Brexit was delivered. Indeed, this is important in understanding how Brexit talks induced uncertainty and ultimately impacted the economy and UK financial markets.

Following the referendum decision to leave the EU, the UK was plunged into a state of economic uncertainty and political chaos. Indeed, Brexit was only delivered three prime ministers later in 2020 as part of the Trade and Cooperation Agreement. The end state was a far removed trade deal than many would have anticipated in 2016, with the UK unilaterally engineering the biggest trade disruption in its economic history with a "hard Brexit".

The decade since the Brexit referendum can be divided into two distinct phases – each with their own economic and political characteristics.

PHASE I	PHASE II
2016–2019	2020–2025
Referendum Shock/Constitutional Crisis	Pandemic/TAC/Energy Shock

Phase 1 entailed the referendum shock, including a constitutional crisis.

The immediate aftermath of the referendum result was messy.

The UK saw an immediate constitutional convulsion on how actually to deliver Brexit, raising uncertainty (politically and economically). The resignation of PM Cameron ushered in a new government under PM May, where she set out twelve Brexit principles to guide future negotiations – including establishing the infamous 'red-lines' in securing a deal with the EU.

To shore up support and strengthen her hand, PM Theresa May called a snap general election in 2017, which ended up stripping the Conservatives of their working majority and creating dependency on the Northern Irish Democratic Unionist Party (DUP) via a confidence and supply agreement.

Over the next several quarters, the UK was plunged into a constitutional crisis. This in turn saw renewed political uncertainty, with investment and business decisions coming to a standstill. The core political paradox was that Parliament had a majority against every available Brexit outcome simultaneously: against no-deal, against May's deal, against remaining, and against any second referendum.

Eventually, PM Theresa May called time on her premiership, opening the path towards a new Prime Minister. Boris Johnson's premiership from July 2019 brought resolution through confrontation. PM Johnson prorogued Parliament (declared unlawful), threatened a no-deal Brexit, expelled 21 Tory MPs who voted against him, and ultimately secured a renegotiated Withdrawal Agreement by solving the



Irish border through a customs border in the Irish Sea. To then obtain a parliamentary majority for his deal, PM Johnson called a general election for December 2019, in which the Conservatives won a large majority.

The UK formally left the EU under the Withdrawal agreement at the end of January 2020, but was to stay in the single market and customs union during a one-year transition period in which a full trade deal was to be negotiated.

Phase 2 was underscored by a global pandemic – with the final rush to secure a trade deal with the EU.

Despite the pandemic, trade negotiations between the UK and EU continued, but under extraordinary conditions. With the Covid-19 pandemic consuming essentially 100% of legislative bandwidth from March 2020 to mid-2021, the UK-EU Trade and Cooperation Agreement (TCA) – the most important trade negotiation in the country's post-war history – was conducted entirely via video conference.

After years of negotiation and deliberation, the culmination of the Brexit journey – at least as things stand – was the TCA being agreed on Christmas Eve 2020 – 1,645 days after the UK voted to leave the EU. For the avoidance of doubt, the TCA represented a marked, "hard Brexit," departure from the UK's previous relationship with the EU. [Figure 1](#) highlights all the key dates on this multi-year journey.

Figure 1: Key events in the run up to Brexit

Date	Key Political Event
Phase 1: The Referendum Shock	
23-Jun-16	Referendum: 51.9% Leave, 48.1% Remain. Turnout 72.2%.
24-Jun-16	David Cameron announces resignation
13-Jul-16	Theresa May becomes Prime Minister.
17-Jan-17	Lancaster House speech: May sets 12 Brexit principles.
29-Mar-17	Article 50 formally triggered. Two-year countdown begins.
08-Jun-17	Snap general election. Conservatives lose majority: 318 seats (–13).
6–9 Jul 2018	Chequers Agreement. Cabinet 'white paper' Brexit plan agreed.
14-Nov-18	Draft Withdrawal Agreement published.
10-Apr-19	Article 50 extended to 31 October 2019. UK must hold European Parliament elections.
24-May-19	May announces resignation. Cites failure to deliver Brexit. Conservative leadership contest: 10 candidates. Boris Johnson wins.
24-Jul-19	Boris Johnson becomes PM.
17-Oct-19	Revised Withdrawal Agreement agreed with EU. Irish Sea border accepted.
12-Dec-19	General election. Conservatives win 365 seats: 80-seat majority.
Phase 2: The Deal And Pandemic Collision	
31-Jan-20	UK formally leaves the European Union at 11:00pm. Transition period begins (to year end).
24-Dec-20	Trade and Cooperation Agreement (TCA) agreed. Christmas Eve deal.
01-Jan-21	TCA in force. UK customs border operational for first time.

Source : Deutsche Bank, News Sources



The UK Scorecard: economic performance relative to the G7

How did the UK fare relative to its counterparts in the G7 over the past decade? In this section, we build on the UK's economic performance since its decision to leave the European Union via a UK scorecard. We group this into three time periods.

The first (2010-2015) is the pre-referendum period – which in itself encapsulated its own economic issues as the UK dealt with the aftermath of the global financial crisis (GFC) and the government's austerity.

The second period is the referendum shock and the constitutional crisis (2016-2019) as the UK's Withdrawal Agreement with the EU was negotiated amidst domestic political turbulence.

And the third period looks at the pandemic period and beyond (2020-2025), where the UK and EU negotiated and implemented its TCA during a global pandemic, and shortly after which saw the Russian invasion of Ukraine and energy crisis of 2022.

We briefly set out some headline statistics as part of an economic scorecard looking at GDP growth, business investment, trade, employment and inflation ([Figure 2](#)).

Figure 2: The Broad UK Economic Scorecard

UK Economic Scorecard			
	2010-2015	2016-2019	2020-2025
Real GDP Growth	1.9%	2.0%	1.1%
% annual growth	(4)	(3)	(4)
Gross Fixed Capital Formation	3.8%	2.9%	2.3%
% annual growth	(2)	(7)	(2)
Current Account Balance, % of GDP	-3.6%	-3.7%	-2.5%
	(7)	(7)	(6)
Employment Rate	71.6%	75.2%	75.0%
	(3)	(2)	(3)
CPI Inflation	2.4%	1.9%	4.3%
% annual rate	(1)	(2)	(1)
*G7 Rank In Parantheses (1= highest, 7=lowest)			

Source : Deutsche Bank, IMF, OECD

GDP

On the surface, UK real GDP has expanded sizably since the referendum – especially in 2016 and 2017. During the early post-referendum years, UK GDP annual growth averaged 2% – above the G7 average and behind only the USA and Canada. From 2020 onwards, UK GDP growth slipped meaningfully to 1.1% – converging to the G7 average ([Figure 3](#)).



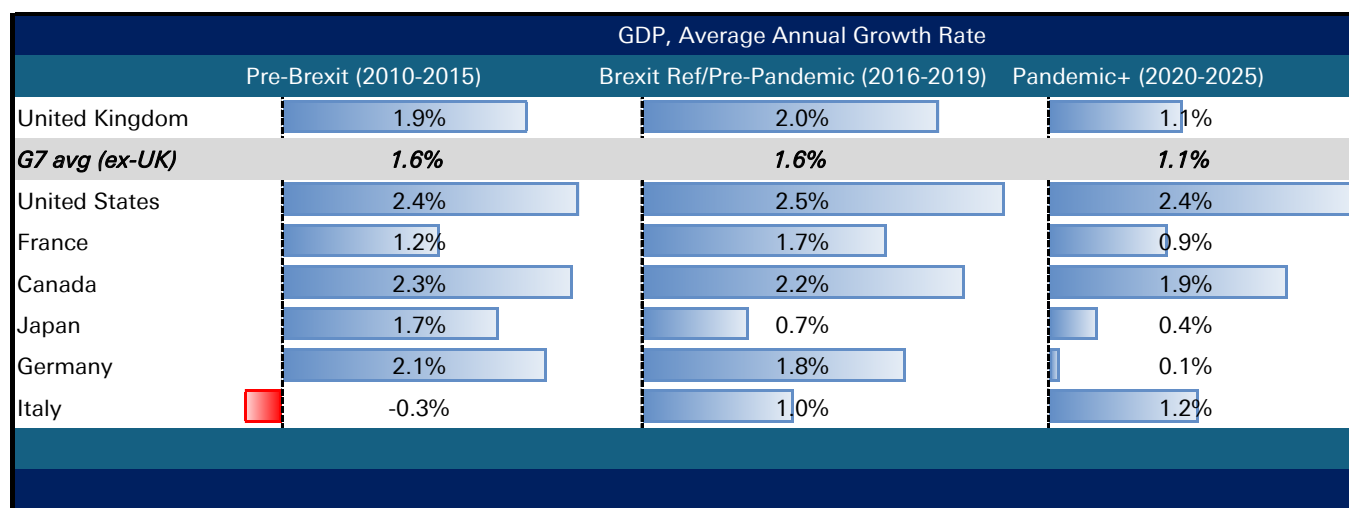
What drove some of the upward momentum in GDP growth? Part of the reason for the outperformance could be linked to easier monetary policy. Indeed, the Bank of England did its best to support economic activity by unveiling a new package of liquidity measures immediately following the referendum result, cutting Bank Rate to 0.25% in August 2016, and pushing through a new quantitative easing package comprising GBP 60bn of government bond purchases and GBP 10bn of corporate bond purchases.

Stockpiling also helped support activity. Businesses began stockpiling inventories ahead of the referendum with Q4-15 seeing the largest single quarterly inventories on record. And then between Q4-2018 and Q1-2019, inventories shot up again by a cumulative GBP 11.1bn, supporting activity as the first Brexit deadline drew closer.

Real GDP *per capita*, however, tells a slightly different story. Much of the jump in growth – unsurprisingly – was concentrated rather than distributed across the population. While real GDP per capita annual growth sat slightly above its G7 comparators for much of the decade preceding the pandemic and Brexit deal, the UK then fell well behind its peers from 2020, averaging 0.3% compared to a G7 average of 0.7%. Part of this was a reflection of a deeper pandemic drop, while it may have been the case that Brexit slowed the UK's catchup with its peers.

To be sure, the GDP gap cannot be solely attributed to Brexit and its uncertainty effect. Post 2010, the UK was suffering from a cocktail of cyclical and structural issues, including a chronically low savings rate, the hangover of the GFC, weak manufacturing productivity, and regional economic concentration – all of which may have constrained long-run growth.

Figure 3: UK GDP Scorecard



Source : Deutsche Bank, IMF; bars represent relative strength for each country across the three periods.

Business Investment

If there was a single macroeconomic variable that most reflected the Brexit effect – separated from pandemic noise, global inflation, and cyclical forces – it was business investment.



Why was investment impacted even before Brexit happened? Despite no change in trading relations between the UK and EU, political uncertainty was starting to have far reaching implications on economic uncertainty – with businesses uncertain around the future trading outlook. Indeed, it became difficult for firms to plan supply chains, staffing, and/or regulatory compliance.

Investment decisions, in the end, were delayed or shelved entirely with firms grappling with uncertainty around future trade arrangements, regulatory environments, and labour market conditions.

The impact of deferred investment cannot be overstated, as its effects compound over time, impacting growth and living standards. Put simply, foregone investment meant foregone productive capacity, foregone technology adoption, and foregone productivity growth.

The investment paralysis was deep and broad. UK business investment fell in four of the eight quarters between Q1-2018 and the eve of the pandemic, Q4-2019. This was the largest sustained investment shortfall in the post-war UK record outside of the 2008-09 financial crisis. As our scorecard highlights, gross fixed capital formation slowed dramatically since the Brexit referendum – moving from a 3.8% annualised pace to 2.9% per annum between 2016 and 2019 and slowing further in the 2020s (2.3% per annum) ([Figure 4](#)). As a share of GDP, UK investment relative to its G7 peers continued to underperform. And on the World Bank's own scoring measure of 'Government Effectiveness', the UK went from 'above average' to 'below average'.

What impacted investment plans? Constant threats of a 'no-deal' Brexit left just-in-time sectors particularly exposed – automotive, pharmaceuticals, food manufacturing, etc. Several auto manufacturers pre-announced production shutdowns timed to avoid a no-deal Brexit scenario. SMMT [estimated](#) that the collective cost of these shutdowns amounted to GBP 130m per day of lost production.

In numbers, the OBR [estimated](#) that Brexit uncertainty was costing approximately 0.5pp of annual GDP growth, equivalent to c. GBP 10-12bn of foregone output per year. A separate paper by [Bloom et al. \(2019\)](#) estimated that Brexit anticipation itself gradually cut investment by 11% and lowered productivity by 2-5%.

Major corporates deferred plans as a direct consequence of the paralysis: the Honda Swindon closure (February 2019, 3,500 jobs); Dyson HQ moved to Singapore; Goldman Sachs, JP Morgan, HSBC, Barclays, and UBS all moved substantial EU-facing operations to Dublin, Paris, Frankfurt, or Amsterdam. EY's [Brexit Tracker](#) (Q4-19) noted that 7k+ financial services positions had formally relocated to EU cities. Productivity growth effectively dropped to 0% with UK total factor productivity moving into negative territory.

It's not all bad news though. As our own scorecard shows, FDI and R&D expenditure has picked up since the referendum. Net FDI inflows jumped from 1.8% of GDP in the pre-Brexit period, rising to 4.1% between 2016 and 2019, before slowing to 2.4% of GDP since 2020. R&D expenditure (as a share of GDP) also improved over the past decade or so, pushing from below the G7 (ex-UK) average to above it since 2016. So while business investment dropped, international investors still sought to maintain exposure to UK markets and its role as a domestic innovator.



Figure 4: UK Investment Scorecard

Investment Scorecard			
	Pre-Brexit (2010-2015)	Brexit Ref/Pre-Pandemic (2016-2019)	Pandemic+ (2020-2025)
Gross Fixed Capital Formation, % Annual Growth	3.8%	2.9%	2.3%
G7 ex UK	2.5%	2.5%	1.5%
Gross Fixed Capital Formation, % of GDP	16.8%	18.8%	18.7%
G7 ex UK	19.9%	20.4%	21.3%
FDI (Net), % of GDP	1.8%	4.1%	2.4%
G7 ex UK	1.5%	1.8%	1.7%
R&D Expenditure, % of GDP	1.8%	2.5%	2.8%
G7 ex UK	2.3%	2.4%	2.6%
Government Effectiveness (std dev from world avg)	1.53	1.54	1.26
G7 ex UK	1.4	1.5	1.5

Source : Deutsche Bank, OECD, IMF; bars represent relative strength for each variable across the three periods

Employment

The UK labour market has witnessed three distinct phases since the GFC. The pre-Brexit period saw the UK jobless rate peak at a multi-decade high of over 8% in late 2011. By the time the EU referendum came around, the jobless rate had receded to around 5%.

Post-referendum, the UK saw a new wave of EU migration – the share of EU-born workers in the labour force jumped from 5.5% to 7.2% (as EU migrants likely sought to bypass any potential future immigration restrictions). But by the time the pandemic rolled around and Brexit was enacted, the share of EU workers then dropped. In fact, since the pandemic began, the average annual rate change in the EU born population in the UK has been -0.2%.

Relative to the G7, the UK maintained a higher employment rate (over the last decade +), with the jobless rate tracking below its peers – a reflection of its more flexible labour market. Unit labour cost growth slowed during the pandemic and widened dramatically during the pandemic/TCA period (Figure 5).

The headline statistics, however, fail to capture the impact of Brexit. The replacement of EU free movement with a points-based immigration system fundamentally altered the supply of workers in sectors that previously relied on low-to-medium skilled EU nationals: hospitality, social care, agriculture, haulage, and food processing. The resulting shortages have had two consequences – wage growth at the bottom of the distribution (a genuine distributional gain for low-paid UK workers) and persistent vacancies in those low-wage sectors.

Empirically, it's likely that in the absence of Brexit, employment levels would have been higher. As investment plans were shelved and/or staff were relocated elsewhere, the case for higher employment should the UK have remained in the EU is strong. Breinlich & Magli in 2024 [noted](#) that UK services firms dodged Brexit barriers by changing how they served the EU – increasingly via local affiliate sales (commercial presence) rather than cross-border exports from the UK. This protected firm-level services exports but at the cost of lower domestic employment – jobs and activity grew at the new EU affiliates instead of in the UK.



Figure 5: UK Labour Market Scorecard

Labour Market Scorecard			
	Pre-Brexit (2010-2015)	Brexit Ref/Pre-Pandemic (2016-2019)	Pandemic+ (2020-2025)
Employment Rate (16-64)	71.6%	75.2%	75.0%
G7 ex UK	67.3%	70.0%	70.9%
Unemployment Rate (16+)	7.2%	4.3%	4.4%
G7 ex UK	7.5%	6.1%	5.5%
Vacancy Rate	1.9%	2.5%	2.7%
G7 ex UK	1.7%	2.6%	3.2%
Unit Labour Costs, % Growth	0.7%	2.1%	0.5%
G7 ex UK	0.8%	1.5%	3.1%
EU Nationals, % of UK Labour Force	5.5%	7.2%	6.7%
Non-EU Nationals, % of UK Labour Force	9.4%	10.3%	12.7%

Source : Deutsche Bank, IMF, OECD; bars represent relative strength for each variable across the three periods

Prices

UK inflation since 2016 has been shaped by four overlapping forces: a Brexit-specific component (sterling depreciation pass-through, non-tariff barrier (NTB) cost pass-through, labour market restructuring); a global pandemic component (supply chain disruption, goods demand surge); a global energy component (Russia-Ukraine, gas prices); and a domestic demand component (tight labour market, fiscal support).

Unsurprisingly, UK inflation scored well above its G7 peers through the last several years. The first inflation wave came just after the Brexit referendum, when trade-weighted sterling collapsed by around 7%. The fall in sterling fed directly into import prices with non-energy industrial goods prices jumping over 200bps compared to 2016. Food inflation, over the same period, pushed over 400bps. Breinlich et al. (2017) [estimated](#) that the sterling depreciation cost the average household GBP 404 a year by mid-2017.

The gap between the UK and the G7 widened the most after the onset of the pandemic, when the UK officially left the EU ([Figure 6](#)). While the aftermath of the pandemic and Russia-Ukraine war led to a big spike in inflationary pressures, the effects of Brexit were hard felt too. Structural gaps in the labour market – owing to a re-jigging of EU labour – added to wage pressures. And most directly, the UK's exposure to goods/food inflation was laid bare by rises in non-tariff costs. A [study](#) from the LSE's CEP identified a 6% cumulative increase in food import prices attributable to sanitary and phytosanitary checks and customs friction, which they estimate contributed approximately 0.3-0.5pp to headline CPI on a sustained basis. Reduced import competition may have also shifted prices further. With fewer EU producers competing freely in the UK market, domestic services pricing faced less competitive constraint.



Figure 6: UK Prices Scorecard

Inflation Scorecard			
	Pre-Brexit (2010-2015)	Brexit Ref/Pre-Pandemic (2016-2019)	Pandemic+ (2020-2025)
Headline CPI, YoY	2.4%	1.9%	4.3%
G7 ex UK	1.4%	1.3%	3.2%
Unit Labour Costs, YoY	2.2%	1.9%	3.9%
G7 ex UK	1.2%	1.2%	2.6%
Import Prices, YoY	0.0%	3.5%	3.2%
G7 ex UK	2.6%	1.3%	5.1%

Source : Deutsche Bank, OECD, IMF; bars represent relative strength for each variable across the three periods

Trade and external accounts

Brexit was perhaps first and foremost a trade shock, with the UK breaking from decades of closer integration with its largest trading partner. Ten years on, however, the scorecard on the UK's external *balances* is relatively favourable, which may be surprising to some. Strip out erratics like gold, and the UK's current account balance improved notably. This improvement was seen across both the trade balance and the income balance components. To be sure, the average G7 current account balance also improved over the same time period – but the UK's improvement was particularly welcome given its very weak external position at the time of the referendum ([Figure 7](#)).

Figure 7: UK external balance improved in the decade since Brexit

UK trade scorecard (%GDP unless otherwise specified)			
	Pre-Brexit (2010-2015)	Brexit Ref/Pre-Pandemic (2016-2019)	Pandemic+ (2020-2025)
Current account balance	-3.1%	-3.9%	-2.4%
G7 ex UK	-0.7%	1.6%	0.9%
Trade balance	-1.0%	-1.3%	-0.6%
Of which goods	-6.2%	-6.3%	-6.8%
Of which services	5.2%	5.0%	6.2%
Of which balance with EU	-2.3%	-3.6%	-2.7%
Of which balance with non-EU countries	1.1%	2.1%	2.3%
Income balance	-2.1%	-2.6%	-1.9%
Goods exports growth (cumulative, real terms)	18%	-5%	-11%
G7 ex UK	20%	4%	-4%

Source : Deutsche Bank, Haver Analytics; bars represent relative strength for each variable across the three periods

On the trade balance, improvement came from continued growth in services exports, which more than offset a widening goods balance – something we explore in more detail later on. Geographically, the UK's trade surplus with non-EU countries increased, more than offsetting a slight deterioration in the balance with the EU vs the pre-referendum period. To be sure, the trade scorecard still contains some dark marks. Perhaps most strikingly, growth in UK goods' exports volumes underperformed the rest of the G7 over the past ten years.



Empirically, many studies have set out the effects of Brexit. Freeman et al. in 2022, [noted](#) that the UK saw a sharp drop in trade once new barriers set in from January 2021. Many smaller UK firms simply stopped exporting to the EU because they could not absorb the fixed compliance costs (rules of origin, SPS paperwork, etc.). Larger firms coped; the long tail of small exporters did not.

Financial markets

Financial markets are in many ways a real time [scorecard](#) of economic performance. The past decade for the UK does not make for pretty reading on this front ([Figure 8](#)).

The pound is perhaps the single cleanest, catch-all measure of relative UK financial market performance, memorably falling over 7% (in trade-weighted terms) as the referendum results came in. Since then, over the past decade the pound has gradually recovered in nominal terms. And combined with higher inflation in the UK, in real terms the pound is now actually slightly stronger than right before the vote, albeit still below its average from the few years preceding the referendum.

Perhaps the main takeaway from looking at the currency, however, is its relative stability over the past ten years, with far narrower ranges than have historically been the norm. To be sure, that's consistent with a global pattern of broadly lower currency volatility over the past ten years, but in the UK's case, it is still notable given what at times has felt like a continued stream of headlines and events.

Figure 8: Higher relative borrowing costs, weaker pound, underperforming equities relative to pre-referendum period

Financial market scorecard			
	Pre-Brexit (2010-2015)	Brexit Ref/Pre-Pandemic (2016-2019)	Pandemic+ (2020-2025)
Govt. borrowing costs: 10y gilt yield average (%)	2.5%	1.0%	2.7%
G7 ex UK	2.2%	1.1%	2.0%
Major equities: FTSE 100 cumulative total return (%)	43%	43%	63%
G7 ex UK	64%	42%	116%
Midcap equities: FTSE 250 cumulative total return (%)	120%	40%	21%
G7 ex UK	102%	37%	67%
Purchasing power of a pound: GBP FX Index average	80.6	74.0	74.2

Source : Deutsche Bank, Bloomberg Finance LP, Haver Analytics; bars represent relative strength for each variable across the three periods

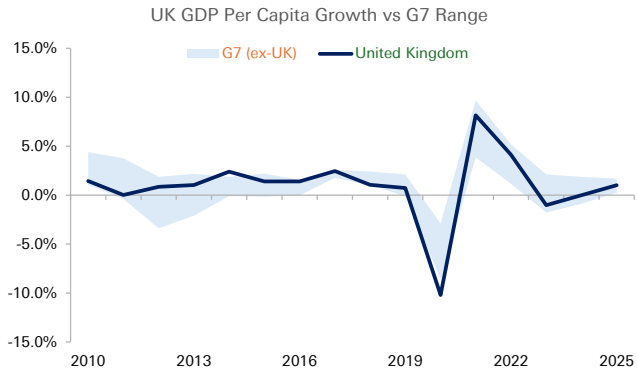
In the fixed income space, the UK government's cost of borrowing has risen sharply in the past few years and more so than in other G7 countries ([Figure 18](#)). The decision to leave the EU was far from a primary force at play here, with the dominating factors being the monetary policy responses to the inflation shocks of recent years, looser fiscal policy, and also a more [aggressive programme](#) of active quantitative tightening.

Finally, the underperformance of UK equities since the pandemic is noteworthy. The FTSE 100 having a lower weighting on sectors which have outperformed globally, (such as tech), may be one factor at play and is explored in more detail later. Meanwhile, for the more domestically focused FTSE 250, the larger increase in UK borrowing costs vs peers may also be playing a role. In any case, it has provided only one-third of the return vs the G7 average since 2020.



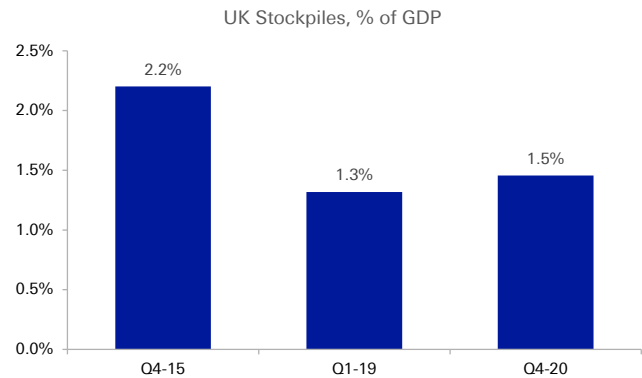
UK Scorecards in Charts

Figure 9: UK GDP per capita falling to bottom of G7 pack since the referendum



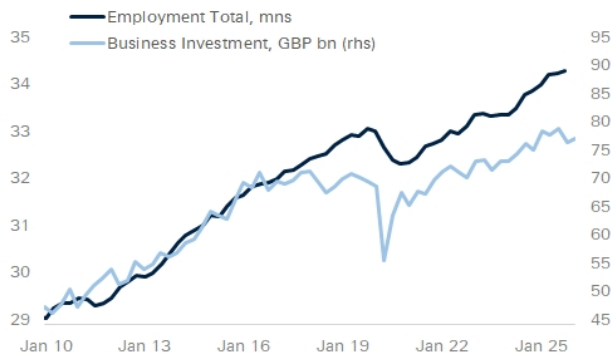
Source : Deutsche Bank, IMF

Figure 10: Stockpiling added materially during period of uncertainty through the pre-Brexit period



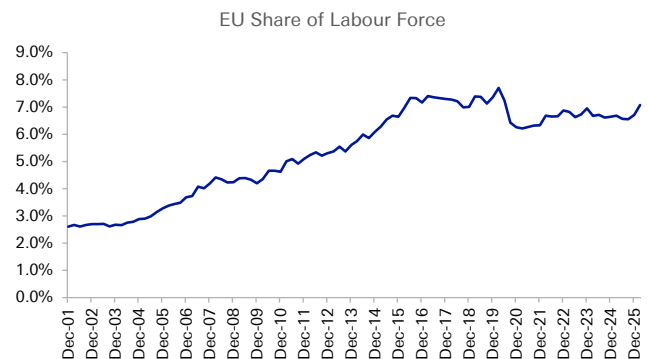
Source : Deutsche Bank, Macrobond

Figure 11: Employment and business investment decoupled as firms substituted labour for capital



Source : Deutsche Bank, Macrobond

Figure 12: Share of EU nationals in the UK saw small spike just after the referendum but has fallen meaningfully since



Source : Deutsche Bank, ONS

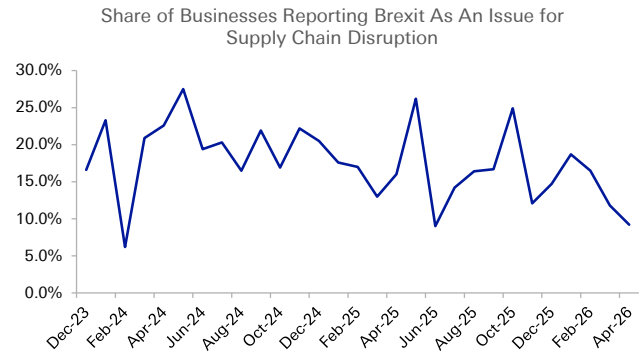


Figure 13: UK goods exports to the EU have fallen in real terms since the vote



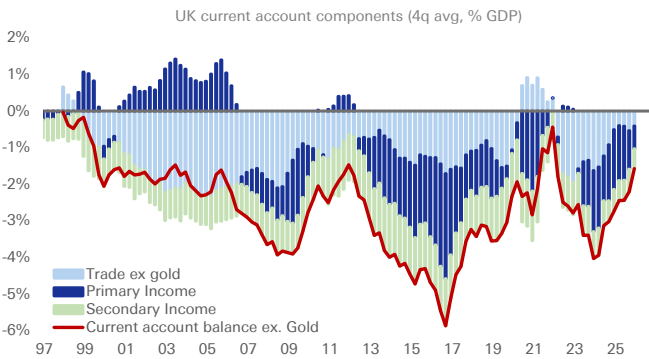
Source : Deutsche Bank, Macrobond

Figure 14: Around 10% of firms still blame Brexit for supply chain disruptions



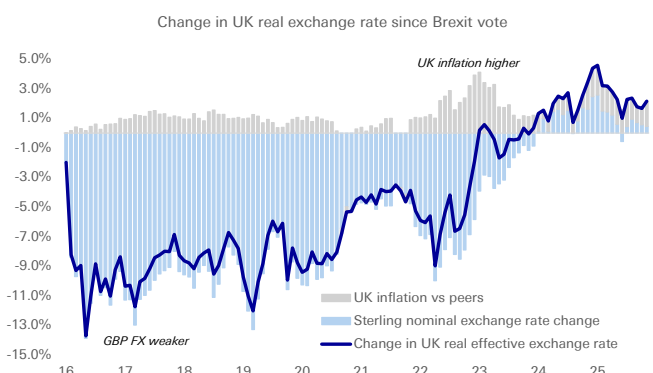
Source : Deutsche Bank, ONS

Figure 15: UK current account deficit improved over the past decade



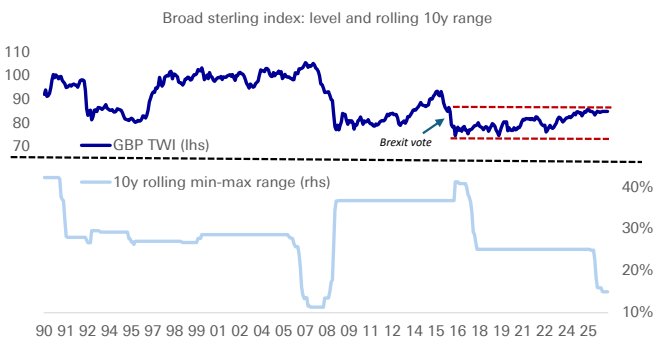
Source : Deutsche Bank, Bloomberg Finance LP, Haver Analytics

Figure 16: Pound in real terms is marginally stronger in real terms than before vote....



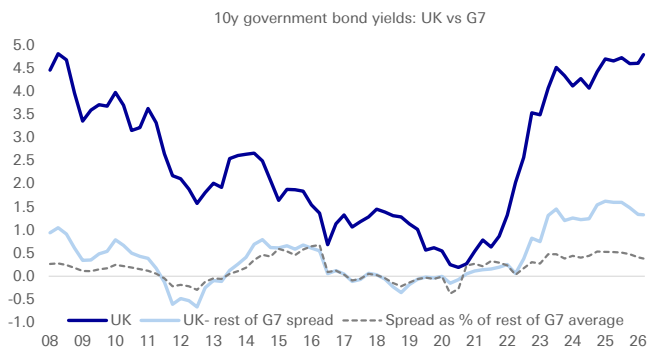
Source : Deutsche Bank, Bloomberg Finance LP, Haver Analytics

Figure 17: ..and has been very stable by historical standards since initial Brexit shock



Source : Deutsche Bank, Bloomberg Finance LP, Haver Analytics

Figure 18: Cost of UK borrowing firmly higher than peers but hard to isolate Brexit effect



Source : Deutsche Bank, Bloomberg Finance LP, Haver Analytics



Marking to market the Brexit impact: a doppelganger analysis

Real GDP -4.0% GBP 1,700 per person	Employment -2.0% 680k jobs	Prices 0.7% CPI, 4q
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Deutsche Bank SCM Analyses

The preceding two sections set out what the UK economy actually produced, invested, traded, employed, and priced over the post-referendum decade. But they cannot, on their own, answer the question that matters most to policy-makers and investors: how much of this was Brexit?

Indeed, several factors have shaped the economic outlook over the past decade. The UK experienced Brexit, a pandemic, an energy crisis, and a sharp monetary tightening cycle essentially simultaneously. Every metric examined above was influenced by all four shocks in different proportions. Disentangling them requires a counterfactual — a credible picture of what the UK economy would have looked like absent Brexit, controlling for the other shocks.

Turning to a Synthetic Control

As mentioned above, the UK — alongside other economies — experienced several shocks at the same time. Resorting to a doppelganger type framework (SCM) has several advantages.

First, it addresses the 'no simple control group' problem. Indeed, we cannot observe a 'UK without Brexit' in principle. Every conventional comparison — UK vs eurozone, UK vs G7 average, UK vs pre-referendum trend — conflates the Brexit effect with differences in economic structure, pandemic policy (furlough generosity, lockdown timing), energy exposure, monetary policy frameworks, and starting conditions. Simple comparisons, therefore, fail to estimate a 'genuine' Brexit impact.

Second, there are overlapping shocks. As noted above, Brexit was not the only problem facing the UK. The pandemic hit in 2020; the TCA came into force in 2021; monetary tightening began in late 2021; and energy prices spiked in 2022. Each of these shocks interacts with the Brexit structural effect — amplifying it in some cases (HGV shortage = Brexit × COVID), and attenuating it in others (pandemic emergency spending temporarily offsetting investment gap).

Third, phase dependency. As highlighted separately, the impact of Brexit can be traced through various phases. What began via consistent political uncertainty, led to economic disruption, which in turn led to the biggest single re-write of trade relations in UK history. Put differently, an aggregate 10-year average conceals the timing and magnitude of Brexit's effect. A good counterfactual method must be able to show when divergence began, how it evolved, and whether it has persisted.



Fourth, the self-selection problem. The UK did not 'randomly' Brexit. The vote was correlated with pre-existing economic characteristics. A counterfactual that ignores this correlation will either overestimate or underestimate the causal Brexit effect.

The synthetic control method (SCM) best solves for these issues. So, what is a SCM?

The SCM constructs a 'Synthetic UK' — a weighted composite of economies that most closely matched the UK on GDP, investment, trade, labour market, and inflation metrics in the pre-referendum period. Post-2016 divergence between the UK and its synthetic twin is then our best available estimate of the Brexit causal effect — not perfect, but transparent, visually intuitive, and methodologically rigorous. The [appendix](#) presents more detail on the SCM, including its limitations.

How do we construct the UK synthetic control?

We measure the Brexit effect on UK GDP, employment and prices by comparing the UK to a counterfactual "doppelganger" built from a donor pool of comparable economies. Here, we rely on a donor pool from OECD economies.

For GDP, employment, and prices, we use the standard Synthetic Control Method (SCM): a weighted blend of donor economies chosen to track the UK closely before the 2016 referendum, with the post-2016 divergence as the estimated Brexit effect. The SCM's weights are constrained to be positive and to sum to one, so the counterfactual is always a plausible combination of real economies.

Figure 19: SCM Controls

SCM Controls	
	Control Variables
GDP	Gross Fixed Capital Formation, Exports, Trade Openness, Government Consumption, Net FDI, Years of Schooling, R&D Investment, Employment, Professional/Science Ratio to GDP
Employment	Consumption, Gross Fixed Capital Formation, Unit Labour Costs, Years of Schooling, Trade Openness, GDP, Professional/Science Ratio to GDP, Net FDI
Prices	Unit Labour Costs, Import Prices, Export Prices, Policy Rates, GDP, Trade Openness, Employment

Source : Deutsche Bank

Impact of Brexit: results from a synthetic control

GDP: As expected, the UK economy underperformed relative to our 'UK doppelganger'. Our synthetic control doppelganger showed GDP tracking 4% stronger, relative to the UK's actual output (2025) ([Figure 20](#)). In pound terms, we estimate this to be equal to just around GBP 120bn – worth roughly GBP 1,700 per capita (in 2020 terms).

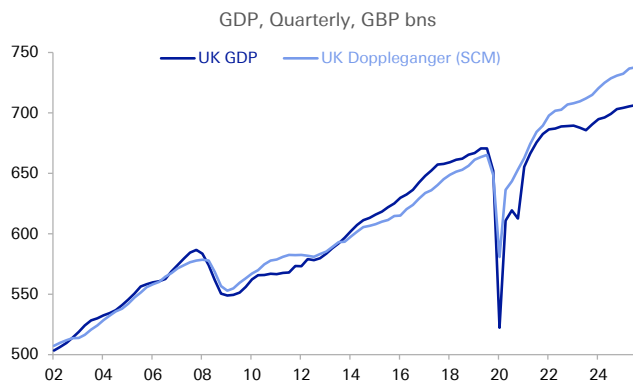
Our estimated output gap sits slightly better than other external estimates of a 6-8% GDP loss due to Brexit. However, note the breakdown in performance. The initial Brexit years, which encapsulated both the Brexit referendum shock and the period



of constitutional crisis, saw UK GDP actually outperform its doppelganger – likely a function of early monetary stimulus, competitive exports (via the FX depreciation), higher net immigration, and some stockpiling.

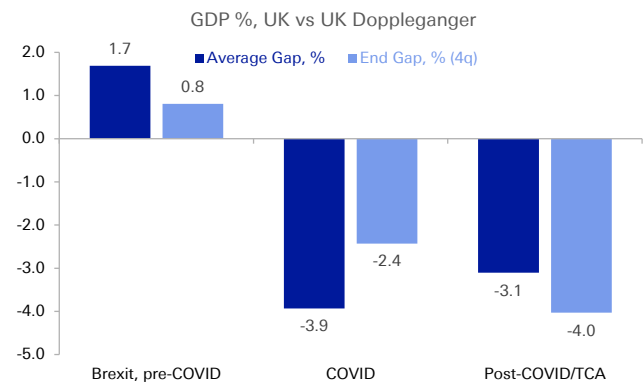
It wasn't until the pandemic began and the UK officially left the EU, that GDP started to materially underperform from its doppelganger. As of the most recent data point available, we estimate the GDP gap to have widened to 4.2% (Q4-25). It's worth noting, however, that on average since the Q2-2016, the GDP gap sits at a smaller -1.3% relative to its doppelganger – underscoring the wider variation in GDP trajectory over the last decade ([Figure 21](#)).

Figure 20: UK GDP sits around 4% lower relative to its doppelganger



Source : Deutsche Bank, Macrobond

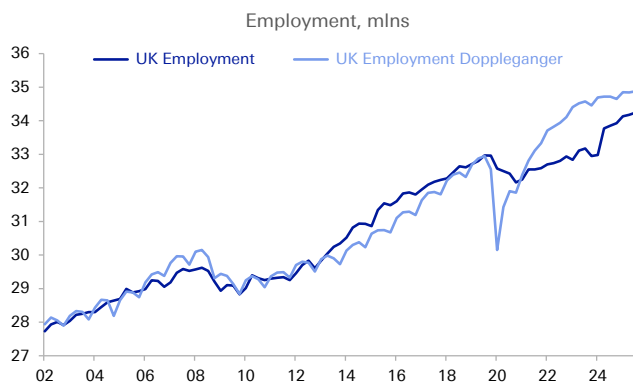
Figure 21: There is a wide variation in performance across the decade, with an output gap emerging only after 2019



Source : Deutsche Bank

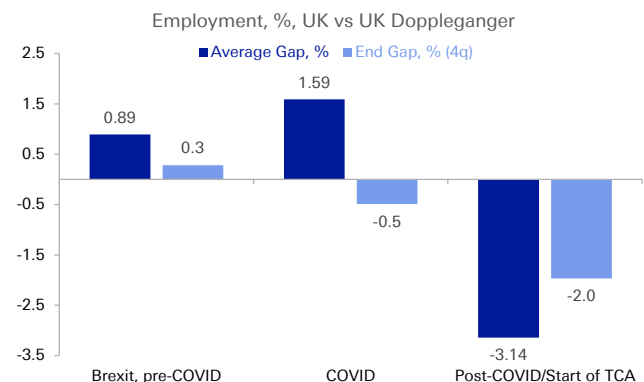
Employment: With GDP lower, employment is also likely to be lower. We estimate that the UK employment gap at the tail end of the sample (2025) sits closer to -2%. In numbers, that works out to just over 685k jobs ([Figure 22](#)).

Figure 22: We estimate the UK employment gap to be around 2%



Source : Deutsche Bank, Macrobond

Figure 23: The employment gap only really turned negative after the pandemic and the implementation of the TCA



Source : Deutsche Bank



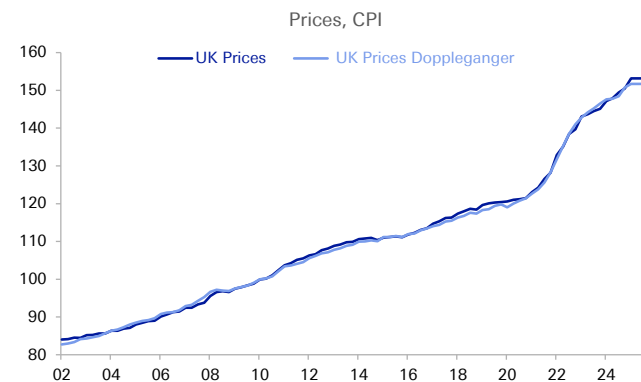
Again, we highlight the differences between the different phases of Brexit. By the end of the pre-pandemic period, employment levels were 0.3% above its estimated UK doppelganger. This, in our minds, chimes with the large inflows of workers seen just after referendum and in the run up to the pandemic where firms substituted away from capital and into cheaper labour, driving up employment. It wasn't until the pandemic period (2020-2021) where UK employment levels fell below the estimated synthetic (-0.5%). And in the TCA period (2022-2025), the gap between the UK and its doppelganger reached its widest levels, with both the average and end point showing a markedly bigger net fall ([Figure 23](#)).

Prices: Perhaps the most transparent impact of Brexit was inflation. On our SCM model, we estimate that UK prices (CPI) sits 0.7% above where it would have otherwise have been (i.e. in level terms) ([Figure 24](#)).

The end state, however, masks several phases of Brexit. The initial aftermath of the referendum saw sterling plummet, pushing CPI nearly 1% above its doppelganger. By the time the pandemic hit and the UK officially left the EU, prices were only marginally higher (0.4%) compared to the constructed UK synthetic.

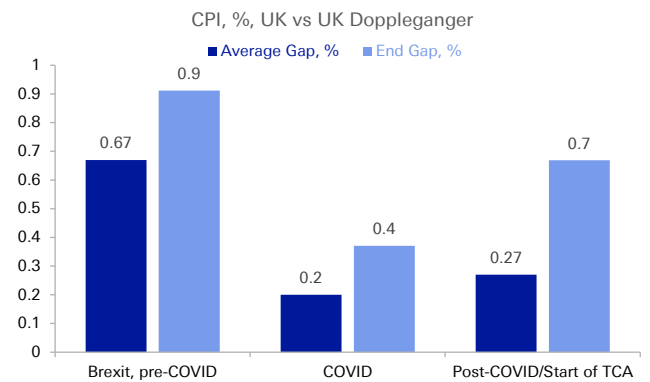
By the latest data point (2025), however, the gap between the UK and its doppelganger widened to around 0.7% ([Figure 25](#)). Overall, we estimate that over the last decade, the CPI level has tracked 0.5% higher, on average, due to Brexit forces with trade frictions, an initially weaker sterling, and structural changes to the UK labour market all likely playing some role in driving inflation higher, relative to the counterfactual.

Figure 24: CPI sits around 0.7% higher over the last year, compared to its doppelganger



Source : Deutsche Bank, Macrobond

Figure 25: CPI was stronger across all three time periods following the referendum



Source : Deutsche Bank



The Brexit Balance Sheet: what's worked and what hasn't

From setting out the UK scorecard on some of the effects of Brexit, we turn to the Brexit Balance Sheet, looking at what has worked and what hasn't. Indeed, while Brexit may not have been the success some claimed it would be, some things have worked better – while many others have not. We set them out in more detail below.

What's worked?

1/ Current account deficit has improved. In the trade scorecard we noted how the overall trade deficit had narrowed over the past decade. The income side has also improved, in part thanks to Brexit. Contributions to the EU budget ceased at the end of the transition period in 2020, but the UK's divorce payments - agreed as part of the Withdrawal Agreement - have also now tapered down to a rounding error from a balance of payments perspective. Indeed, net payments to the EU totaled nearly EUR 11bn in 2020 but by last year had fallen to under EUR 1bn. Only EUR 6bn of obligations for the next forty years remains [outstanding](#). As such, the lesser explored secondary income component of the UK current account has materially improved from a decade ago - from averaging 1.3% of GDP in the 5 years before the Brexit vote to just over 0.5% of GDP in 2025 ([Figure 27](#)).

2/ Free trade across Irish border preserved. [Figure 28](#) shows how trade between the Republic of Ireland and Northern Ireland jumped up immediately once the TCA came into effect in January 2021. With Northern Ireland de facto remaining part of the EU customs union with a border instead erected across the Irish Sea, goods trade on the island has thrived and is much higher than before the referendum.

3/ Regulatory Autonomy. The UK's freedom to set its own regulatory environment is the most significant theoretical advantage of Brexit. This has allowed the UK to move faster than its European counterparts in several areas, including:

- **Financial Services:** The Edinburgh Reforms (2022-23) introduced a more outcomes-based, less prescriptive prudential and conduct framework. The removal of the EU bonus cap for bankers (2023) was symbolically significant; more substantively, the UK has maintained its lead in derivatives trading volumes (LCH/LME), with the EU's attempts to force euro-denominated clearing to the continent only partially successful.
- **Life Sciences:** The MHRA's independence from the EMA has allowed faster clinical trial approvals and novel regulatory frameworks for advanced therapeutics (ATMP). The COVID vaccine programme was the most visible example; the MHRA's Innovative Licensing and Access Pathway (ILAP) has since attracted several early launches ahead of EU approval.
- **Artificial Intelligence:** The UK has deliberately positioned itself as a lighter-touch regulatory jurisdiction relative to the EU's AI Act. The 2023 AI Safety Summit at Bletchley Park and the establishment of the AI Security Institute have given the UK a soft-power position in global AI governance. This has allowed the UK to take a leading role in Europe when it comes to AI deployment and innovation.



- **Green Finance:** The UK's post-Brexit Sustainability Disclosure Standards diverge from EU SFDR in ways that some institutional investors consider more workable, potentially attracting more ESG capital.

4/ Independent Trade Policy. Post-Brexit, the UK has signed or rolled over approximately 70+ trade agreements covering economies representing around 67% of UK exports.

New agreements include the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP, formally joined May 2023) and updated deals with Australia and New Zealand (UKFTA 2023). The economic scale of these agreements is, to be direct, modest in aggregate — the CPTPP is estimated to add approximately 0.08% to UK GDP by the late 2030s.

Figure 26: The UK has been busy signing trade deals

Trade Deals				
Trade Agreement	Type	Status	GDP Impact (est)	Key Sectors
EU-UK TCA	Goods FTA / Services partial	In force Jan 2021	Net negative vs EU membership	All sectors
CPTPP	Regional trade agreement	In force May 2023	+0.08% GDP (long run)	Agri, autos, services
UK-Australia FTA	Bilateral FTA	In force May 2023	+0.08% GDP (long run)	Agriculture, finance
UK-New Zealand FTA	Bilateral FTA	In force Feb 2024	+0.03% GDP (long run)	Agri, digital
UK-India FTA	Bilateral FTA	Negotiations ongoing	+0.2% GDP (est.)	IT services, pharma
UK-US FTA	Bilateral FTA	Stalled / under review	Contested	Food, pharma, finance

Source : Deutsche Bank, UK Government

5/ Subsidy Flexibility. Outside EU state aid rules, the UK has been able to deploy targeted industrial subsidies at scale. The Freeport programme (10 English freeports operational by 2025) provides tax incentives and customs flexibility that replicate some single market benefits in a localised way. Net investment linked to freeport locations has been estimated at several billion over the programme period. Additionally, the [UK's Semiconductor Strategy](#) and advanced manufacturing support programmes benefit from freedom to exceed EU de-minimis thresholds.

What's not worked?

1/ UK trade is more unbalanced between goods and services than ever before.

Increased frictions with the EU have decreased overall levels of trade in goods between the two blocs. From a deficit perspective, the trade balance in goods has widened, most notably in manufactured goods and food and drink. On the flip side, UK services trade continues to boom, especially telecoms and IT as well as "other business services" which captures things like law, management consultancy, and accountancy. Geographically, the UK's services surpluses with both the US and the EU have increased materially in recent years ([Figure 29](#)). At a high level, since leaving the EU, the UK has leaned further into its comparative advantage on services rather than rebalancing towards manufacturing and goods trade ([Figure 30](#)). Indeed, the UK is now far more unbalanced than countries of a similar wealth ([Figure 31](#)).

2/ Non-tariff barriers and trade friction. The TCA delivered zero tariffs and zero quotas – but raised substantial costs via non-tariff barriers (NTBs) on UK-EU trade. The TCA eliminated tariffs on goods meeting rules of origin requirements, but the reintroduction of customs declarations, regulatory border checks, and rules of



origin compliance has imposed significant costs on UK-EU goods trade. Academic work from the LSE CEP, CEPR, and the UK Trade Policy Observatory [saw](#) a 15-25% reduction in UK goods trade intensity with the EU attributable to these NTBs, after controlling for the pandemic and other factors.

A recent [paper](#) by the Productivity Institute suggested that Brexit led to a drop in imports of 23.7% and exports by 18.6%, with business investment also suffering by 2.5% (as a result of the drop-off in trade intensity). A separate [study](#) showed that non-tariff barriers raised prices by 6% (implying a pass-through of 50-80%) with households losing nearly GBP 5.8bn in added welfare costs.

3/ Regional policy. EU regional policy was a major, long-running transfer to the UK's poorer regions: between 1975 and 2020 the UK [received](#) around GBP 66 billion from Brussels – historically supporting west Wales, Cornwall, the Scottish Highlands, Merseyside and south Yorkshire. Had the UK stayed in, external [analysis](#) suggests it would have been entitled to around EUR 13 billion of structural funds for 2021–27 – a 22% increase on the prior period, as worsening regional inequality pushed more areas into the “less developed” category. The replacement – the UK Shared Prosperity Fund – has been widely judged to fall short: the Welsh Government estimates it alone faces a [shortfall](#) of around GBP 772 million against what EU funds would have delivered over 2021–25.

To add to this, empirical work show that UK regions underperformed their counterparts elsewhere. Fetzer & Wang (2020) [showed](#), via local synthetic controls (local doppelgangers) that in 2016–2019 many areas grew more slowly than their no-Brexit counterfactual. Interestingly, they also found that costs tended to be higher in areas that voted to leave the EU, had larger manufacturing sectors and more low-skilled workers – i.e. the places promised the most gains bore disproportionate early costs.

The grey zone: neither worked nor didn't work

1/ Some equity sectors have continued to outperform, some have continued to lag In the market's scorecard we highlighted how in aggregate both UK major and midcap equity indices have underperformed peers, most notably over the past few years. However, the sectoral breakdown presents a more nuanced picture, with a different story comparing UK sectors to their European peers and to their US equivalents ([Figure 32](#)). Interestingly, against European peers, some sectors that outperformed during the uncertainty period have largely continued this trend since 2020 (most notably materials), though others such as financials have lagged. Perhaps most striking of all, however, is the underperformance of technology shares (despite strong contributions to UK GDP growth), in both of our two Brexit periods and relative to both European and US equivalents.

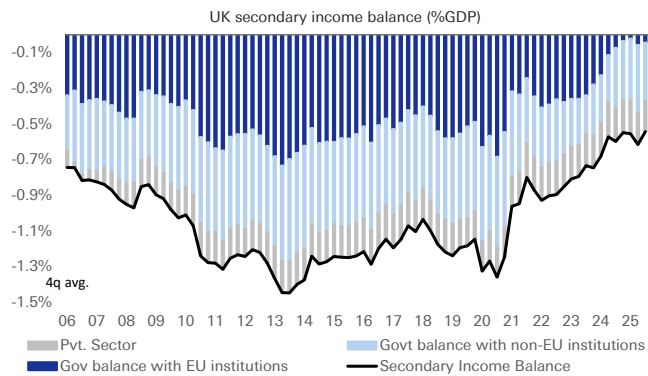
2/ Defence and security. For most of the post-Brexit period defence barely registered on the EU–UK agenda, with NATO and bilateral frameworks (such as the 2010 Lancaster House treaty with France) serving as the default. The geopolitics of 2025 changed that abruptly: the May 2025 EU–UK summit produced the first-ever Security and Defence Partnership, establishing structured dialogue across maritime, space, cyber and hybrid-threat domains. In November 2025, talks on UK participation in the EU's EUR 150 billion “Security Action for Europe” (SAFE) defence-procurement programme broke down, exposing persistent post-Brexit friction even as third countries like Canada moved to associate. The UK's eventual



role as a defence leader in Europe remains in question, and arguably has left the UK more exposed in a post-Brexit world.

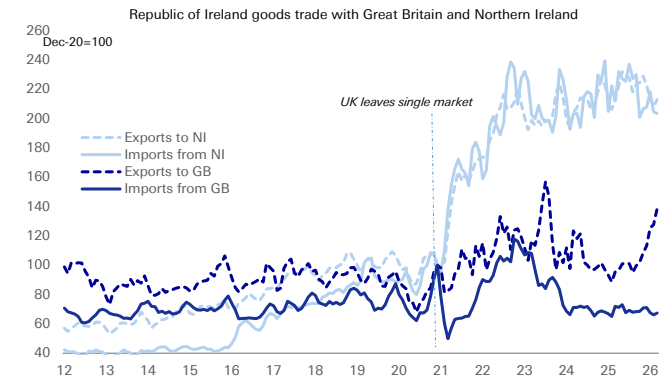
3/ Immigration. Ending free movement with the EU was meant to cut migration; instead, overall net migration rose to a peak of over 900,000 in 2023 before falling back to 431,000 in 2024. The CER–Portes counterfactual [found](#) that by 2024, Brexit had reduced EU-born workers by about 2.3% of the workforce, offset by a slightly larger rise in non-EU workers of around 2.95%, for a net increase of roughly 207,000 foreign-born workers — about 0.6% of the total workforce.

Figure 27: UK saves c. 0.6% of GDP on payments to EU with divorce payments having now tapered down



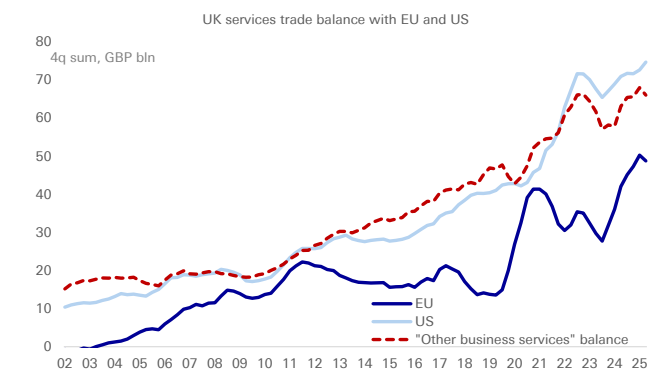
Source : Deutsche Bank, Haver Analytics

Figure 28: Brexit deals have successfully preserved free movement of goods across the Irish border



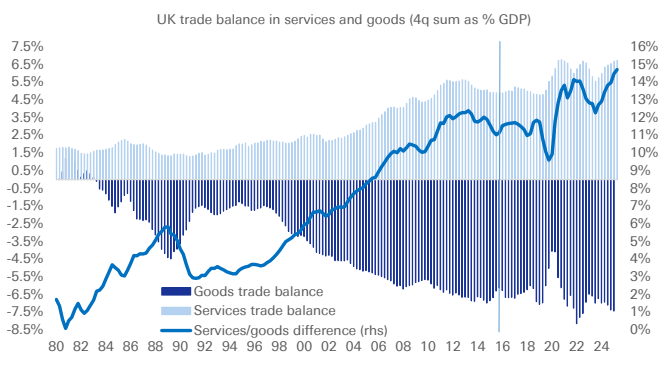
Source : Deutsche Bank, Haver Analytics

Figure 29: UK services surplus has grown with both the EU and the US....



Source : Deutsche Bank, Haver Analytics

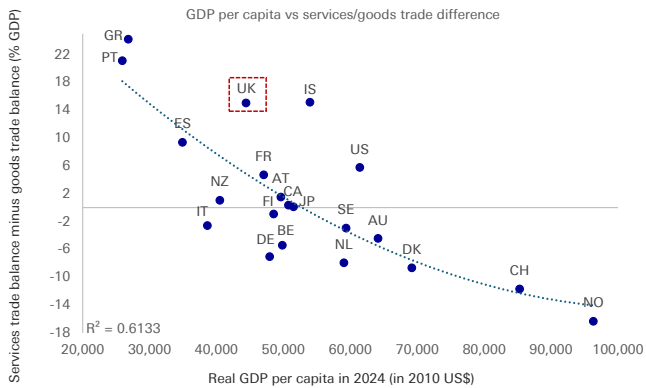
Figure 30: ..but overall UK trade more unbalanced than ever between service surplus and goods deficit



Source : Deutsche Bank, Haver Analytics

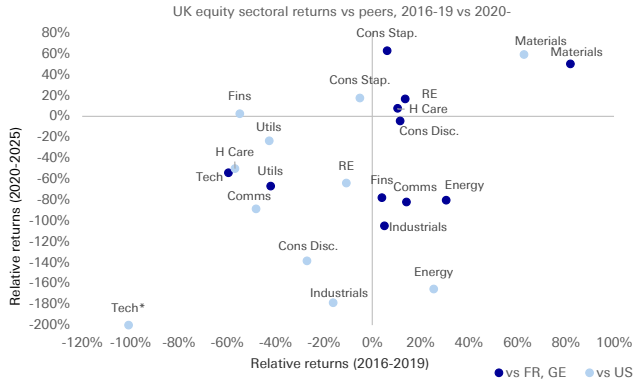


Figure 31: UK far more unbalanced between goods and services trade than nations of comparable wealth



Source : Deutsche Bank, OECD, Haver Analytics

Figure 32: Sectoral breakdown shows a mixed picture vs Europe and the US over the past decade



Source : Deutsche Bank, Bloomberg Finance LP
*tech vs US truncated for chart visibility



The Look Ahead: What's Next?

The impact of Brexit – though uncertain in scale – remains clear: the UK has lost output, employment, and has seen a larger cost of living shock.

So, what next? In this section, we look at the potential paths ahead for the UK. First, we look at how the UK can build on the current TCA deal. Second, we ask the big medium-term question: is there a path back for the UK into the EU? And, if so, what form could this take? Finally, we end with some thoughts on what (and how) the UK should consider as it rethinks its relationship with its largest trading neighbour.

The near-term questions: how can the UK improve on the TCA?

While policy debates have centered around the bigger (perhaps politically challenging) questions, there's more the UK can do to improve on the current TCA.

Recent policy proposals have focused on reducing trade and non-trade barriers between the UK and EU. Progress on these could have a measurable impact on UK GDP. Such 'Brexit reset' proposals can be summarised below:

Figure 33: Brexit reset proposals

Improving the TCA		
Policy proposal	Description	Main barrier reduced
Sanitary and Phytosanitary agreement (SPS)	Simplifies border checks and provides for dynamic alignment of food safety and animal welfare standards with the EU.	Agricultural and food product trade barriers
Visa and transport barriers for touring artists	Introduces targeted visa waivers for performers and technical crews and reduces transport frictions for longer tours.	Cross-border mobility
Mutual recognition of professional qualifications	Supports recognition of regulated professional qualifications, including in sectors such as nursing and legal services.	Labour mobility
Youth Mobility Scheme	Allows UK and EU 18–30-year-olds to move for work, study or other purposes for up to four years, with proposals also covering domestic tuition-fee treatment.	Labour, education and mobility barriers
Emissions Trading Scheme linkage	Links the UK and EU carbon markets, reducing carbon-price-related frictions and potential CBAM-related costs for covered trade.	Trade frictions
Product safety and metrology	Aligns UK product-safety and metrology rules with EU requirements, reducing duplicate compliance checks and easing market access for traded goods.	Regulatory barriers

Source : Deutsche Bank

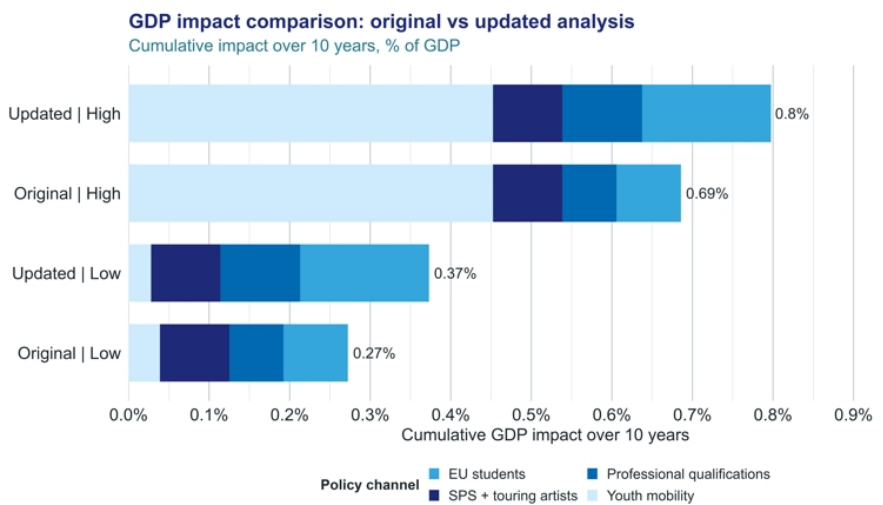


We quantify the GDP impact of the first four policy proposals in the table. Following Springford (2024), we update the underlying data and present a range of estimates for their effect on UK GDP. These calculations are approximate and draw on available OBR estimates and relevant academic literature. The estimates measure the cumulative impact on UK GDP over a 10-year horizon, relative to a counterfactual in which the policies are not implemented.

Overall, we estimate a cumulative GDP impact of +0.4–0.8% over 10 years. Our updated estimates are slightly higher at the top of the range, mainly because of the larger assumed effect from mutual recognition of professional qualifications. If the UK were treated as re-entering the relevant EU labour market, UK workers would account for around 15% of regulated professionals. Combined with higher assumed inflows of EU regulated professionals than outflows of UK regulated workers, this produces a somewhat stronger effect than in the original estimates. The low and high estimates vary primarily with assumptions about take-up of the Youth Mobility Scheme (see the appendix for more detail).

The chart below compares Springford’s original estimates with our updated estimates:

Figure 34: Estimated impact of a Brexit reset



Source : Deutsche Bank, Springford (2024)

Because the Youth Mobility Scheme is an important source of uncertainty, we test two alternative assumptions: first, we assume a diminishing annual growth rate providing a more realistic projection of scheme take-up; and second, we vary the productivity assumption using OBR scenarios in which additional migration has a positive, negative or neutral effect on potential GDP. The effects of GDP are broadly negligible here, widening the overall impact to 0.33-0.84%.

Overall, the results suggest that the GDP effects are material, although the exercise does not capture wider general-equilibrium effects. The estimates should therefore be interpreted as a partial assessment of the potential impact of Brexit reset policies.



Finally, these estimates *exclude* the potential effects of UK-EU ETS linkage and product safety and metrology proposals. The table below summarises the main channels through which these policies could affect the UK economy:

Figure 35: UK-EU ETS and product safety changes can also boost GDP materially

Other policies to consider		
Policy proposal	Channels	Impact on GDP
UK-EU ETS Linkage	1. Lower UK carbon prices than otherwise, reflecting increased liquidity and access to the EU ETS market. 2. Lower costs for UK exporters by removing the need to pay a carbon price differential.	1. Lower intermediate input costs due to lower carbon credit prices, with potential pass-through to consumer prices and GDP. 2. Support for export competitiveness in affected sectors, raising export receipts and GDP growth.
Product safety and metrology	Reduces duplicate compliance and conformity-assessment costs for firms trading goods with the EU.	Lower compliance costs could improve exporter competitiveness and support GDP growth.

Source : Deutsche Bank

Including these additional proposals would likely raise the estimated GDP impact, as both ETS linkage and product safety and metrology could reduce costs for firms and support trade competitiveness.

Overall, the estimates point to a positive but partial view of the GDP impact from the Brexit reset proposals. However, they should provide some indication of the overall effects of adopting each of these proposals.

The big medium-term questions: is there a path back?

Beyond the very near term, the bigger question remains: is there a path towards closer alignment with the EU? In this section, we set out some of the thinking around the UK's options. Here, there are two: one, building on the TCA and pushing towards either a customs union or single market; or two, rejoining the EU.

Building on the TCA: customs union or single market?

The previous section's balance sheet pointed to a single asymmetry: Brexit's costs are concentrated at the core — via re-erected trade and regulatory friction with the UK's largest market — while its gains sit at the margin.

Any forward path therefore is based on how much of that core friction can be removed, and at what price in sovereignty. The two structural options on the table are a customs union and single-market membership (or a close emulation of it). They are not the same thing, they remove different frictions, and the empirical literature attaches very different magnitudes to each.

Customs Union: A customs union would align the UK's external tariff with the EU's and remove the customs border between them, which does two concrete things.



First, it eliminates rules-of-origin compliance. This is the requirement to prove what share of a product originated in the UK or EU to qualify for zero tariffs under the current TCA. Such compliance costs are [estimated](#) to add 2–8% to the cost of exporting goods, the burden falling heaviest on heavy industry and integrated manufacturing supply chains. The friction is real and quantified: in 2021, the TCA [reduced](#) the count of product-destination export relationships with EU countries per quarter by around 30%.

Second, a customs union would restore frictionless participation in European supply chains, where components cross the Channel multiple times before final assembly — automotive and chemicals being the textbook cases. Under a customs union those repeated crossings no longer trigger origin checks. A UK–EU mutual-recognition agreement on conformity assessment — a natural addition to a customs union — has been [estimated](#) to raise UK goods exports to the EU by 10% on average, and by up to 25% in chemicals and machinery.

What's the biggest limitation of a customs union? A customs union is a goods-only instrument. It does nothing for services. It also removes the UK's freedom to strike independent goods-trade deals, since the common external tariff would be set in Brussels. And it does not, by itself, remove SPS (food-safety) border checks, which require a separate veterinary agreement (something that would need to be agreed separately).

Single Market: The single market is a regulatory union: product and service regulations are harmonised or mutually recognised, so that non-tariff barriers between members fall to virtually zero. Re-entry would remove the bulk of the friction Brexit created, and crucially it would do so for services — the UK's structural strength.

Beyond the direct trade channel, single-market membership delivers a productivity dividend that a customs union cannot. Deeper market integration intensifies competition. OBR [estimates](#) suggest that the biggest cost from Brexit in the long run can be attributed to a productivity channel — a 4% long-run productivity loss. Re-entry would also restore the labour-supply flexibility, investment certainty and FDI attractiveness that the uncertainty channel of Brexit eroded.

What would a single market entry cost? The single market's price is sovereignty. Membership requires accepting the four freedoms — including freedom of movement — and submitting to EU regulation designed without a UK seat at the table. It also implies a budget contribution. These are precisely the red lines that successive governments have ruled out, which is why the single market is the larger economic prize but the smaller political probability.

To rejoin the EU or not to?

A key consideration when assessing the UK's longer-term relationship with the EU is public support, and what the UK public thinks about Brexit and its future.

This [piece](#) by polling expert John Curtice nicely summarises the state of play. When asked how they would vote in a hypothetical second referendum, polls on average this year have shown around 60% support for being in the EU, vs 40% for staying out. Brexit is seen by a majority of voters to have had a negative impact on both the economy and immigration. Interestingly, given the large scale of non-EU net migration into the EU over the past few years noted above, the view that it has had a negative impact on the view towards immigration is equally held by those who

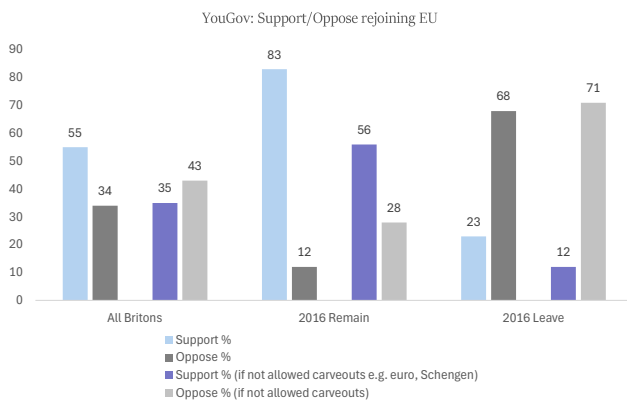


voted Remain as those who voted Leave in 2016.

To be sure, when given more context about the [potential preconditions](#) for rejoining the EU, the picture amongst voters becomes more split. For instance, if rejoining the EU had to be accompanied by the UK adopting the euro, polling suggests UK public support drops materially per [YouGov](#), to 35% ([Figure 36](#)). On the economic side, UK fiscal picture would have to improve to meet the convergence criteria for the euro, both in terms of running a narrower fiscal deficit and lowering (or illustrating credible path towards lowering) debt/GDP ([Figure 37](#)).

Here, the wider continental context is important. It is not inconceivable that the next decade or two sees a variety of countries across northern and eastern Europe vote on either joining the EU (e.g. [Norway](#), [Iceland](#)) or on adopting the euro (e.g. [Sweden](#), [Hungary](#), Czechia). By the time that the UK were to hold any hypothetical vote on closer ties, the bar for the EU to allow the UK to rejoin on its pre-Brexit terms would likely be even higher. Even now, the relationship with the UK does not feature near the top of the EU's priority list. We explore this and the wider view of Brexit from across the channel in the next section.

Figure 36: UK public supports rejoining EU, but only on similar pre-Brexit terms



Source : Deutsche Bank, YouGov

Figure 37: UK would likely need material fiscal consolidation if euro membership was made a condition to rejoin

Euro convergence criteria			
	Rule	UK latest reading	EU reference rate or target
Inflation	Less than 1.5% above 3 lowest EMU member states	3.0%	2.9%
Long-term yields	No more than 2 ppts above three best EMU member states	4.6%	3.4%
Fiscal deficit	Not exceed 3% GDP	4.2%	3.0%
Debt/GDP	Gross debt to GDP below 60% or diminishing at satisfactory pace	94.0%	60.0%
Currency volatility	2 years in ERM II where EUR/GBP trades in +/- 15% range of agreed central rate	7.3%*	+/- 15% range from central rate

Source : Deutsche Bank, Haver Analytics, Bloomberg Finance LP

What about the EU side?

Pragmatic rapprochement without reintegration. The UK's relationship with the EU ten years after the 23 June 2016 referendum is best understood as a shift from membership to structured third-country cooperation. Institutionally, the relationship is now more structured than it was immediately after Brexit, with annual summitry and multiple negotiation tracks under the 2025 "reset" (see [here](#)). Substantively, however, it remains selective and transactional rather than integrated. The most likely near-term outlook from an EU perspective is incremental deepening in sectors where geopolitical pressure and economic interdependence are strongest.

The UK-EU reset has produced real, but limited results so far. The reset in UK-EU relations has produced real but limited results—primarily on defence, food trade and energy. Stumbling blocks on further progress are disagreements on financial contributions by the UK in return for access to EU defence loan programs and



access to the joint electricity market.

- On defence, the first joint UK-EU [summit](#) after Brexit in May 2025 produced a Security and Defence Partnership. However, the fact that both sides could not find a compromise on the level of the UK's financial contribution for allowing British firms to participate in the EU's EUR 150bn SAFE defence loan program, is a setback in defence integration.
- On food trade, the UK and EU have agreed to work towards a Sanitary and Phytosanitary agreement on food and agricultural standards, which the UK hopes to be in play by 2027.
- And on energy, the EU Council green-lighted talks on electricity market integration in early 2026. However, the EU will likely insist that in return for access to the EU single market for electricity, the UK should contribute to EU cohesion payments, like [Norway](#) and [Switzerland](#).

UK-EU relations do not feature prominently at the top of the EU agenda. The reality is while the UK continues to debate what its future with the EU looks like, the EU has other more pressing issues.

The EU Commission's key policy priorities are centred around managing trade relations with the US and China, supporting Ukraine and improving its competitiveness (see our recent note [here](#)). As such, improving UK-EU relations are of lower-order importance, although deeper defence, electricity and trade integration would be beneficial for both sides (e.g. by better reaping the potential of North Sea wind). However, there is little in UK-EU relations so far that offers tangible benefits for national leaders in EU capitals and thus, negotiations remain at the technical rather than strategic level.

Relationship remains constrained by UK red lines and EU caution. The political will exists on both sides to further deepen ties as global geopolitical developments strongly favour selective rapprochement. But the relationship remains constrained by UK red lines (not rejoining of the single market or the customs union, not reopening freedom of movement) and EU caution and fear of cherry-picking. Both sides can be described as transactional in their approach limiting the willingness to be creative in improving their relationship.¹

Next critical milestones to watch. The TCA review and the upcoming 22 July 2026 summit are the next critical milestones. Legally, the 2026 review is anchored in Article 776 TCA, which provides for a review of implementation five years after entry into force and every five years thereafter. The review does not constitute a commitment to reopen the TCA and the EU prefers a "light-touch" assessment focused on implementation. EU council decisions on the youth experience scheme (see [here](#)), a common sanitary and phytosanitary area (see [here](#)) and emission trading (see [here](#)) show that the preferred EU-UK method in the reset phase is not to reopen the TCA, but instead, to build targeted new agreements alongside it. For areas where the EU thinks that the UK benefits more than the EU from new agreements, other package deals are likely to be asked for.

¹ See also CER, EU-UK relations - Will 2026 be the year to reset the reset?, February 2026



A hypothetical path ahead

The path ahead for the UK remains riddled with uncertainty. Politically, the push for further integration with the EU remains slow and disparate. Economically, the case for integration remains strong.

Social perception of Brexit has also changed in the past decade. More people view Brexit as a mistake. And the latest polls suggest that the UK would command a majority to rejoin the EU – but only under the right circumstances.

The main question, however, is at what cost would further integration come? If the UK could retain some of the exemptions that it benefitted from prior to its departure from the EU alongside maintaining some of the trade, regulatory and immigration freedoms, the case to rejoin the EU would be strong.

The reality, however, is that any future deal with the EU won't have as many exemptions. Politically, as well, no major political party has pushed a mandate for wholesale changes to the UK's relationship with the EU. Moreover, as we have noted, it takes two (or really 28 countries) to tango. Fears around any rapprochement of UK-EU reintegration due to fragmented politics (and remaining EU scepticism) will likely hinder any progress on a better deal.

So, what should the UK do? A sensible approach could be for a two-staged approach. The first, given the political constraints, would be to build on the current TCA to close some of the gaps in non-tariff barriers left as part of the original Brexit deal. This would include things like implementing the agri-food deal, establishing a Youth Experience Scheme, discounting university prices for EU students, reducing rules of origin checks, and agreeing on mutual recognition of qualifications. Our own analysis suggests that this could boost output meaningfully, scraping back some of the lost output in GDP. Doing so could even give the UK enough of a growth uplift to partially offset its diminishing fiscal headroom.

The second-stage would involve broadening out the UK's integration with the EU on things like defence, AI, and financial services. Should there be political appetite to push further towards the EU via either joining the Single Market, customs union, or re-joining the EU, no move should be made without a clear public mandate in order to avoid the political paralysis that followed after the Brexit referendum. The UK should also look to prioritise carve outs on its existing trade deals (though we think this could be difficult for the EU to agree to).

To be sure, in the current global ecosystem, the UK should prioritise its ability to retain flexibility in sectors of the future such as AI. Its ever growing reliance on services relative to goods - far larger than nations of comparable wealth - means that the UK may still benefit from use of its increased sovereignty and flexibility, which Brexit in theory has allowed for.



Appendix

A. The Synthetic Control Method

A1. The counterfactual problem

The impact of Brexit cannot be simply measured by a before-and-after comparison. Why? Because the UK simultaneously experienced a pandemic, an energy crisis and a sharp monetary tightening. Isolating the Brexit-specific effect requires a credible counterfactual — an estimate of what UK GDP, employment and prices would have done had the 2016 vote gone the other way, holding the other shocks constant. The synthetic control family constructs that counterfactual from a donor pool of economies that did not undergo Brexit.

A2. Standard SCM (used for GDP)

The Synthetic Control Method (Abadie, Diamond & Hainmueller, 2010) builds a "synthetic UK" as a weighted average of donor economies. The weights are chosen so the synthetic UK matches the actual UK as closely as possible across the pre-referendum period, on both the outcome's own trajectory and a set of economic predictors. Two constraints are central: the weights are non-negative and sum to one. This makes the synthetic UK a convex combination of real economies — it can never lie outside the range of the donor pool, so the counterfactual is always economically plausible and the method cannot extrapolate to an implausible level.

The post-2016 gap between the actual UK and its synthetic counterpart is the estimated Brexit effect. Because the outcomes are modelled in logs, this gap reads directly as a percentage. Statistical credibility rests on small pre-treatment fit error (RMSPE), confirming the synthetic tracks the UK before treatment.

Why is the SCM the right tool? The UK exhibits a tight convex match on GDP, employment, and prices (low pre-RMSPE), for one. Additionally, GDP's principal post-treatment confound (the pandemic) enters largely as a level effect that the convex blend handles well. The SCM's constrained form — its core strength — is not a binding limitation here.

A3. SCM Weights

We include our estimated weights for our SCM (including factor-augmented) below for our key three variables: GDP, employment, prices.



Figure 38:SCM weights

SCM Weights			
	GDP	Employment	Prices
Australia	39.2%	4.0%	33.7%
Austria		0.6%	54.7%
Canada		11.3%	0.3%
Denmark	0.1%	0.5%	3.7%
France		1.1%	0.1%
Germany			0.1%
Greece		0.5%	0.1%
Hungary	17.9%	0.8%	5.9%
Ireland		1.2%	
Italy	15.9%	0.7%	0.3%
Japan		0.3%	
Netherlands		21.8%	
New Zealand		4.5%	0.1%
Norway		0.8%	0.2%
Portugal	4.3%	0.6%	0.1%
South Korea		0.5%	
Spain		2.3%	0.2%
Sweden		1.6%	0.3%
Switzerland	22.3%	2.5%	
United States		44.4%	

Source : Deutsche Bank

References: Abadie, Diamond & Hainmueller (2010), JASA; Abadie (2021), JEL; Xu (2017), Political Analysis; Bai (2009), Econometrica.

A6. Limitations of SCM

Our approach to estimating the impact of Brexit (thus far) has its own limitations. We set them out below.

Synthetic control. The synthetic control method, for all its transparency, rests on assumptions that aren't perfect.

The counterfactual is only as good as the donor pool: it assumes some weighted combination of economies can credibly reproduce the UK's pre-referendum trajectory, and that those donors were not themselves materially affected by Brexit or by idiosyncratic shocks of their own over the estimation window.

The method also assumes no anticipation — that the UK's pre-2016 path was not distorted by the prospect of the referendum — and that the relationship between the UK and its synthetic twin, calibrated before 2016, would have held absent the treatment. A close pre-treatment fit makes these assumptions more plausible but



cannot prove them.

Inference is another constraint: the synthetic control has no conventional standard errors. Estimates are also sensitive to specification choices — the donor set, the predictor list, and the treatment date — and the further the analysis runs beyond the referendum, the more the accumulating overlay of unrelated shocks (a pandemic, an energy crisis, a monetary tightening cycle). This erodes the cleanliness of the counterfactual. We mitigate these concerns by ensuring a tight pre-treatment fit, but they cannot be eliminated.

Overall, use of a SCM does not deliver a definitive causal estimate; it merely provides a disciplined, transparent counterfactual under stated assumptions. We regard the convergence of our results with independent external estimates as more informative than any single point estimate. All figures should be read as central estimates surrounded by genuine, if unquantified, uncertainty.

B. Detailed workings on TCA adjustments

This appendix sets out the sensitivity analysis used to estimate the GDP impact of selected Brexit reset proposals. The approach updates Springford (2024) with more recent data and tests how the results vary under alternative assumptions. We refer to the original article for the full workings behind these calculations.

The quantified policies are the first four proposals listed in the main table.

- ETS linkage and product safety and metrology are treated qualitatively, given the broader range of assumptions and data required for a robust estimate.
- The recent data used to update the estimates are as follows:
- SPS: Updated UK agrifood exports and imports to and from the EU to 2025.
- EU students: Updated the number of EU-domiciled students in the UK to 2024/25. We assume that universities can accommodate the relevant increase in demand fully (given the drop in student visa applications).
- Mutual recognition of professional qualifications: Updated total intra-EU recognition decisions to 2024.
- Youth Mobility Scheme: Updated Eurostat data on young people in employment in the EU, HMRC data on EU young people in payrolled employment, and UN data on UK young people in the EU to 2024/25.

The scenario assumptions are as follows:

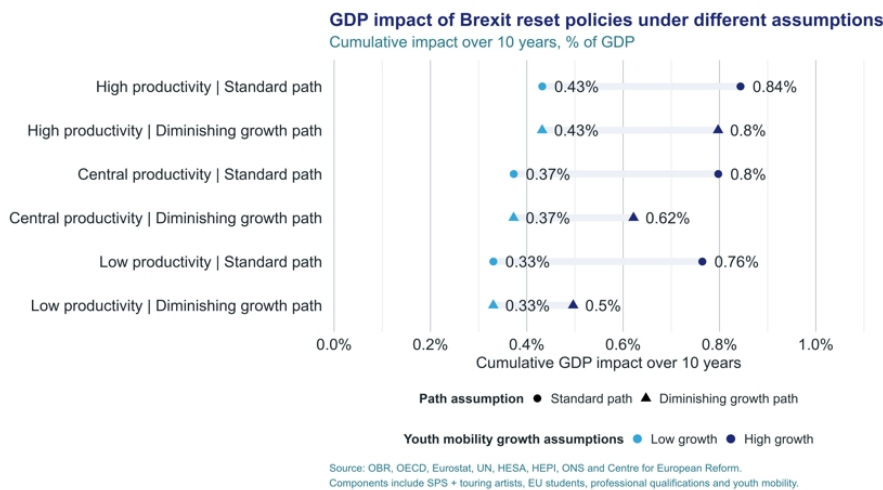
- Path assumptions: Instead of applying a constant average annual growth rate, we use a diminishing growth rate for projected take-up of the Youth Mobility Scheme. We model this using a simple AR(1) process that converges to a steady growth rate.
- Productivity assumptions: We apply OBR estimates for the impact of additional migration on potential GDP.
 - The central assumption assumes additional migrants have the same productivity per worker as the existing population, for example if capital per worker is unchanged or any dilution is offset by higher total factor productivity growth. Under this assumption, GDP is around 1.5% higher.



- The low assumption assumes lower overall productivity, for example due to lower capital per worker, resulting in a smaller GDP increase of around 1%.
- The high assumption assumes higher productivity, reflecting additional migrants having a much higher participation rate than the adult UK population. Under this assumption, GDP is around 2.2% higher.

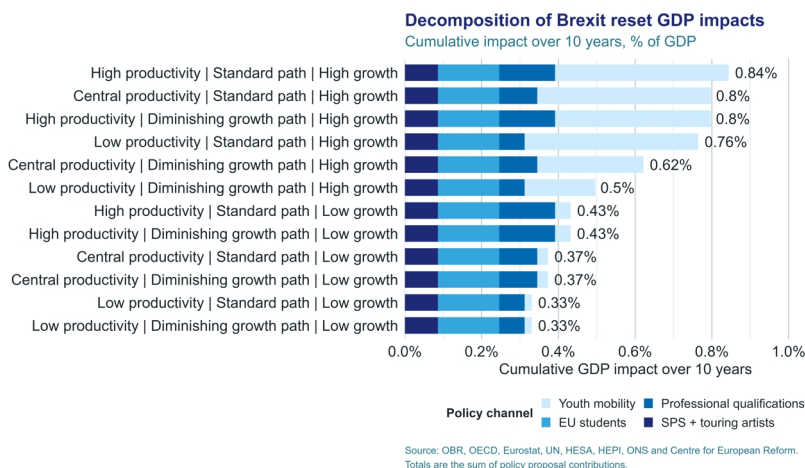
The figures below provide a more detailed overview of the policy proposals and their contributions to the estimates.

Figure 39: Brexit reset policies under different assumptions



Source : Deutsche Bank

Figure 40: Decomposition of Brexit reset GDP impacts



Source : Deutsche Bank



Appendix 1

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